

me 338a
continuum mechanics

lecture notes

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lecture notes 01

tuesday, january 05th, 2010

1 Tensor calculus

1.1 Tensor algebra

1.1.1 Vector algebra

Notation

- Einstein's summation convention

$$u_i = \sum_{j=1}^3 A_{ij} x_j + b_i = A_{ij} x_j + b_i \quad (1.1.1)$$

summation over indices that appear twice in a term or symbol, with silent (dummy) index j and free index i , and thus

$$\begin{aligned} u_1 &= A_{11} x_1 + A_{12} x_2 + A_{13} x_3 + b_1 \\ u_2 &= A_{21} x_1 + A_{22} x_2 + A_{23} x_3 + b_2 \\ u_3 &= A_{31} x_1 + A_{32} x_2 + A_{33} x_3 + b_3 \end{aligned} \quad (1.1.2)$$

- Kronecker symbol δ_{ij}

$$\delta_{ij} = \begin{cases} 1 & \text{for } i = j \\ 0 & \text{for } i \neq j \end{cases} \quad (1.1.3)$$

multiplication with Kronecker symbol corresponds to exchange of silent index with free index of Kronecker symbol

$$u_i = \delta_{ij} u_j \quad (1.1.4)$$

- permutation symbol e_{ijk}^3

$$e_{ijk}^3 = \begin{cases} 1 & \text{for } \{i, j, k\} \dots \text{even permutation} \\ -1 & \text{for } \{i, j, k\} \dots \text{odd permutation} \\ 0 & \text{for } \dots \text{else} \end{cases} \quad (1.1.5)$$

Euclidian vector space

- consider linear vector space $\mathcal{V}^3$ characterized through addition of its elements u, v and multiplication with real scalars α, β

$$\alpha, \beta \in \mathcal{R} \quad \mathcal{R} \dots \text{real numbers}$$

$$u, v \in \mathcal{V}^3 \quad \mathcal{V}^3 \dots \text{linear vector space}$$

definition of linear vector space $\mathcal{V}^3$ through the following axioms

$$\begin{aligned} \alpha (u + v) &= \alpha u + \alpha v \\ (\alpha + \beta) u &= \alpha u + \beta u \\ (\alpha \beta) u &= \alpha (\beta u) \end{aligned} \quad (1.1.6)$$

zero element and identity

$$0 u = 0 \quad 1 u = u \quad (1.1.7)$$

linear independence of elements $e_1, e_2, e_3 \in \mathcal{V}^3$ if $\alpha_1 = \alpha_2 = \alpha_3 = 0$ is the only (trivial) solution to

$$\alpha_i e_i = 0 \quad (1.1.8)$$

- consider linear vector space $\mathcal{V}^3$ equipped with a norm $n(u)$ mapping elements of the linear vector space $\mathcal{V}^3$ to the space

of real numbers $\mathcal{R}$

$$n : \mathcal{V}^3 \rightarrow \mathcal{R} \quad \text{norm} \quad (1.1.9)$$

definition of norm through the following axioms

$$\begin{aligned} n(\mathbf{u}) &\geq 0 & n(\mathbf{u}) &= 0 \Leftrightarrow \mathbf{u} = \mathbf{0} \\ n(\alpha \mathbf{u}) &= |\alpha| n(\mathbf{u}) \\ n(\mathbf{u} + \mathbf{v}) &\leq n(\mathbf{u}) + n(\mathbf{v}) \\ n^2(\mathbf{u} + \mathbf{v}) + n^2(\mathbf{u} - \mathbf{v}) &= 2 [n^2(\mathbf{u}) + n^2(\mathbf{v})] \end{aligned} \quad (1.1.10)$$

• consider Euclidian vector space $\mathcal{E}^3$ equipped with the Euclidian norm

$$n(\mathbf{u}) = \|\mathbf{u}\| = \sqrt{\mathbf{u} \cdot \mathbf{u}} = [u_1^2 + u_2^2 + u_3^2]^{1/2} \quad (1.1.11)$$

mapping elements of the Euclidian vector space $\mathcal{E}^3$ to the space of real numbers $\mathcal{R}$

$$n : \mathcal{E}^3 \rightarrow \mathcal{R} \quad \text{Euclidian norm} \quad (1.1.12)$$

representation of three-dimensional vector $\mathbf{a} \in \mathcal{E}^3$

$$\mathbf{a} = a_i \mathbf{e}_i = a_1 \mathbf{e}_1 + a_2 \mathbf{e}_2 + a_3 \mathbf{e}_3 \quad (1.1.13)$$

with a_1, a_2, a_3 coordinates (components) of $\mathbf{a}$ relative to the basis $\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3$

$$\mathbf{a} = [a_1, a_2, a_3]^t \quad (1.1.14)$$

Scalar product

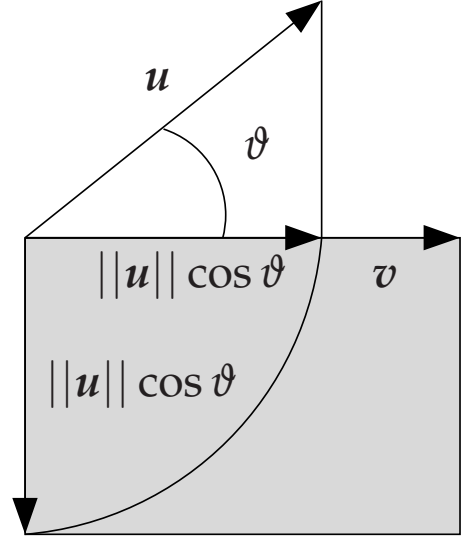
Euclidian norm enables the definition of scalar (inner) product between two vectors $\mathbf{u}$, $\mathbf{v}$ and introduces a scalar $\alpha \in \mathcal{R}$

$$\mathbf{u} \cdot \mathbf{v} = \alpha \quad (1.1.15)$$

geometric interpretation with $0 \leq \vartheta \leq \pi$ being the angle enclosed by the vectors $\mathbf{u}$ and $\mathbf{v}$, then $\|\mathbf{u}\| \cos \vartheta$ can be interpreted as the projection of $\mathbf{u}$ onto the direction of $\mathbf{v}$ and

$$\mathbf{u} \cdot \mathbf{v} = \|\mathbf{u}\| \|\mathbf{v}\| \cos \vartheta$$

corresponds to the grey area in the picture



with the above interpretation with $0 \leq \vartheta \leq \pi$, obviously

$$\|\mathbf{u} \cdot \mathbf{v}\| \leq \|\mathbf{u}\| \|\mathbf{v}\| \quad (1.1.16)$$

properties of scalar product

$$\mathbf{u} \cdot \mathbf{v} = \mathbf{v} \cdot \mathbf{u}$$

$$(\alpha \mathbf{u} + \beta \mathbf{v}) \cdot \mathbf{w} = \alpha (\mathbf{u} \cdot \mathbf{w}) + \beta (\mathbf{v} \cdot \mathbf{w}) \quad (1.1.17)$$

$$\mathbf{w} \cdot (\alpha \mathbf{u} + \beta \mathbf{v}) = \alpha (\mathbf{w} \cdot \mathbf{u}) + \beta (\mathbf{w} \cdot \mathbf{v})$$

positive definiteness of scalar product

$$\mathbf{u} \cdot \mathbf{u} \geq 0, \quad \mathbf{u} \cdot \mathbf{u} = 0 \Leftrightarrow \mathbf{u} = \mathbf{0} \quad (1.1.18)$$

orthogonal vectors $\mathbf{u}$ and $\mathbf{v}$

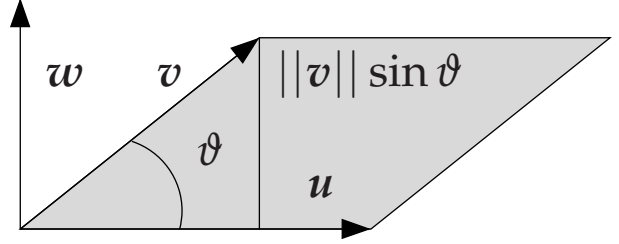
$$\mathbf{u} \cdot \mathbf{v} = 0 \quad \Leftrightarrow \quad \mathbf{u} \perp \mathbf{v} \quad (1.1.19)$$

Vector product

vector product of two vectors $\mathbf{u}, \mathbf{v}$ defines a new vector $\mathbf{w} \in \mathcal{E}^3$

$$\mathbf{u} \times \mathbf{v} = \mathbf{w} \quad (1.1.20)$$

geometric interpretation
with $0 \leq \vartheta \leq \pi$ being the
angle enclosed by the vec-
tors $\mathbf{u}$ and $\mathbf{v}$, then $\|\mathbf{v}\| \sin \vartheta$
can be interpreted as the height of the grey polygon and



$$\mathbf{u} \times \mathbf{v} = \|\mathbf{u}\| \|\mathbf{v}\| \sin \vartheta \mathbf{n}$$

introduces the vector $\mathbf{w}$ orthogonal to $\mathbf{u}$ and $\mathbf{v}$ whereby its
length corresponds to the grey area

with the above interpretation, obviously $\mathbf{u}$ parallel to $\mathbf{v}$ if

$$\mathbf{u} \times \mathbf{v} = \mathbf{0} \quad \Leftrightarrow \quad \mathbf{u} \parallel \mathbf{v} \quad (1.1.21)$$

index representation of $\mathbf{w} = \mathbf{u} \times \mathbf{v}$

$$\begin{bmatrix} w_1 \\ w_2 \\ w_3 \end{bmatrix} = \begin{bmatrix} u_2 v_3 - u_3 v_2 \\ u_3 v_1 - u_1 v_3 \\ u_1 v_2 - u_2 v_1 \end{bmatrix} \quad (1.1.22)$$

properties of vector product

$$\mathbf{u} \times \mathbf{v} = -\mathbf{v} \times \mathbf{u}$$

$$(\alpha \mathbf{u} + \beta \mathbf{v}) \times \mathbf{w} = \alpha (\mathbf{u} \times \mathbf{w}) + \beta (\mathbf{v} \times \mathbf{w}) \quad (1.1.23)$$

$$\mathbf{u} \cdot (\mathbf{u} \times \mathbf{v}) = 0$$

$$(\mathbf{u} \times \mathbf{v}) \cdot (\mathbf{u} \times \mathbf{v}) = (\mathbf{u} \cdot \mathbf{u}) (\mathbf{v} \cdot \mathbf{v}) - (\mathbf{u} \cdot \mathbf{v})^2$$

Scalar triple vector product

scalar triple vector product of three vectors u, v, w introduces a scalar $\alpha \in \mathcal{R}$

$$[u, v, w] = u \cdot (v \times w) = \alpha \quad (1.1.24)$$

geometric interpretation
with vector product

$$v \times w = ||v|| ||w|| \sin \vartheta n$$

defining area of ground surface

$$[u, v, w] = u \cdot (v \times w)$$

defines volume of parallelepiped

obviously

$$\begin{aligned} \alpha &= u \cdot (v \times w) = v \cdot (w \times u) = w \cdot (u \times v) \\ -\alpha &= u \cdot (w \times v) = v \cdot (u \times w) = w \cdot (v \times u) \end{aligned} \quad (1.1.25)$$

index representation of $\alpha = [u, v, w]$

$$\alpha = u_1(v_2w_3 - v_3w_2) + u_2(v_3w_1 - v_1w_3) + u_3(v_1w_2 - v_2w_1) \quad (1.1.26)$$

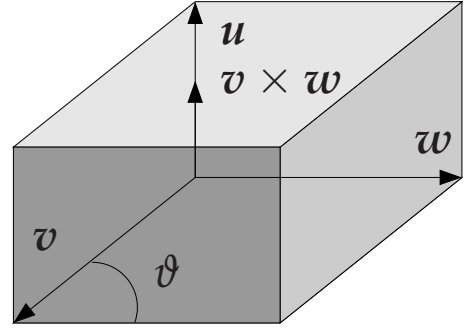
properties of scalar triple product

$$\begin{aligned} [u, v, w] &= [v, w, u] = [w, u, v] \\ &= -[u, w, v] = -[v, u, w] = -[w, v, u] \end{aligned} \quad (1.1.27)$$

$$[\alpha u + \beta v, w, d] = \alpha [u, w, d] + \beta [v, w, d]$$

three vectors u, v, w are linearly independent if

$$[u, v, w] \neq 0 \quad (1.1.28)$$



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lecture notes 02

thursday, january 07th, 2010

1.1.2 Tensor algebra

Notation

Second order tensors

tensor (dyadic) product $\mathbf{u} \otimes \mathbf{v}$ of two vectors $\mathbf{u}$ and $\mathbf{v}$ introduces a second order tensor A

$$A = \mathbf{u} \otimes \mathbf{v} \quad (1.1.29)$$

introducing $\mathbf{u} = u_i \mathbf{e}_i$ and $\mathbf{v} = v_j \mathbf{e}_j$ yields index representation of three-dimensional second order tensor A

$$A = A_{ij} \mathbf{e}_i \otimes \mathbf{e}_j \quad (1.1.30)$$

with $A_{ij} = u_i v_j$ coordinates (components) of A relative to the tensor basis $\mathbf{e}_i \otimes \mathbf{e}_j$, matrix representation of coordinates

$$[A_{ij}] = \begin{bmatrix} A_{11} & A_{12} & A_{13} \\ A_{21} & A_{22} & A_{23} \\ A_{31} & A_{32} & A_{33} \end{bmatrix} \quad (1.1.31)$$

transpose of second order tensor A^t

$$A^t = (\mathbf{u} \otimes \mathbf{v})^t = \mathbf{v} \otimes \mathbf{u} \quad (1.1.32)$$

introducing $\mathbf{u} = u_i \mathbf{e}_i$ and $\mathbf{v} = v_j \mathbf{e}_j$ yields index representation of transpose of second order tensor A^t

$$A^t = A_{ji} \mathbf{e}_i \otimes \mathbf{e}_j \quad (1.1.33)$$

with $A_{ji} = v_j u_i$ coordinates (components) of A^t relative to the tensor basis $\mathbf{e}_i \otimes \mathbf{e}_j$, matrix representation of coordinates

$$[A_{ji}] = \begin{bmatrix} A_{11} & A_{21} & A_{31} \\ A_{12} & A_{22} & A_{32} \\ A_{13} & A_{23} & A_{33} \end{bmatrix} \quad (1.1.34)$$

second order unit tensor $\mathbf{I}$ in terms of Kronecker symbol δ_{ij}

$$\mathbf{I} = \delta_{ij} \mathbf{e}_i \otimes \mathbf{e}_j \quad (1.1.35)$$

matrix representation of coordinates δ_{ij}

$$[\delta_{ij}] = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad (1.1.36)$$

Third order tensors

tensor (dyadic) product $\mathbf{A} \otimes \mathbf{u}$ of second order tensor $\mathbf{A}$ and vectors $\mathbf{u}$ introduces a third order tensor $\overset{3}{\mathbf{a}}$

$$\overset{3}{\mathbf{a}} = \mathbf{A} \otimes \mathbf{u} \quad (1.1.37)$$

introducing $\mathbf{A} = A_{ij} \mathbf{e}_i \otimes \mathbf{e}_j$ and $\mathbf{u} = u_k \mathbf{e}_k$ yields index representation of three-dimensional third order tensor $\overset{3}{\mathbf{a}}$

$$\overset{3}{\mathbf{a}} = \overset{3}{a}_{ijk} \mathbf{e}_i \otimes \mathbf{e}_j \otimes \mathbf{e}_k \quad (1.1.38)$$

with $\overset{3}{a}_{ijk} = A_{ij} u_k$ coordinates (components) of $\overset{3}{\mathbf{a}}$ relative to the tensor basis $\mathbf{e}_i \otimes \mathbf{e}_j \otimes \mathbf{e}_k$

third order permutation tensor $\overset{3}{\mathbf{e}}$ in terms of permutation symbol $\overset{3}{e}_{ijk}$

$$\overset{3}{\mathbf{e}} = \overset{3}{e}_{ijk} \mathbf{e}_i \otimes \mathbf{e}_j \otimes \mathbf{e}_k \quad (1.1.39)$$

Fourth order tensors

tensor (dyadic) product $A \otimes B$ of two second order tensors A and B introduces a fourth order tensor $\mathbb{A}$

$$\mathbb{A} = A \otimes B \quad (1.1.40)$$

introducing $A = A_{ij} \mathbf{e}_i \otimes \mathbf{e}_j$ and $B = B_{kl} \mathbf{e}_k \otimes \mathbf{e}_l$ yields index representation of three-dimensional fourth order tensor $\mathbb{A}$

$$\mathbb{A} = A_{ijkl} \mathbf{e}_i \otimes \mathbf{e}_j \otimes \mathbf{e}_k \otimes \mathbf{e}_l \quad (1.1.41)$$

with $A_{ijkl} = A_{ij} B_{kl}$ coordinates (components) of $\mathbb{A}$ relative to the tensor basis $\mathbf{e}_i \otimes \mathbf{e}_j \otimes \mathbf{e}_k \otimes \mathbf{e}_l$

fourth order unit tensor $\mathbb{I}$

$$\mathbb{I} = \delta_{ik} \delta_{jl} \mathbf{e}_i \otimes \mathbf{e}_j \otimes \mathbf{e}_k \otimes \mathbf{e}_l \quad (1.1.42)$$

transpose fourth order unit tensor $\mathbb{I}^t$

$$\mathbb{I}^t = \delta_{il} \delta_{jk} \mathbf{e}_i \otimes \mathbf{e}_j \otimes \mathbf{e}_k \otimes \mathbf{e}_l \quad (1.1.43)$$

symmetric fourth order unit tensor $\mathbb{I}^{\text{sym}}$

$$\mathbb{I}^{\text{sym}} = \frac{1}{2} [\delta_{ik} \delta_{jl} + \delta_{il} \delta_{jk}] \mathbf{e}_i \otimes \mathbf{e}_j \otimes \mathbf{e}_k \otimes \mathbf{e}_l \quad (1.1.44)$$

skew-symmetric fourth order unit tensor $\mathbb{I}^{\text{skw}}$

$$\mathbb{I}^{\text{skw}} = \frac{1}{2} [\delta_{ik} \delta_{jl} - \delta_{il} \delta_{jk}] \mathbf{e}_i \otimes \mathbf{e}_j \otimes \mathbf{e}_k \otimes \mathbf{e}_l \quad (1.1.45)$$

volumetric fourth order unit tensor $\mathbb{I}^{\text{vol}}$

$$\mathbb{I}^{\text{vol}} = \frac{1}{3} \delta_{ij} \delta_{kl} \mathbf{e}_i \otimes \mathbf{e}_j \otimes \mathbf{e}_k \otimes \mathbf{e}_l \quad (1.1.46)$$

deviatoric fourth order unit tensor $\mathbb{I}^{\text{dev}}$

$$\mathbb{I}^{\text{dev}} = [\frac{1}{2} \delta_{ik} \delta_{jl} + \frac{1}{2} \delta_{il} \delta_{jk} - \frac{1}{3} \delta_{ij} \delta_{kl}] \mathbf{e}_i \otimes \mathbf{e}_j \otimes \mathbf{e}_k \otimes \mathbf{e}_l \quad (1.1.47)$$

Scalar products

- scalar product $A \cdot u$ between second order tensor A and vector u defines a new vector $v \in \mathcal{E}^3$

$$\begin{aligned} A \cdot u &= (A_{ij} e_i \otimes e_j) \cdot (u_k e_k) \\ &= A_{ij} u_k \delta_{jk} e_i = A_{ij} u_j e_i = v_i e_i = v \end{aligned} \quad (1.1.48)$$

second order zero tensor 0 , second order identity tensor I

$$0 \cdot a = 0 \quad I \cdot a = a \quad (1.1.49)$$

positive semi-definiteness of second order tensor A

$$a \cdot A \cdot a \geq 0 \quad (1.1.50)$$

positive definiteness of second order tensor A

$$a \cdot A \cdot a > 0 \quad (1.1.51)$$

properties of scalar product

$$\begin{aligned} A \cdot (\alpha a + \beta b) &= \alpha (A \cdot a) + \beta (A \cdot b) \\ (A + B) \cdot a &= A \cdot a + B \cdot a \\ (\alpha A) \cdot a &= \alpha (A \cdot a) \end{aligned} \quad (1.1.52)$$

- scalar product $A \cdot B$ between two second order tensors A and B defines a second order tensor C

$$\begin{aligned} A \cdot B &= (A_{ij} e_i \otimes e_j) \cdot (B_{kl} e_k \otimes e_l) \\ &= A_{ij} B_{kl} \delta_{jk} e_i \otimes e_l \\ &= A_{ij} B_{jl} e_i \otimes e_l = C_{il} e_i \otimes e_l = C \end{aligned} \quad (1.1.53)$$

second order zero tensor 0 , second order identity tensor I

$$0 \cdot A = 0 \quad I \cdot A = A \quad (1.1.54)$$

properties of scalar product

$$\begin{aligned}
 \alpha (A \cdot B) &= (\alpha A) \cdot B = A \cdot (\alpha B) \\
 A \cdot (B + C) &= A \cdot B + A \cdot C \\
 (A + B) \cdot C &= A \cdot C + B \cdot C
 \end{aligned} \tag{1.1.55}$$

properties in terms of transpose A^t of a tensor A

$$\begin{aligned}
 a \cdot (A^t \cdot b) &= b \cdot (A \cdot a) \\
 (\alpha A + \beta B)^t &= \alpha A^t + \beta B^t \\
 (A \cdot B)^t &= B^t \cdot A^t
 \end{aligned} \tag{1.1.56}$$

• scalar product $A : B$ between two second order tensors A and B defines a scalar $\alpha \in \mathcal{R}$

$$\begin{aligned}
 A : B &= (A_{ij} e_i \otimes e_j) : (B_{kl} e_k \otimes e_l) \\
 &= A_{ij} B_{kl} \delta_{ik} \delta_{jl} = A_{ij} B_{ij} = \alpha
 \end{aligned} \tag{1.1.57}$$

• scalar product $\mathbb{A} : B$ between fourth order tensor $\mathbb{A}$ and second order tensor B defines a new second order tensor C

$$\begin{aligned}
 \mathbb{A} : B &= (A_{ijkl} e_i \otimes e_j \otimes e_k \otimes e_l) : (B_{mn} e_m \otimes e_n) \\
 &= A_{ijkl} B_{mn} \delta_{km} \delta_{ln} e_i \otimes e_j \\
 &= A_{ijkl} B_{kl} e_i \otimes e_j = C_{ij} e_i \otimes e_j = C
 \end{aligned} \tag{1.1.58}$$

Dyadic product

- tensor (dyadic) product $\mathbf{u} \otimes \mathbf{v}$ of two vectors $\mathbf{u}$ and $\mathbf{v}$ introduces a second order tensor $\mathbf{A}$

$$\mathbf{A} = \mathbf{u} \otimes \mathbf{v} = u_i \mathbf{e}_i \otimes v_j \mathbf{e}_j = u_i v_j \mathbf{e}_i \otimes \mathbf{e}_j = A_{ij} \mathbf{e}_i \otimes \mathbf{e}_j \quad (1.1.59)$$

properties of dyadic product

$$\begin{aligned} (\mathbf{u} \otimes \mathbf{v}) \cdot \mathbf{w} &= (\mathbf{v} \cdot \mathbf{w}) \mathbf{u} \\ (\alpha \mathbf{u} + \beta \mathbf{v}) \otimes \mathbf{w} &= \alpha (\mathbf{u} \otimes \mathbf{w}) + \beta (\mathbf{v} \otimes \mathbf{w}) \\ \mathbf{u} \otimes (\alpha \mathbf{v} + \beta \mathbf{w}) &= \alpha (\mathbf{u} \otimes \mathbf{v}) + \beta (\mathbf{u} \otimes \mathbf{w}) \\ (\mathbf{u} \otimes \mathbf{v}) \cdot (\mathbf{w} \otimes \mathbf{x}) &= (\mathbf{v} \cdot \mathbf{w}) (\mathbf{u} \otimes \mathbf{x}) \\ \mathbf{A} \cdot (\mathbf{u} \otimes \mathbf{v}) &= (\mathbf{A} \cdot \mathbf{u}) \otimes \mathbf{v} \\ (\mathbf{u} \otimes \mathbf{v}) \cdot \mathbf{A} &= \mathbf{u} \otimes (\mathbf{A}^t \cdot \mathbf{v}) \end{aligned} \quad (1.1.60)$$

or in index notation

$$\begin{aligned} (u_i v_j) w_j &= (v_j w_j) u_i \\ (\alpha u_i + \beta v_i) w_j &= \alpha (u_i w_j) + \beta (v_i w_j) \\ u_i (\alpha v_j + \beta w_j) &= \alpha (u_i v_j) + \beta (u_i w_j) \\ (u_i v_j) (w_j x_k) &= (v_j w_j) (u_i x_k) \\ A_{ij} (u_j v_k) &= (A_{ij} u_j) v_k \\ (u_i v_j) A_{jk} &= u_i (A_{kj} v_j) \end{aligned} \quad (1.1.61)$$

Scalar triple vector product

consider the set of Cartesian base vectors $\{e_i\}_{i=1,2,3}$ and an arbitrary second set of base vectors $\{u, v, w\}$ with scalar triple product $[u, v, w]$, with arbitrary second order tensor A , evaluate

$$[[A \cdot u, v, w] + [u, A \cdot v, w] + [u, v, A \cdot w]] / [u, v, w] \quad (1.1.62)$$

with index representation of each term according to

$$[A \cdot u, v, w] = [A \cdot (u_i e_i), (v_j e_j), (w_k e_k)] = u_i v_j w_k [A \cdot e_i, e_j, e_k] \quad (1.1.63)$$

expression (1.1.62) can be rewritten as

$$u_i v_j w_k [[A \cdot e_1, e_2, e_3] + [e_1, A \cdot e_2, e_3] + [e_1, e_2, A \cdot e_3]] / [u, v, w] \quad (1.1.64)$$

term in brackets remains unchanged upon cyclic permutation of $\{e_i\}_{i=1,2,3}$, its sign reverses upon non-cyclic permutations, thus

$$\begin{aligned} & u_i v_j w_k \overset{3}{e_{ijk}} [[A \cdot e_1, e_2, e_3] + [e_1, A \cdot e_2, e_3] + [e_1, e_2, A \cdot e_3]] / [u, v, w] \\ &= \overset{+}{-} [[A \cdot e_1, e_2, e_3] + [e_1, A \cdot e_2, e_3] + [e_1, e_2, A \cdot e_3]] \end{aligned} \quad (1.1.65)$$

the above expression according to (1.1.62) is thus invariant under the choice of base system, it yields the same scalar value I_A for arbitrary base systems

$$\begin{aligned} I_A &= [[A \cdot u, v, w] + [u, A \cdot v, w] + [u, v, A \cdot w]] / [u, v, w] \\ &= [[A \cdot e_1, e_2, e_3] + [e_1, A \cdot e_2, e_3] + [e_1, e_2, A \cdot e_3]] / [e_1, e_2, e_3] \end{aligned} \quad (1.1.66)$$

I_A is called the first invariant of the second order tensor A

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tuesday, january 12th, 2010

Invariants of second order tensors

the following property of the scalar triple product

$$[u, v, w] = u \cdot (v \times w) \quad (1.1.67)$$

introduces three scalar-valued quantities I_A , II_A , III_A associated with the second order tensor A

$$\begin{aligned} [A \cdot u, v, w] + [u, A \cdot v, w] + [u, v, A \cdot w] &= I_A [u, v, w] \\ [u, A \cdot v, A \cdot w] + [A \cdot u, v, A \cdot w] + [A \cdot u, A \cdot v, w] &= II_A [u, v, w] \\ [A \cdot u, A \cdot v, A \cdot w] &= III_A [u, v, w] \end{aligned} \quad (1.1.68)$$

the proof of II_A , III_A being invariant for different base systems $\{u, v, w\}$ is similar to the one for I_A

I_A , II_A , III_A are called the three principal invariants of A which can be expressed as

$$\begin{aligned} I_A &= \text{tr}(A) & \partial_A I_A &= I \\ II_A &= \frac{1}{2} [\text{tr}^2(A) - \text{tr}(A^2)] & \partial_A II_A &= I_A I - A \\ III_A &= \det(A) & \partial_A III_A &= III_A A^{-t} \end{aligned} \quad (1.1.69)$$

alternatively, we could work with the three basic invariants $\bar{I}_A$, $\bar{II}_A$, $\bar{III}_A$ of A which are more common in the context of anisotropy

$$\begin{aligned} \bar{I}_A &= A^1 : I \\ \bar{II}_A &= A^2 : I \\ \bar{III}_A &= A^3 : I \end{aligned} \quad (1.1.70)$$

Trace of second order tensors

trace $\text{tr}(A)$ of a second order tensor $A = \boldsymbol{u} \otimes \boldsymbol{v}$ introduces a scalar $\text{tr}(A) \in \mathcal{R}$

$$\text{tr}(\boldsymbol{u} \otimes \boldsymbol{v}) = \boldsymbol{u} \cdot \boldsymbol{v} \quad (1.1.71)$$

such that $\text{tr}(A)$ is the sum of the diagonal entries A_{ii} of A

$$\begin{aligned} \text{tr}(A) &= \text{tr}(A_{ij} \boldsymbol{e}_i \otimes \boldsymbol{e}_j) \\ &= A_{ij} \text{tr}(\boldsymbol{e}_i \otimes \boldsymbol{e}_j) = A_{ij} \boldsymbol{e}_i \cdot \boldsymbol{e}_j \\ &= A_{ij} \delta_{ij} = A_{ii} = A_{11} + A_{22} + A_{33} \end{aligned} \quad (1.1.72)$$

with

$$I_A = \bar{I}_A = \text{tr}(A) \quad (1.1.73)$$

properties of the trace of second order tensors

$$\begin{aligned} \text{tr}(I) &= 3 \\ \text{tr}(A^t) &= \text{tr}(A) \\ \text{tr}(A \cdot B) &= \text{tr}(B \cdot A) \\ \text{tr}(\alpha A + \beta B) &= \alpha \text{tr}(A) + \beta \text{tr}(B) \\ \text{tr}(A \cdot B^t) &= A : B \\ \text{tr}(A) &= \text{tr}(A \cdot I) = A : I \end{aligned} \quad (1.1.74)$$

Determinant of second order tensors

determinant $\det(A)$ of second order tensor A introduces a scalar $\det(A) \in \mathcal{R}$

$$\begin{aligned}\det(A) &= \det(A_{ij}) = \frac{1}{6} e_{ijk} e_{abc} A_{ia} A_{jb} A_{kc} \\ &= A_{11}A_{22}A_{33} + A_{21}A_{32}A_{13} + A_{31}A_{12}A_{23} \\ &\quad - A_{11}A_{23}A_{32} - A_{22}A_{31}A_{13} - A_{33}A_{12}A_{21}\end{aligned}\quad (1.1.75)$$

with

$$III_A = \det(A) \quad (1.1.76)$$

determinant defining vector product $\mathbf{u} \times \mathbf{v}$

$$\mathbf{u} \times \mathbf{v} = \det \begin{bmatrix} u_1 & v_1 & \mathbf{e}_1 \\ u_2 & v_2 & \mathbf{e}_2 \\ u_3 & v_3 & \mathbf{e}_3 \end{bmatrix} = \begin{bmatrix} u_2 v_3 - u_3 v_2 \\ u_3 v_1 - u_1 v_3 \\ u_1 v_2 - u_2 v_1 \end{bmatrix} \quad (1.1.77)$$

determinant defining scalar triple vector product $[\mathbf{u}, \mathbf{v}, \mathbf{w}]$

$$[\mathbf{u}, \mathbf{v}, \mathbf{w}] = (\mathbf{u} \times \mathbf{v}) \cdot \mathbf{w} = \det \begin{bmatrix} u_1 & v_1 & w_1 \\ u_2 & v_2 & w_2 \\ u_3 & v_3 & w_3 \end{bmatrix} \quad (1.1.78)$$

properties of determinant of a second order tensors

$$\begin{aligned}\det(I) &= 1 \\ \det(A^t) &= \det(A) \\ \det(\alpha A) &= \alpha^3 \det(A) \\ \det(A \cdot B) &= \det(A) \det(B) \\ \det(\mathbf{u} \otimes \mathbf{v}) &= 0\end{aligned}\quad (1.1.79)$$

Inverse of second order tensors

if $\det(A) \neq 0$

existence of inverse A^{-1} of second order tensor A

$$A \cdot A^{-1} = A^{-1} \cdot A = I \quad (1.1.80)$$

in particular

$$v = A \cdot u \quad A^{-1} \cdot v = u \quad (1.1.81)$$

properties of inverse of two second order tensors

$$\begin{aligned} (A^{-1})^{-1} &= A \\ (\alpha A^{-1})^{-1} &= \alpha^{-1} A \\ (A \cdot B)^{-1} &= B^{-1} \cdot A^{-1} \end{aligned} \quad (1.1.82)$$

determinant $\det(A^{-1})$ of inverse of A

$$\det(A^{-1}) = 1/\det(A) \quad (1.1.83)$$

adjoint A^{adj} of a second order tensor A

$$A^{\text{adj}} = \det(A) A^{-1} \quad (1.1.84)$$

cofactor A^{cof} of a second order tensor A

$$A^{\text{cof}} = \det(A) A^{-t} = (A^{\text{adj}})^t \quad (1.1.85)$$

with

$$\partial_A \det(A) = \det(A) A^{-t} = III_A A^{-t} = A^{\text{cof}} \quad (1.1.86)$$

1.1.3 Spectral decomposition

eigenvalue problem of arbitrary second order tensor A

$$A \cdot \mathbf{n}_A = \lambda_A \mathbf{n}_A \quad [A - \lambda_A I] \cdot \mathbf{n}_A = \mathbf{0} \quad (1.1.87)$$

solution introduces eigenvector(s) $\mathbf{n}_{Ai}$ and eigenvalue(s) λ_{Ai}

$$\det(A - \lambda_A I) = 0 \quad (1.1.88)$$

alternative representation in terms of scalar triple product

$$[A \cdot \mathbf{u} - \lambda_A \mathbf{u}, A \cdot \mathbf{v} - \lambda_A \mathbf{v}, A \cdot \mathbf{w} - \lambda_A \mathbf{w}] = 0 \quad (1.1.89)$$

removal of arbitrary factor $[\mathbf{u}, \mathbf{v}, \mathbf{w}]$ yields characteristic equation

$$\lambda_A^3 - I_A \lambda_A^2 + II_A \lambda_A - III_A = 0 \quad (1.1.90)$$

roots of characteristic equations are principal invariants of A

$$\begin{aligned} I_A &= \text{tr}(A) \\ II_A &= \frac{1}{2} [\text{tr}^2(A) - \text{tr}(A^2)] \\ III_A &= \det(A) \end{aligned} \quad (1.1.91)$$

spectral decomposition of A

$$A = \sum_{i=1}^3 \lambda_{Ai} \mathbf{n}_{Ai} \otimes \mathbf{n}_{Ai} \quad (1.1.92)$$

Cayleigh–Hamilton theorem:

a tensor A satisfies its own characteristic equation

$$A^3 - I_A A^2 + II_A A - III_A I = \mathbf{0} \quad (1.1.93)$$

1.1.4 Symmetric – skew-symmetric decomposition

symmetric – skew-symmetric decomposition of second order tensor A

$$A = \frac{1}{2}[A + A^t] + \frac{1}{2}[A - A^t] = A^{\text{sym}} + A^{\text{skw}} \quad (1.1.94)$$

with symmetric and skew-symmetric second order tensor A^{sym} and A^{skw}

$$A^{\text{sym}} = (A^{\text{sym}})^t \quad A^{\text{skw}} = -(A^{\text{skw}})^t \quad (1.1.95)$$

- symmetric second order tensor A^{sym}

$$A^{\text{sym}} = \frac{1}{2}[A + A^t] = \mathbb{I}^{\text{sym}} : A \quad (1.1.96)$$

upon double contraction symmetric fourth order unit tensor $\mathbb{I}^{\text{sym}}$ extracts symmetric part of second order tensor

$$\begin{aligned} \mathbb{I}^{\text{sym}} &= \frac{1}{2} [\mathbb{I} + \mathbb{I}^t] \\ \mathbb{I}^{\text{sym}} &= \frac{1}{2} [\delta_{ik}\delta_{jl} + \delta_{il}\delta_{jk}] \mathbf{e}_i \otimes \mathbf{e}_j \otimes \mathbf{e}_k \otimes \mathbf{e}_l \end{aligned} \quad (1.1.97)$$

- skew-symmetric second order tensor A^{skw}

$$A^{\text{skw}} = \frac{1}{2}[A - A^t] = \mathbb{I}^{\text{skw}} : A \quad (1.1.98)$$

upon double contraction skew-symmetric fourth order unit tensor $\mathbb{I}^{\text{skw}}$ extracts skew-symmetric part of second order tensor

$$\begin{aligned} \mathbb{I}^{\text{skw}} &= \frac{1}{2} [\mathbb{I} - \mathbb{I}^t] \\ \mathbb{I}^{\text{skw}} &= \frac{1}{2} [\delta_{ik}\delta_{jl} - \delta_{il}\delta_{jk}] \mathbf{e}_i \otimes \mathbf{e}_j \otimes \mathbf{e}_k \otimes \mathbf{e}_l \end{aligned} \quad (1.1.99)$$

Symmetric tensors

symmetric part A^{sym} of a second order tensor A

$$A^{\text{sym}} = \frac{1}{2}[A + A^t] \quad A^{\text{sym}} = (A^{\text{sym}})^t \quad (1.1.100)$$

alternative representation

$$A^{\text{sym}} = \mathbb{I}^{\text{sym}} : A \quad (1.1.101)$$

whereby symmetric fourth order tensor $\mathbb{I}^{\text{sym}}$ extracts symmetric part A^{sym} of second order tensor A

a symmetric second order tensor $S = A^{\text{sym}}$ processes three real eigenvalues $\{\lambda_{Si}\}_{i=1,2,3}$ and three corresponding orthogonal eigenvectors $\{\mathbf{n}_{Si}\}_{i=1,2,3}$, such that the spectral representation of S takes the following form

$$S = \sum_{i=1}^3 \lambda_{Si} (\mathbf{n}_{Si} \otimes \mathbf{n}_{Si}) \quad (1.1.102)$$

three invariants I_S, II_S, III_S of symmetric tensor $S = A^{\text{sym}}$

$$\begin{aligned} I_S &= \lambda_{S1} + \lambda_{S2} + \lambda_{S3} \\ II_S &= \lambda_{S2} \lambda_{S3} + \lambda_{S3} \lambda_{S1} + \lambda_{S1} \lambda_{S2} \\ III_S &= \lambda_{S1} \lambda_{S2} \lambda_{S3} \end{aligned} \quad (1.1.103)$$

square root $\sqrt{S}$, inverse S^{-1} , exponent $\exp(S)$ and logarithm $\ln(S)$, of positive semi-definite symmetric tensor S for which $\lambda_{Si} \geq 0$

$$\begin{aligned} \sqrt{S} &= \sum_{i=1}^3 \sqrt{\lambda_{Si}} (\mathbf{n}_{Si} \otimes \mathbf{n}_{Si}) \\ S^{-1} &= \sum_{i=1}^3 \lambda_{Si}^{-1} (\mathbf{n}_{Si} \otimes \mathbf{n}_{Si}) \\ \exp(S) &= \sum_{i=1}^3 \exp(\lambda_{Si}) (\mathbf{n}_{Si} \otimes \mathbf{n}_{Si}) \\ \ln(S) &= \sum_{i=1}^3 \ln(\lambda_{Si}) (\mathbf{n}_{Si} \otimes \mathbf{n}_{Si}) \end{aligned} \quad (1.1.104)$$

Skew-symmetric tensors

skew-symmetric part A^{skw} of a second order tensor A

$$A^{\text{skw}} = \frac{1}{2}[A - A^t] \quad A^{\text{skw}} = -(A^{\text{skw}})^t \quad (1.1.105)$$

alternative representation

$$A^{\text{skw}} = \mathbb{I}^{\text{skw}} : A \quad (1.1.106)$$

whereby skew-symmetric fourth order tensor $\mathbb{I}^{\text{skw}}$ extracts skew-symmetric part A^{skw} of second order tensor A

a skew-symmetric second order tensor $W = A^{\text{skw}}$ possesses three independent entries, three entries vanish identically, three are equal to the negative of the independent entries, these define the axial vector w

$$w = -\frac{1}{2} \overset{3}{e} : W \quad w = -\overset{3}{e} \cdot w \quad (1.1.107)$$

associated with each skew-symmetric tensor $W = A^{\text{skw}}$

$$W \cdot p = w \times p \quad (1.1.108)$$

three invariants I_W, II_W, III_W of skew-symmetric tensor W

$$\begin{aligned} I_W &= \text{tr}(W) = 0 \\ II_W &= w \cdot w \\ III_W &= \det(W) = 0 \end{aligned} \quad (1.1.109)$$

1.1.5 Volumetric – deviatoric decomposition

volumetric – deviatoric decomposition of second order tensor A

$$A = A^{\text{vol}} + A^{\text{dev}} \quad (1.1.110)$$

with volumetric and deviatoric second order tensor A^{vol} and A^{dev}

$$\text{tr}(A^{\text{vol}}) = \text{tr}(A) \quad \text{tr}(A^{\text{dev}}) = 0 \quad (1.1.111)$$

- volumetric second order tensor A^{vol}

$$A^{\text{vol}} = \frac{1}{3}[A : I] I = \mathbb{I}^{\text{vol}} : A \quad (1.1.112)$$

upon double contraction volumetric fourth order unit tensor $\mathbb{I}^{\text{vol}}$ extracts volumetric part of second order tensor

$$\begin{aligned} \mathbb{I}^{\text{vol}} &= \frac{1}{3} I \otimes I \\ \mathbb{I}^{\text{vol}} &= \frac{1}{3} \delta_{ij} \delta_{kl} \mathbf{e}_i \otimes \mathbf{e}_j \otimes \mathbf{e}_k \otimes \mathbf{e}_l \end{aligned} \quad (1.1.113)$$

- deviatoric second order tensor A^{dev}

$$A^{\text{dev}} = A - \frac{1}{3}[A : I] I = \mathbb{I}^{\text{dev}} : A = 0 \quad (1.1.114)$$

upon double contraction deviatoric fourth order unit tensor $\mathbb{I}^{\text{dev}}$ extracts deviatoric part of second order tensor

$$\begin{aligned} \mathbb{I}^{\text{dev}} &= \mathbb{I}^{\text{sym}} - \mathbb{I}^{\text{vol}} = \mathbb{I}^{\text{sym}} - \frac{1}{3} I \otimes I \\ \mathbb{I}^{\text{dev}} &= \left[\frac{1}{2} \delta_{ik} \delta_{jl} + \frac{1}{2} \delta_{il} \delta_{jk} - \frac{1}{3} \delta_{ij} \delta_{kl} \right] \mathbf{e}_i \otimes \mathbf{e}_j \otimes \mathbf{e}_k \otimes \mathbf{e}_l \end{aligned} \quad (1.1.115)$$

1.1.6 Orthogonal tensors

a second order tensor $\mathbf{Q}$ is called orthogonal if its inverse $\mathbf{Q}^{-1}$ is identical to its transpose $\mathbf{Q}^t$

$$\mathbf{Q}^{-1} = \mathbf{Q}^t \quad \Leftrightarrow \quad \mathbf{Q}^t \cdot \mathbf{Q} = \mathbf{Q} \cdot \mathbf{Q}^t = \mathbf{I} \quad (1.1.116)$$

a second order tensor $\mathbf{A}$ can be decomposed multiplicatively into a positive definite symmetric tensor $\mathbf{U}^t = \mathbf{U}$ or $\mathbf{V}^t = \mathbf{V}$ with $\mathbf{a} \cdot \mathbf{U} \cdot \mathbf{a} \geq 0$ and $\mathbf{a} \cdot \mathbf{V} \cdot \mathbf{a} \geq 0$ and an orthogonal tensor $\mathbf{Q}^t = \mathbf{Q}^{-1}$ as

$$\mathbf{A} = \mathbf{Q} \cdot \mathbf{U} = \mathbf{V} \cdot \mathbf{Q} \quad (1.1.117)$$

with $S0(3)$ being the special orthogonal group, $\mathbf{Q} \in S0(3)$ if $\det(\mathbf{Q}) = +1$, then $\mathbf{Q}$ is called proper orthogonal

a proper orthogonal tensor $\mathbf{Q} \in S0(3)$ has an eigenvalue equal to one $\lambda_Q = 1$ introducing an eigenvector $\mathbf{n}_Q$ such that

$$\mathbf{Q} \cdot \mathbf{n}_Q = \mathbf{n}_Q \quad (1.1.118)$$

let $\{\mathbf{n}_{Qi}\}_{i=1,2,3}$ be a Cartesian basis containing the vector $\mathbf{n}_Q$, then matrix representation of coordinates Q_{ij}

$$[Q_{ij}] = \begin{bmatrix} 1 & 0 & 0 \\ 0 & +\cos \varphi & +\sin \varphi \\ 0 & -\sin \varphi & +\cos \varphi \end{bmatrix} \quad (1.1.119)$$

geometric interpretation: $\mathbf{Q}$ characterizes a finite rotation around the axis $\mathbf{n}_Q$ with $\mathbf{Q} \cdot \mathbf{n}_Q = \mathbf{n}_Q$, i.e. associated with $\lambda_Q = 1$

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1.2 Tensor analysis

1.2.1 Derivatives

consider smooth, differentiable scalar field Φ with

- scalar argument $\Phi : \mathcal{R} \rightarrow \mathcal{R}; \quad \Phi(x) = \alpha$
- vectorial argument $\Phi : \mathcal{R}^3 \rightarrow \mathcal{R}; \quad \Phi(\mathbf{x}) = \alpha$
- tensorial argument $\Phi : \mathcal{R}^3 \times \mathcal{R}^3 \rightarrow \mathcal{R}; \quad \Phi(\mathbf{X}) = \alpha$

Frechet derivative

- scalar $D\Phi(x) = \frac{\partial \Phi(x)}{\partial x} = \partial_x \Phi(x)$
 - vectorial $D\Phi(\mathbf{x}) = \frac{\partial \Phi(\mathbf{x})}{\partial \mathbf{x}} = \partial_{\mathbf{x}} \Phi(\mathbf{x})$
 - tensorial $D\Phi(\mathbf{X}) = \frac{\partial \Phi(\mathbf{X})}{\partial \mathbf{X}} = \partial_{\mathbf{X}} \Phi(\mathbf{X})$
- (1.2.1)

Gateaux derivative

Gateaux derivative as particular Frechet derivative with respect to directions u , $\mathbf{u}$ and $\mathbf{U}$

- scalar $D\Phi(x) \cdot u = \frac{d}{d\epsilon} \Phi(x + \epsilon u) |_{\epsilon=0} \quad \forall u \in \mathcal{R}$
 - vectorial $D\Phi(\mathbf{x}) \cdot \mathbf{u} = \frac{d}{d\epsilon} \Phi(\mathbf{x} + \epsilon \mathbf{u}) |_{\epsilon=0} \quad \forall \mathbf{u} \in \mathcal{R}^3$
 - tensorial $D\Phi(\mathbf{X}) : \mathbf{U} = \frac{d}{d\epsilon} \Phi(\mathbf{X} + \epsilon \mathbf{U}) |_{\epsilon=0} \quad \forall \mathbf{U} \in \mathcal{R}^3 \otimes \mathcal{R}^3$
- (1.2.2)

in what follows in particular vectorial arguments, e.g., point position $\mathbf{x}$ or displacement $\mathbf{u}$

the Gateaux derivative is a simple mechanism to determine derivatives with respect to vectors or tensor, by rewriting them as derivatives with respect to a scalar ϵ

example: derivatives of invariants

use the Gateaux derivative

$$D\Phi(A) : \Delta A = \frac{\partial \Phi(A)}{\partial A} : \Delta A = \frac{d}{d\epsilon} \Phi(A + \epsilon \Delta A) \Big|_{\epsilon=0}$$

to determine the derivatives of the three invariants I_A, II_A, III_A of the second order tensor A with respect to A itself!

in words: the Gateaux derivative D of a scalar field Φ (in this case the invariant I_A, II_A, III_A) along a given direction $\mathbf{U}$ (in this case ΔA) is the derivative of the field Φ at position $\mathbf{X}$ (in this case A) "perturbed" by the direction $\epsilon \mathbf{U}$ (in this case $\epsilon \Delta A$) with respect to ϵ evaluated at $\epsilon = 0$

setting $\epsilon = 0$ filters out the linear terms, i.e., all higher order terms are set to zero, the Gateaux derivative is therefore also referred to as linearization, it corresponds to the first term of a Taylor series

derivative of first invariant $I_A = \text{tr}(A) = A : I$

$$\begin{aligned} DI_A(A) : \Delta A &= \frac{d}{d\epsilon} \text{tr}(A + \epsilon \Delta A)|_{\epsilon=0} \\ &= \frac{d}{d\epsilon} [A + \epsilon \Delta A] : I|_{\epsilon=0} \\ &= \Delta A : I|_{\epsilon=0} = I : \Delta A \end{aligned}$$

$$DI_A(A) = \frac{\partial I_A(A)}{\partial A} = \underline{I}$$

remark: I_A is linear in A , its Gateaux derivative is constant, there are no terms in ϵ once we take the derivative $d/d\epsilon$

derivative of second invariant $II_A = \frac{1}{2} [[A : I]^2 + A : A^t]$

$$\begin{aligned} DII_A(A) : \Delta A &= \frac{d}{d\epsilon} \frac{1}{2} \text{tr}^2(A + \epsilon \Delta A) - \frac{1}{2} \text{tr}(A + \epsilon \Delta A)^2|_{\epsilon=0} \\ &= \frac{d}{d\epsilon} \frac{1}{2} [[A + \epsilon \Delta A] : I]^2 \\ &\quad - \frac{1}{2} [A + \epsilon \Delta A] : [A^t + \epsilon \Delta A^t]|_{\epsilon=0} \\ &= [A + \epsilon \Delta A] : I \Delta A : I \\ &\quad - \frac{1}{2} \Delta A : A^t - \frac{1}{2} A : \Delta A^t - \epsilon \Delta A : \Delta A^t|_{\epsilon=0} \\ &= [\text{tr}(A) I - A^t] : \Delta A \end{aligned}$$

$$DII_A(A) = \frac{\partial II_A(A)}{\partial A} = \underline{\underline{\text{tr}(A) I - A^t}}$$

here, we have used the following identity

$$\text{tr}(A^2) = (A \cdot A) : I = A : A^t$$

remark: II_A is quadratic in A , its Gateaux derivative is linear, once we took the derivative $d/d\epsilon$, the higher order term in ϵ is filtered out by setting $\epsilon = 0$

derivative of third invariant $III_A = \det(A)$

$$\begin{aligned}
 DIII_A(A) : \Delta A &= \frac{d}{d\epsilon} \det(A + \epsilon \Delta A) \big|_{\epsilon=0} \\
 &= \frac{d}{d\epsilon} \det(A \cdot [I + A^{-1} \cdot \epsilon \Delta A]) \big|_{\epsilon=0} \\
 &= \frac{d}{d\epsilon} \det(A) \cdot \det(\epsilon A^{-1} \cdot \Delta A + I) \big|_{\epsilon=0} \\
 &= \frac{d}{d\epsilon} \det(A) \cdot [(\epsilon \lambda_{A^{-1} \cdot \Delta A 1} + 1) \\
 &\quad (\epsilon \lambda_{A^{-1} \cdot \Delta A 2} + 1)(\epsilon \lambda_{A^{-1} \cdot \Delta A 3} + 1)] \big|_{\epsilon=0} \\
 &= \det(A) \cdot [\lambda_{A^{-1} \cdot \Delta A 1} + \lambda_{A^{-1} \cdot \Delta A 2} + \lambda_{A^{-1} \cdot \Delta A 3}] \\
 &= \det(A) \cdot \text{tr}(A^{-1} \cdot \Delta A) \\
 &= \det(A) \cdot A^{-t} : \Delta A
 \end{aligned}$$

$$DIII_A(A) = \frac{\partial III_A(A)}{\partial A} = \underline{\underline{\det(A) \cdot A^{-t}}}$$

here, we have used the following expression for the determinant of $(\epsilon A^{-1} \cdot \Delta A)$ expressed through the characteristic polynomial for the eigenvalue $\lambda = -1$

$$\begin{aligned}
 \det(\epsilon A^{-1} \cdot \Delta A - \lambda I) &= (\epsilon \lambda_{A^{-1} \cdot \Delta A 1} - \lambda) \\
 &\quad (\epsilon \lambda_{A^{-1} \cdot \Delta A 2} - \lambda)(\epsilon \lambda_{A^{-1} \cdot \Delta A 3} - \lambda)
 \end{aligned}$$

reformulation with the help of index notation

$$\begin{aligned}
 \text{tr}(A^{-1} \cdot \Delta A) &= (A^{-1} \cdot \Delta A) : I \\
 &= (A_{ij}^{-1} \mathbf{e}_i \otimes \mathbf{e}_j) \cdot (\Delta A_{kl} \mathbf{e}_k \otimes \mathbf{e}_l) : (\delta_{mn} \mathbf{e}_m \otimes \mathbf{e}_n) \\
 &= (A_{ij}^{-1} \Delta A_{jl} \mathbf{e}_i \otimes \mathbf{e}_l) : (\delta_{mn} \mathbf{e}_m \otimes \mathbf{e}_n) \\
 &= A_{ij}^{-1} \Delta A_{jl} \delta_{im} \delta_{ln} \delta_{mn} = \underline{\underline{A^{-t} : \Delta A}}
 \end{aligned}$$

remark: III_A is cubic in A

derivative of function with tensorial argument show that the derivative of the function $\Phi(A)$ with respect to its tensorial argument A is given through the component-wise derivative with respect to the individual tensorial entries A_{ij}

index representation of second order tensor A

$$A = A_{ij} \mathbf{e}_i \otimes \mathbf{e}_j$$

$$\begin{aligned} A = & A_{11}\mathbf{e}_1 \otimes \mathbf{e}_1 + A_{12}\mathbf{e}_1 \otimes \mathbf{e}_2 + A_{13}\mathbf{e}_1 \otimes \mathbf{e}_3 \\ & + A_{21}\mathbf{e}_2 \otimes \mathbf{e}_1 + A_{22}\mathbf{e}_2 \otimes \mathbf{e}_2 + A_{23}\mathbf{e}_2 \otimes \mathbf{e}_3 \\ & + A_{31}\mathbf{e}_3 \otimes \mathbf{e}_1 + A_{32}\mathbf{e}_3 \otimes \mathbf{e}_2 + A_{33}\mathbf{e}_3 \otimes \mathbf{e}_3 \end{aligned}$$

extraction of individual components A_{ij} through "projection" onto the base vectors $\mathbf{e}_i$ and $\mathbf{e}_j$

$$A_{ij} = \mathbf{e}_i \cdot A \cdot \mathbf{e}_j = \mathbf{e}_i \cdot (A_{kl} \mathbf{e}_k \otimes \mathbf{e}_l) \cdot \mathbf{e}_j = \delta_{ik} A_{kl} \delta_{lj} = A_{ij}$$

and thus

$$\frac{\partial \Phi(A)}{\partial A} = \frac{\partial \Phi}{\partial A_{ij}} \frac{\partial A_{ij}}{\partial A} = \frac{\partial \Phi}{\partial A_{ij}} \frac{\partial \mathbf{e}_i \cdot A \cdot \mathbf{e}_j}{\partial A} = \frac{\partial \Phi}{\partial A_{ij}} \mathbf{e}_i \otimes \mathbf{e}_j$$

$$\begin{aligned} \frac{\partial \Phi(A)}{\partial A} = & \frac{\partial \Phi}{\partial A_{11}} \mathbf{e}_1 \otimes \mathbf{e}_1 + \frac{\partial \Phi}{\partial A_{12}} \mathbf{e}_1 \otimes \mathbf{e}_2 + \frac{\partial \Phi}{\partial A_{13}} \mathbf{e}_1 \otimes \mathbf{e}_3 \\ & + \frac{\partial \Phi}{\partial A_{21}} \mathbf{e}_2 \otimes \mathbf{e}_1 + \frac{\partial \Phi}{\partial A_{22}} \mathbf{e}_2 \otimes \mathbf{e}_2 + \frac{\partial \Phi}{\partial A_{23}} \mathbf{e}_2 \otimes \mathbf{e}_3 \\ & + \frac{\partial \Phi}{\partial A_{31}} \mathbf{e}_3 \otimes \mathbf{e}_1 + \frac{\partial \Phi}{\partial A_{32}} \mathbf{e}_3 \otimes \mathbf{e}_2 + \frac{\partial \Phi}{\partial A_{33}} \mathbf{e}_3 \otimes \mathbf{e}_3 \end{aligned}$$

remark: derivatives of functions with respect to tensors can be derived individually for each component

derivative of first invariant I_A wrt A use the component-wise derivation to determine the derivative of the first invariant I_A with respect to its tensor and validate the result derived previously

$$I_A = \text{tr}A = A_{11} + A_{22} + A_{33}$$

$$\frac{\partial I_A(A)}{\partial A} = \left[\frac{\partial I_A}{\partial A_{ij}} \right] \mathbf{e}_i \otimes \mathbf{e}_j$$

$$\left[\frac{\partial I_A}{\partial A_{ij}} \right] = \begin{bmatrix} \frac{\partial I_A}{\partial A_{11}} & \frac{\partial I_A}{\partial A_{12}} & \frac{\partial I_A}{\partial A_{13}} \\ \frac{\partial I_A}{\partial A_{21}} & \frac{\partial I_A}{\partial A_{22}} & \frac{\partial I_A}{\partial A_{23}} \\ \frac{\partial I_A}{\partial A_{31}} & \frac{\partial I_A}{\partial A_{32}} & \frac{\partial I_A}{\partial A_{33}} \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\underline{\underline{\frac{\partial I_A(A)}{\partial A} = I}}$$

remark: derivatives of functions with respect to tensors can be derived individually for each component

derivative of A wrt itself

$$A = A_{ij} \mathbf{e}_i \otimes \mathbf{e}_j$$

$$\frac{\partial A}{\partial A} = \left[\frac{\partial A_{ij}}{\partial A_{kl}} \right] \mathbf{e}_i \otimes \mathbf{e}_j \otimes \mathbf{e}_k \otimes \mathbf{e}_l \quad \left[\frac{\partial A_{ij}}{\partial A_{kl}} \right] = \delta_{ik} \delta_{jl}$$

$$\underline{\underline{\frac{\partial A}{\partial A} = I \otimes I = \mathbb{I}}}$$

remark: always keep in mind that the derivative of A with respect to itself is not(!) 1 but the fourth order unity tensor $\mathbb{I}$

derivative of inverse A^{-1} wrt A determine the derivative of the inverse of A wrt A by using the definition of the inverse $A^{-1} \cdot A = I$

$$\frac{\partial A^{-1}}{\partial A} = \left[\frac{\partial A_{ij}^{-1}}{\partial A_{kl}} \right] \mathbf{e}_i \otimes \mathbf{e}_j \otimes \mathbf{e}_k \otimes \mathbf{e}_l$$

derivative of identity with respect to A

$$\frac{\partial \delta_{im}}{\partial A_{kl}} = \frac{\partial A_{ij}^{-1} A_{jm}}{\partial A_{kl}} = \frac{\partial A_{ij}^{-1}}{\partial A_{kl}} A_{jm} + A_{ij}^{-1} \frac{\partial A_{jm}}{\partial A_{kl}} = \mathbb{O}$$

and thus

$$\left[\frac{\partial A_{ij}^{-1}}{\partial A_{kl}} \right] = -A_{ij}^{-1} \frac{\partial A_{jm}}{\partial A_{kl}} A_{mj}^{-1} = -A_{ij}^{-1} \delta_{jk} \delta_{ml} A_{jm}^{-t} = -A_{ik}^{-1} A_{jl}^{-t}$$

$$\underline{\underline{\frac{\partial A^{-1}}{\partial A} = -A^{-1} \overline{\otimes} A^{-t}}}$$

remark: the derivative of the inverse of a tensor wrt the tensor itself is negative. it can be determined with a trick by using the definition of the inverse.

1.2.2 Gradient

consider vector valued scalar and vector field $f(\mathbf{x})$ and $\mathbf{f}(\mathbf{x})$ on domain $\mathcal{B} \in \mathcal{R}^3$

$$\begin{aligned} f : \mathcal{B} &\rightarrow \mathcal{R} & f : \mathbf{x} &\rightarrow f(\mathbf{x}) \\ f : \mathcal{B} &\rightarrow \mathcal{R}^3 & \mathbf{f} : \mathbf{x} &\rightarrow \mathbf{f}(\mathbf{x}) \end{aligned}$$

Gradient of a scalar field

gradient $\nabla f(\mathbf{x})$ of vector valued scalar field $f(\mathbf{x})$

$$\nabla f(\mathbf{x}) = \frac{\partial f(\mathbf{x})}{\partial x_i} = f_{,i}(\mathbf{x}) \mathbf{e}_i \quad (1.2.3)$$

and thus

$$\nabla f(\mathbf{x}) = \begin{bmatrix} f_{,1} \\ f_{,2} \\ f_{,3} \end{bmatrix} \quad (1.2.4)$$

gradient of scalar field renders a vector field

Gradient of a vector field

gradient $\nabla \mathbf{f}(\mathbf{x})$ of vector valued vector field $\mathbf{f}(\mathbf{x})$

$$\nabla \mathbf{f}(\mathbf{x}) = \frac{\partial f_i(\mathbf{x})}{\partial x_j} = f_{i,j}(\mathbf{x}) \mathbf{e}_i \otimes \mathbf{e}_j \quad (1.2.5)$$

and thus

$$\nabla \mathbf{f}(\mathbf{x}) = \begin{bmatrix} f_{1,1} & f_{1,2} & f_{1,3} \\ f_{2,1} & f_{2,2} & f_{2,3} \\ f_{3,1} & f_{3,2} & f_{3,3} \end{bmatrix} \quad (1.2.6)$$

gradient of vector field renders a (second order) tensor field

1.2.3 Divergence

consider vector valued vector and tensor field $f(x)$ and $F(x)$ on domain $\mathcal{B} \in \mathcal{R}^3$

$$f : \mathcal{B} \rightarrow \mathcal{R}^3 \quad f : x \rightarrow f(x)$$

$$F : \mathcal{B} \rightarrow \mathcal{R}^3 \otimes \mathcal{R}^3 \quad F : x \rightarrow F(x)$$

Divergence of a vector field

divergence $\nabla f(x)$ of vector valued vector field $f(x)$

$$\text{div}(f(x)) = \text{tr}(\nabla f(x)) = \nabla f(x) : I \quad (1.2.7)$$

with $\nabla f(x) = f_{i,j}(x) e_i \otimes e_j$

$$\text{div}(f(x)) = f_{i,i}(x) = f_{1,1} + f_{2,2} + f_{3,3} \quad (1.2.8)$$

divergence of a vector field renders a scalar field

Divergence of a tensor field

divergence $\nabla F(x)$ of vector valued tensor field $F(x)$

$$\text{div}(F(x)) = \text{tr}(\nabla F(x)) = \nabla F(x) : I \quad (1.2.9)$$

with $\nabla F(x) = F_{ij,k}(x) e_i \otimes e_j \otimes e_k$

$$\text{div}(F(x)) = F_{ij,j}(x) = \begin{bmatrix} F_{11,1} + F_{12,2} + F_{13,3} \\ F_{21,1} + F_{22,2} + F_{23,3} \\ F_{31,1} + F_{32,2} + F_{33,3} \end{bmatrix} \quad (1.2.10)$$

divergence of a second order tensor field renders a vector field

1.2.4 Laplace operator

consider vector valued scalar and vector field $f(x)$ and $f(x)$ on domain $\mathcal{B} \in \mathcal{R}^3$

$$f : \mathcal{B} \rightarrow \mathcal{R} \quad f : x \rightarrow f(x)$$

$$f : \mathcal{B} \rightarrow \mathcal{R}^3 \quad f : x \rightarrow f(x)$$

Laplace operator acting on scalar field

Laplace operator $\Delta f(x)$ acting on vector valued scalar field $f(x)$

$$\Delta f(x) = \text{div}(\nabla(f(x))) \quad (1.2.11)$$

and thus

$$\Delta f(x) = f_{,ii} = f_{,11} + f_{,22} + f_{,33} \quad (1.2.12)$$

Laplace operator acting on scalar field renders a scalar field

Laplace operator acting on vector field

Laplace operator $\Delta f(x)$ acting on vector valued vector field $f(x)$

$$\Delta f(x) = \text{div}(\nabla(f(x))) \quad (1.2.13)$$

and thus

$$\Delta f(x) = f_{i,jj} = \begin{bmatrix} f_{1,11} + f_{1,22} + f_{1,33} \\ f_{2,11} + f_{2,22} + f_{2,33} \\ f_{3,11} + f_{3,22} + f_{3,33} \end{bmatrix} \quad (1.2.14)$$

Laplace operator acting on vector field renders a vector field

Useful transformations

consider scalar, vector and second order tensor field $\alpha(x)$, $u(x)$, $v(x)$ and $A(x)$ on domain $\mathcal{B} \in \mathcal{R}^3$

$$\alpha : \mathcal{B} \rightarrow \mathcal{R} \quad \alpha : x \rightarrow \alpha(x)$$

$$u : \mathcal{B} \rightarrow \mathcal{R}^3 \quad u : x \rightarrow u(x)$$

$$v : \mathcal{B} \rightarrow \mathcal{R}^3 \quad v : x \rightarrow v(x)$$

$$A : \mathcal{B} \rightarrow \mathcal{R}^3 \otimes \mathcal{R}^3 \quad A : x \rightarrow A(x)$$

important transformations

$$\begin{aligned} \nabla (\alpha u) &= u \otimes \nabla \alpha + \alpha \nabla u \\ \nabla (u \cdot v) &= u \cdot \nabla v + v \cdot \nabla u \\ \operatorname{div} (\alpha u) &= \alpha \operatorname{div}(u) + u \cdot \nabla \alpha \\ \operatorname{div} (\alpha A) &= \alpha \operatorname{div}(A) + A \cdot \nabla \alpha \\ \operatorname{div} (u \cdot A) &= u \cdot \operatorname{div}(A) + A : \nabla u \\ \operatorname{div} (u \otimes v) &= u \operatorname{div}(v) + v \cdot \nabla u^t \end{aligned} \tag{1.2.15}$$

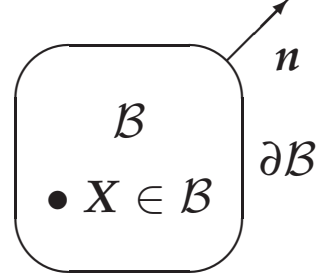
index notation write eqns (1.2.15) in index notation!

$$\begin{aligned} (\alpha u_i)_{,j} &= u_i \alpha_{,j} + \alpha u_{i,j} \\ (u_i v_i)_{,j} &= u_i v_{i,j} + v_i u_{i,j} \\ (\alpha u_i)_{,i} &= \alpha u_{i,i} + u_i \alpha_{,i} \\ (\alpha A_{ij})_{,j} &= \alpha A_{ij,j} + A_{ij} \alpha_{,j} \\ (u_i A_{ij})_{,j} &= u_i A_{ij,j} + A_{ij} u_{i,j} \\ (u_i v_j)_{,j} &= u_i v_{j,j} + v_j u_{i,j} \end{aligned} \tag{1.2.16}$$

remark: sometimes index notation can be more illustrative!

1.2.5 Integral transformations

integral theorems define relations
between surface integral $\int_{\partial\mathcal{B}} \dots dA$
and volume integral $\int_{\mathcal{B}} \dots dV$



consider scalar, vector and second order tensor field $\alpha(\mathbf{x})$, $\mathbf{u}(\mathbf{x})$ and $\mathbf{A}(\mathbf{x})$ on domain $\mathcal{B} \in \mathcal{R}^3$

$$\alpha : \mathcal{B} \rightarrow \mathcal{R} \qquad \alpha : \mathbf{x} \rightarrow \alpha(\mathbf{x})$$

$$\mathbf{u} : \mathcal{B} \rightarrow \mathcal{R}^3 \qquad \mathbf{u} : \mathbf{x} \rightarrow \mathbf{u}(\mathbf{x})$$

$$\mathbf{A} : \mathcal{B} \rightarrow \mathcal{R}^3 \otimes \mathcal{R}^3 \qquad \mathbf{A} : \mathbf{x} \rightarrow \mathbf{A}(\mathbf{x})$$

Integral theorem for scalar fields (Green theorem)

$$\begin{aligned} \int_{\partial\mathcal{B}} \alpha \mathbf{n} \, dA &= \int_{\mathcal{B}} \nabla \alpha \, dV \\ \int_{\partial\mathcal{B}} \alpha n_i \, dA &= \int_{\mathcal{B}} \alpha_{,i} \, dV \end{aligned} \tag{1.2.17}$$

Integral theorem for vector fields (Gauss theorem)

$$\begin{aligned} \int_{\partial\mathcal{B}} \mathbf{u} \cdot \mathbf{n} \, dA &= \int_{\mathcal{B}} \text{div}(\mathbf{u}) \, dV \\ \int_{\partial\mathcal{B}} u_i n_i \, dA &= \int_{\mathcal{B}} u_{i,i} \, dV \end{aligned} \tag{1.2.18}$$

Integral theorem for tensor fields (Gauss theorem)

$$\begin{aligned} \int_{\partial\mathcal{B}} \mathbf{A} \cdot \mathbf{n} \, dA &= \int_{\mathcal{B}} \text{div}(\mathbf{A}) \, dV \\ \int_{\partial\mathcal{B}} A_{ij} n_j \, dA &= \int_{\mathcal{B}} A_{ij,j} \, dV \end{aligned} \tag{1.2.19}$$

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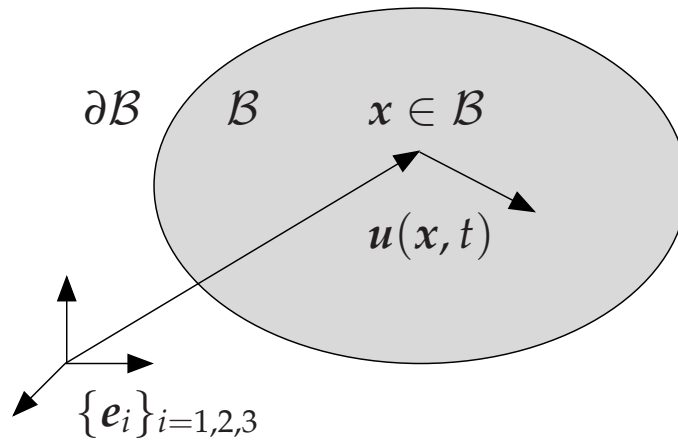
tuesday, january 19th, 2010

2 Kinematics

restriction to geometrically linear theory, valid if local strains remain small

2.1 Motion

consider a material body $\mathcal{B}$ as a simply connected subset of the Euclidian space $\mathcal{R}^3$ as $\mathcal{B} \subset \mathcal{R}^3$, with the boundary being denoted as $\partial\mathcal{B}$, a material point is defined as a point of the body $x \in \mathcal{B}$



motion of a body $\mathcal{B} \subset \mathcal{R}^3$ characterized through time dependent vector field of displacements $u \in \mathcal{R}^3$ parameterized in terms of position $x \in \mathcal{B}$ and time $t \in \mathcal{R}$

$$u : \mathcal{B} \times \mathcal{R} \rightarrow \mathcal{R}^3 \quad u(x, t) = u_i(x, t) e_i \quad (2.1.1)$$

2.2 Rates of kinematic quantities

2.2.1 Velocity

vector field of velocities $\mathbf{v} \in \mathcal{R}^3$ parameterized in terms of position $\mathbf{x} \in \mathcal{B}$ and time $t \in \mathcal{R}$

$$\mathbf{v} : \mathcal{B} \times \mathcal{R} \rightarrow \mathcal{R}^3 \quad \mathbf{v}(\mathbf{x}, t) = v_i(\mathbf{x}, t) \mathbf{e}_i \quad (2.2.1)$$

velocity field $\mathbf{v}$ defined through rate of change of displacement field $\mathbf{u}$

$$\mathbf{v}(\mathbf{x}, t) = \mathbf{D}_t \mathbf{u}(\mathbf{x}, t) = \left. \frac{\partial \mathbf{u}(\mathbf{x}, t)}{\partial t} \right|_{\mathbf{x} \text{ fixed}} \quad (2.2.2)$$

velocity vector

$$\mathbf{v} = [v_1, v_2, v_3]^t = \mathbf{D}_t [u_1, u_2, u_3]^t \quad (2.2.3)$$

common notation in the literature $\mathbf{v} = \dot{\mathbf{u}}$

2.2.2 Acceleration

vector field of accelerations $\mathbf{a} \in \mathcal{R}^3$ parameterized in terms of position $\mathbf{x} \in \mathcal{B}$ and time $t \in \mathcal{R}$

$$\mathbf{a} : \mathcal{B} \times \mathcal{R} \rightarrow \mathcal{R}^3 \quad \mathbf{a}(\mathbf{x}, t) = a_i(\mathbf{x}, t) \mathbf{e}_i \quad (2.2.4)$$

acceleration field $\mathbf{a}$ defined through rate of change of velocity field $\mathbf{v}$

$$\mathbf{a}(\mathbf{x}, t) = \mathbf{D}_t \mathbf{v}(\mathbf{x}, t) = \left. \frac{\partial \mathbf{v}(\mathbf{x}, t)}{\partial t} \right|_{\mathbf{x} \text{ fixed}} = \left. \frac{\partial^2 \mathbf{u}(\mathbf{x}, t)}{\partial t^2} \right|_{\mathbf{x} \text{ fixed}} \quad (2.2.5)$$

acceleration vector

$$\mathbf{a} = [a_1, a_2, a_3]^t = \mathbf{D}_t [v_1, v_2, v_3]^t = \mathbf{D}_t^2 [u_1, u_2, u_3]^t \quad (2.2.6)$$

common notation in the literature $\mathbf{a} = \dot{\mathbf{v}} = \ddot{\mathbf{u}}$

2.3 Gradients of kinematic quantities

2.3.1 Displacement gradient

second order tensor field of displacement gradient $\mathbf{H} \in \mathcal{R}^3 \otimes \mathcal{R}^3$ parameterized in terms of position $\mathbf{x} \in \mathcal{B}$ and time $t \in \mathcal{R}$

$$\mathbf{H} : \mathcal{B} \times \mathcal{R} \rightarrow \mathcal{R}^3 \otimes \mathcal{R}^3 \quad \mathbf{H}(\mathbf{x}, t) = H_{ij}(\mathbf{x}, t) \mathbf{e}_i \otimes \mathbf{e}_j \quad (2.3.1)$$

displacement gradient $\mathbf{H}$ defined through gradient of displacement field $\mathbf{u}$

$$\mathbf{H}(\mathbf{x}, t) = \nabla \mathbf{u}(\mathbf{x}, t) = \left. \frac{\partial \mathbf{u}(\mathbf{x}, t)}{\partial \mathbf{x}} \right|_{t \text{ fixed}} \quad (2.3.2)$$

index representation

$$H_{ij} \mathbf{e}_i \otimes \mathbf{e}_j = \frac{\partial u_i}{\partial X_j} \mathbf{e}_i \otimes \mathbf{e}_j = u_{i,j} \mathbf{e}_i \otimes \mathbf{e}_j \quad (2.3.3)$$

matrix representation of coordinates H_{ij}

$$[H_{ij}] = \begin{bmatrix} H_{11} & H_{12} & H_{13} \\ H_{21} & H_{22} & H_{23} \\ H_{31} & H_{32} & H_{33} \end{bmatrix} = \begin{bmatrix} u_{1,1} & u_{1,2} & u_{1,3} \\ u_{2,1} & u_{2,2} & u_{2,3} \\ u_{3,1} & u_{3,2} & u_{3,3} \end{bmatrix} \quad (2.3.4)$$

- displacement gradient $\mathbf{H} = \nabla \mathbf{u}$ is non-symmetric, $\mathbf{H} \neq \mathbf{H}^t$
- displacement gradient $\mathbf{H} = \nabla \mathbf{u}$ does not vanish for point-wise rigid body motion

Symmetric-skew-symmetric decomposition

symmetric-skew-symmetric decomposition of displacement gradient $\mathbf{H} = \nabla \mathbf{u}$

$$\mathbf{H} = \frac{1}{2}[\mathbf{H} + \mathbf{H}^t] + \frac{1}{2}[\mathbf{H} - \mathbf{H}^t] = \boldsymbol{\epsilon} + \boldsymbol{\omega} \quad (2.3.5)$$

with symmetric and skew-symmetric second order tensor $\boldsymbol{\epsilon} = \mathbf{H}^{\text{sym}}$ and $\boldsymbol{\omega} = \mathbf{H}^{\text{skw}}$

$$\boldsymbol{\epsilon} = \boldsymbol{\epsilon}^t \quad \boldsymbol{\omega} = -\boldsymbol{\omega}^t \quad (2.3.6)$$

- geometrically linear strain tensor $\boldsymbol{\epsilon}$

$$\boldsymbol{\epsilon} = \frac{1}{2}[\mathbf{H} + \mathbf{H}^t] = \frac{1}{2}[\nabla \mathbf{u} + \nabla^t \mathbf{u}] = \nabla^{\text{sym}} \mathbf{u} = \mathbb{I}^{\text{sym}} : \nabla \mathbf{u} \quad (2.3.7)$$

upon double contraction symmetric fourth order unit tensor $\mathbb{I}^{\text{sym}}$ extracts symmetric part of second order tensor

$$\begin{aligned} \mathbb{I}^{\text{sym}} &= \frac{1}{2} [\mathbb{I} + \mathbb{I}^t] \\ \mathbb{I}^{\text{sym}} &= \frac{1}{2} [\delta_{ik}\delta_{jl} + \delta_{il}\delta_{jk}] \mathbf{e}_i \otimes \mathbf{e}_j \otimes \mathbf{e}_k \otimes \mathbf{e}_l \end{aligned} \quad (2.3.8)$$

- geometrically linear rotation tensor $\boldsymbol{\omega}$

$$\boldsymbol{\omega} = \frac{1}{2}[\mathbf{H} - \mathbf{H}^t] = \frac{1}{2}[\nabla \mathbf{u} - \nabla^t \mathbf{u}] = \nabla^{\text{skw}} \mathbf{u} = \mathbb{I}^{\text{skw}} : \nabla \mathbf{u} \quad (2.3.9)$$

upon double contraction skew-symmetric fourth order unit tensor $\mathbb{I}^{\text{skw}}$ extracts skew-symmetric part of second order tensor

$$\begin{aligned} \mathbb{I}^{\text{skw}} &= \frac{1}{2} [\mathbb{I} - \mathbb{I}^t] \\ \mathbb{I}^{\text{skw}} &= \frac{1}{2} [\delta_{ik}\delta_{jl} - \delta_{il}\delta_{jk}] \mathbf{e}_i \otimes \mathbf{e}_j \otimes \mathbf{e}_k \otimes \mathbf{e}_l \end{aligned} \quad (2.3.10)$$

2.3.2 Strain

second order tensor field of (geometrically linear) strains $\epsilon \in \mathcal{R}^3 \otimes \mathcal{R}^3$ parameterized in terms of position $\mathbf{x} \in \mathcal{B}$ and time $t \in \mathcal{R}$

$$\epsilon : \mathcal{B} \times \mathcal{R} \rightarrow \mathcal{R}^3 \otimes \mathcal{R}^3 \quad \epsilon(\mathbf{x}, t) = \epsilon_{ij}(\mathbf{x}, t) \mathbf{e}_i \otimes \mathbf{e}_j \quad (2.3.11)$$

strain tensor ϵ defined through symmetric part of gradient of displacement field $\mathbf{u}$

$$\epsilon(\mathbf{x}, t) = \nabla^{\text{sym}} \mathbf{u}(\mathbf{x}, t) = \left[\frac{\partial \mathbf{u}(\mathbf{x}, t)}{\partial \mathbf{x}} \right]^{\text{sym}} \quad (2.3.12)$$

index representation

$$\epsilon_{ij} \mathbf{e}_i \otimes \mathbf{e}_j = \frac{1}{2} [u_{i,j} + u_{j,i}] \mathbf{e}_i \otimes \mathbf{e}_j \quad (2.3.13)$$

matrix representation of coordinates ϵ_{ij}

$$[\epsilon_{ij}] = \begin{bmatrix} \epsilon_{11} & \epsilon_{12} & \epsilon_{13} \\ \epsilon_{12} & \epsilon_{22} & \epsilon_{23} \\ \epsilon_{13} & \epsilon_{32} & \epsilon_{33} \end{bmatrix} = \frac{1}{2} \begin{bmatrix} 2u_{1,1} & u_{1,2} + u_{2,1} & u_{1,3} + u_{3,1} \\ u_{2,1} + u_{1,2} & 2u_{2,2} & u_{2,3} + u_{3,2} \\ u_{3,1} + u_{1,3} & u_{3,2} + u_{2,3} & 2u_{3,3} \end{bmatrix} \quad (2.3.14)$$

strain tensor $\epsilon = \nabla^{\text{sym}} \mathbf{u}$ is symmetric, $\epsilon = \epsilon^t$

$$[\epsilon_{ij}] = \begin{bmatrix} \epsilon_{11} & \epsilon_{12} & \epsilon_{13} \\ \epsilon_{21} & \epsilon_{22} & \epsilon_{23} \\ \epsilon_{31} & \epsilon_{32} & \epsilon_{33} \end{bmatrix} = \begin{bmatrix} \epsilon_{11} & \epsilon_{21} & \epsilon_{31} \\ \epsilon_{12} & \epsilon_{22} & \epsilon_{32} \\ \epsilon_{13} & \epsilon_{23} & \epsilon_{33} \end{bmatrix} = \epsilon_{ji} \quad (2.3.15)$$

ϵ_{ij} for $i = j$... diagonal entries: normal strain

ϵ_{ij} for $i \neq j$... off-diagonal entries: shear strain

2.3.3 Rotation

second order tensor field of (geometrically linear) rotation $\omega \in \mathcal{R}^3 \otimes \mathcal{R}^3$ parameterized in terms of position $\mathbf{x} \in \mathcal{B}$ and time $t \in \mathcal{R}$

$$\omega : \mathcal{B} \times \mathcal{R} \rightarrow \mathcal{R}^3 \otimes \mathcal{R}^3 \quad \omega(\mathbf{x}, t) = \omega_{ij}(\mathbf{x}, t) \mathbf{e}_i \otimes \mathbf{e}_j \quad (2.3.16)$$

rotation tensor ω defined through skew-symmetric part of gradient of displacement field $\mathbf{u}$

$$\omega(\mathbf{x}, t) = \nabla^{\text{skw}} \mathbf{u}(\mathbf{x}, t) = \left[\frac{\partial \mathbf{u}(\mathbf{x}, t)}{\partial \mathbf{x}} \right]^{\text{skw}} \quad (2.3.17)$$

index representation

$$\omega_{ij} \mathbf{e}_i \otimes \mathbf{e}_j = \frac{1}{2} [u_{i,j} - u_{j,i}] \mathbf{e}_i \otimes \mathbf{e}_j \quad (2.3.18)$$

matrix representation of coordinates ω_{ij}

$$[\omega_{ij}] = \begin{bmatrix} 0 & \omega_{12} & \omega_{13} \\ \omega_{21} & 0 & \omega_{23} \\ \omega_{31} & \omega_{32} & 0 \end{bmatrix} = \frac{1}{2} \begin{bmatrix} 0 & u_{1,2} - u_{2,1} & u_{1,3} - u_{3,1} \\ u_{2,1} - u_{1,2} & 0 & u_{2,3} - u_{3,2} \\ u_{3,1} - u_{1,3} & u_{3,2} - u_{2,3} & 0 \end{bmatrix} \quad (2.3.19)$$

rotation tensor $\omega = \nabla^{\text{skw}} \mathbf{u}$ is skew-symmetric, $\omega = -\omega^t$

$$[\omega_{ij}] = \begin{bmatrix} 0 & \omega_{12} & \omega_{13} \\ \omega_{21} & 0 & \omega_{23} \\ \omega_{31} & \omega_{32} & 0 \end{bmatrix} = - \begin{bmatrix} 0 & \omega_{21} & \omega_{31} \\ \omega_{12} & 0 & \omega_{32} \\ \omega_{13} & \omega_{23} & 0 \end{bmatrix} = -\omega_{ji} \quad (2.3.20)$$

corresponding axial vector $\mathbf{w} = -1/2 \overset{3}{\mathbf{e}} : \omega$

$$\mathbf{w} = [w_1, w_2, w_3]^t = -[\omega_{23}, \omega_{31}, \omega_{12}]^t \quad (2.3.21)$$

Symmetric-skew-symmetric decomposition

symmetric-skew-symmetric decomposition of displacement gradient $\mathbf{H} = \nabla \mathbf{u}$

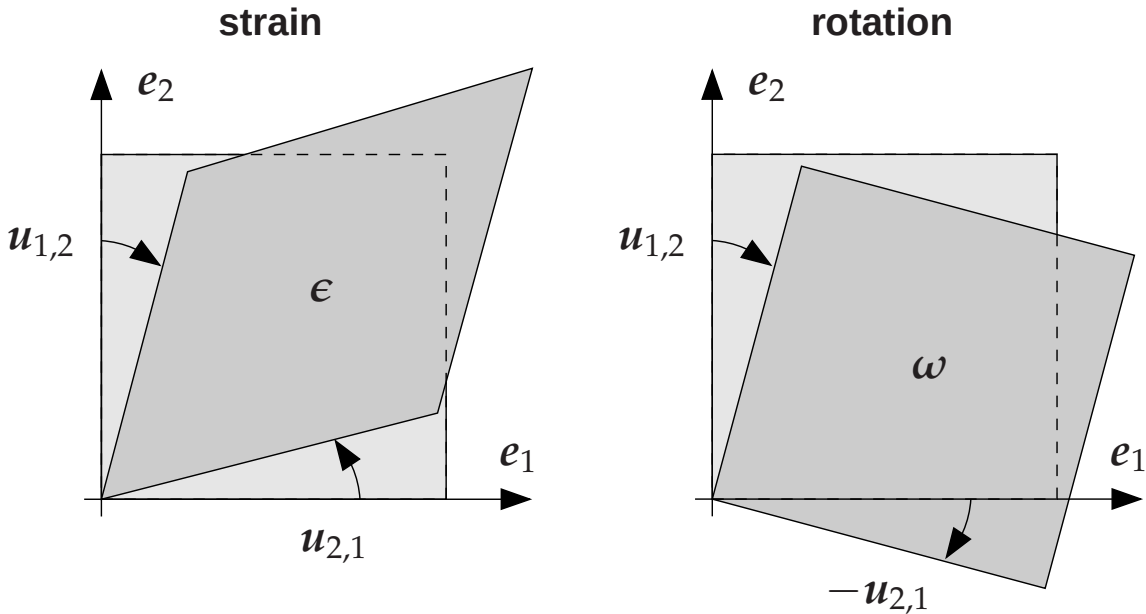
$$\mathbf{H} = \frac{1}{2}[\nabla \mathbf{u} + \nabla^t \mathbf{u}] + \frac{1}{2}[\nabla \mathbf{u} - \nabla^t \mathbf{u}] = \boldsymbol{\epsilon} + \boldsymbol{\omega} \quad (2.3.22)$$

with symmetric and skew-symmetric second order tensor $\boldsymbol{\epsilon} = \nabla^{\text{sym}} \mathbf{u}$ and $\boldsymbol{\omega} = \nabla^{\text{skw}} \mathbf{u}$

$$\boldsymbol{\epsilon} = \boldsymbol{\epsilon}^t \quad \boldsymbol{\omega} = -\boldsymbol{\omega}^t \quad (2.3.23)$$

geometric interpretation:

representation of symmetric and skew-symmetric part of displacement gradient for two-dimensional case



$$\epsilon_{12} = \frac{1}{2} [u_{1,2} + u_{2,1}]$$

$$\omega_{12} = \frac{1}{2} [u_{1,2} - u_{2,1}]$$

symmetric part $\boldsymbol{\epsilon} = \nabla^{\text{sym}} \mathbf{u}$ represents strain while skew-symmetric $\boldsymbol{\omega} = \nabla^{\text{skw}} \mathbf{u}$ part represents rotation

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thursday, january 21st, 2010

2.3.4 Volumetric–deviatoric decomposition

a material volume element can deform volumetrically and deviatorically, volumetric deformation conserves the shape (i.e. no changes in angles, no sliding) while deviatoric (isochoric) deformation conserves the volume

volumetric – deviatoric decomposition of strain tensor ϵ

$$\epsilon = \epsilon^{\text{vol}} + \epsilon^{\text{dev}} \quad (2.3.24)$$

with volumetric and deviatoric strain tensor ϵ^{vol} and ϵ^{dev}

$$\text{tr}(\epsilon^{\text{vol}}) = \text{tr}(\epsilon) \quad \text{tr}(\epsilon^{\text{dev}}) = 0 \quad (2.3.25)$$

- volumetric second order tensor ϵ^{vol}

$$\epsilon^{\text{vol}} = \frac{1}{3}[\epsilon : I] I = \mathbb{I}^{\text{vol}} : \epsilon \quad (2.3.26)$$

upon double contraction volumetric fourth order unit tensor $\mathbb{I}^{\text{vol}}$ extracts volumetric part ϵ^{vol} of strain tensor

$$\begin{aligned} \mathbb{I}^{\text{vol}} &= \frac{1}{3} I \otimes I \\ \mathbb{I}^{\text{vol}} &= \frac{1}{3} \delta_{ij} \delta_{kl} \mathbf{e}_i \otimes \mathbf{e}_j \otimes \mathbf{e}_k \otimes \mathbf{e}_l \end{aligned} \quad (2.3.27)$$

- deviatoric second order tensor ϵ^{dev}

$$\epsilon^{\text{dev}} = \epsilon - \frac{1}{3}[\epsilon : I] I = \mathbb{I}^{\text{dev}} : \epsilon \quad (2.3.28)$$

upon double contraction deviatoric fourth order unit tensor $\mathbb{I}^{\text{dev}}$ extracts deviatoric part of strain tensor

$$\begin{aligned} \mathbb{I}^{\text{dev}} &= \mathbb{I}^{\text{sym}} - \mathbb{I}^{\text{vol}} = \mathbb{I}^{\text{sym}} - \frac{1}{3} I \otimes I \\ \mathbb{I}^{\text{dev}} &= \left[\frac{1}{2} \delta_{ik} \delta_{jl} + \frac{1}{2} \delta_{il} \delta_{jk} - \frac{1}{3} \delta_{ij} \delta_{kl} \right] \mathbf{e}_i \otimes \mathbf{e}_j \otimes \mathbf{e}_k \otimes \mathbf{e}_l \end{aligned} \quad (2.3.29)$$

Volumetric strain

volumetric deformation is characterized through the volume dilatation $e \in \mathcal{R}$, i.e. difference of deformed volume and original volume $dv - dV$ scaled by original volume dV

$$\begin{aligned} e &= \frac{dv - dV}{dV} = (1 + \epsilon_{11})(1 + \epsilon_{22})(1 + \epsilon_{33}) - 1 \\ &= \epsilon_{11} + \epsilon_{22} + \epsilon_{33} + \mathcal{O}(\epsilon_{ij}^2) \end{aligned} \quad (2.3.30)$$

neglection of higher order terms: trace of strain tensor $\text{tr}(\epsilon) = \epsilon : \mathbf{I} \in \mathcal{R}$ as characteristic measure for volume changes

$$e = \text{div } \mathbf{u} = \nabla \mathbf{u} : \mathbf{I} = \epsilon : \mathbf{I} = \text{tr}(\epsilon) \quad (2.3.31)$$

volumetric part ϵ^{vol} of strain tensor ϵ

$$\epsilon^{\text{vol}} = \frac{1}{3} e \mathbf{I} = \frac{1}{3} [\epsilon : \mathbf{I}] \mathbf{I} = \frac{1}{3} [\mathbf{I} \otimes \mathbf{I}] : \epsilon = \mathbb{I}^{\text{vol}} : \epsilon \quad (2.3.32)$$

index representation

$$\epsilon^{\text{vol}} = \epsilon_{ij}^{\text{vol}} \mathbf{e}_i \otimes \mathbf{e}_j \quad (2.3.33)$$

matrix representation of coordinates $[\epsilon_{ij}^{\text{vol}}]$

$$[\epsilon_{ij}^{\text{vol}}] = \frac{1}{3} e \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad e = \text{tr}(\epsilon) \quad (2.3.34)$$

- incompressibility is characterized through $\text{div } \mathbf{u} = 0$
- volumetric strain tensor ϵ^{vol} is a spherical second order tensor as $\epsilon^{\text{vol}} = \frac{1}{3} e \mathbf{I}$
- volumetric strain tensor ϵ^{vol} contains the volume changing, shape preserving part of the total strain tensor ϵ

Deviatoric strain

deviatoric strain tensor ϵ^{dev} preserves the volume and contains the remaining part of the total strain tensor ϵ

deviatoric part ϵ^{dev} of the strain tensor ϵ

$$\epsilon^{\text{dev}} = \epsilon - \epsilon^{\text{vol}} = \epsilon - \frac{1}{3} [\epsilon : \mathbf{I}] \mathbf{I} = \mathbb{I}^{\text{dev}} : \epsilon \quad (2.3.35)$$

index representation

$$\epsilon^{\text{dev}} = \epsilon_{ij}^{\text{dev}} \mathbf{e}_i \otimes \mathbf{e}_j \quad (2.3.36)$$

matrix representation of coordinates $[\epsilon_{ij}^{\text{dev}}]$

$$[\epsilon_{ij}^{\text{dev}}] = \begin{bmatrix} \frac{1}{3}[2\epsilon_{11} - \epsilon_{22} - \epsilon_{33}] & \epsilon_{12} & \epsilon_{13} \\ \epsilon_{21} & \frac{1}{3}[2\epsilon_{22} - \epsilon_{11} - \epsilon_{33}] & \epsilon_{23} \\ \epsilon_{31} & \epsilon_{32} & \frac{1}{3}[2\epsilon_{33} - \epsilon_{11} - \epsilon_{22}] \end{bmatrix} \quad (2.3.37)$$

trace of deviatoric strains $\text{tr}(\epsilon^{\text{dev}})$

$$\begin{aligned} \text{tr}(\epsilon^{\text{dev}}) &= \frac{1}{3}[2\epsilon_{11} - \epsilon_{22} - \epsilon_{33}] \\ &+ \frac{1}{3}[2\epsilon_{22} - \epsilon_{11} - \epsilon_{33}] \\ &+ \frac{1}{3}[2\epsilon_{33} - \epsilon_{11} - \epsilon_{22}] = 0 \end{aligned} \quad (2.3.38)$$

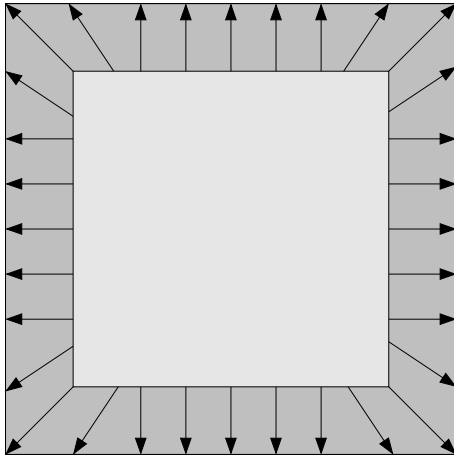
- deviatoric strain tensor ϵ^{dev} is a traceless second order tensor as $\text{tr}(\epsilon^{\text{dev}}) = 0$
- deviatoric strain tensor ϵ^{dev} contains the shape changing, volume preserving part of the total strain tensor ϵ

Volumetric–deviatoric decomposition

- examples of purely volumetric deformation

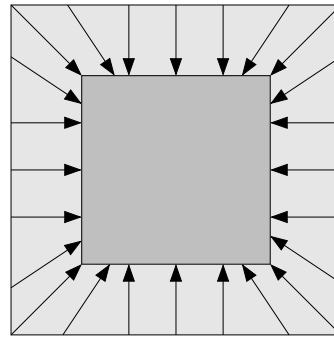
$$\epsilon^{\text{vol}} = \frac{1}{3}[\epsilon : I] I = \mathbb{I}^{\text{vol}} : \epsilon \quad \text{tr}(\epsilon^{\text{vol}}) = \text{tr}(\epsilon) \quad (2.3.39)$$

expansion



$$e > 0 \text{ and } \epsilon^{\text{dev}} = 0$$

compression

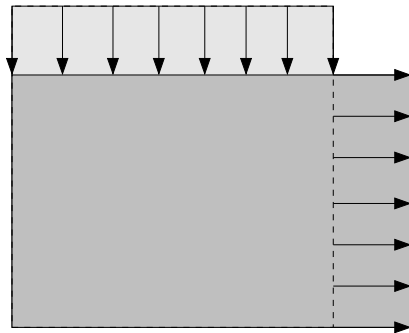


$$e < 0 \text{ and } \epsilon^{\text{dev}} = 0$$

- examples of purely deviatoric deformation

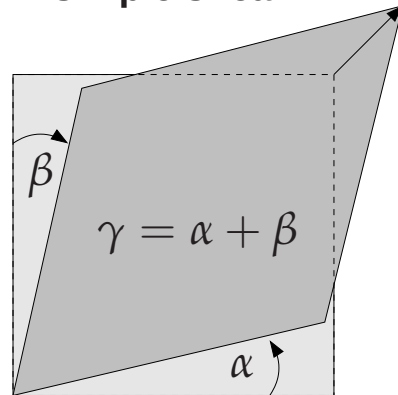
$$\epsilon^{\text{dev}} = \epsilon - \frac{1}{3}[\epsilon : I] I = \mathbb{I}^{\text{dev}} : \epsilon \quad \text{tr}(\epsilon^{\text{dev}}) = 0 \quad (2.3.40)$$

pure shear



$$e = 0 \text{ and } \epsilon^{\text{dev}} \neq 0$$

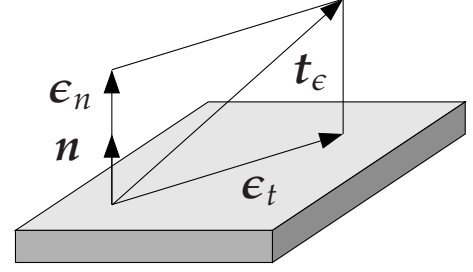
simple shear



$$e = 0 \text{ and } \epsilon^{\text{dev}} \neq 0$$

2.3.5 Strain vector

assume we are interested in strain on a plane characterized through its normal $\mathbf{n}$, strain vector $\mathbf{t}_\epsilon$ acting on plane given through normal projection of strain tensor ϵ



$$\mathbf{t}_\epsilon = \epsilon \cdot \mathbf{n} \quad (2.3.41)$$

index representation

$$\begin{aligned} \mathbf{t}_\epsilon &= (\epsilon_{ij} \mathbf{e}_i \otimes \mathbf{e}_j) \cdot (n_k \mathbf{e}_k) \\ &= \epsilon_{ij} n_k \delta_{jk} \mathbf{e}_i = \epsilon_{ij} n_j \mathbf{e}_i = t_{\epsilon i} \mathbf{e}_i \end{aligned} \quad (2.3.42)$$

representation of coordinates $[t_{\epsilon i}]$

$$\begin{bmatrix} t_{\epsilon 1} \\ t_{\epsilon 2} \\ t_{\epsilon 3} \end{bmatrix} = \begin{bmatrix} \epsilon_{11} n_1 + \epsilon_{12} n_2 + \epsilon_{13} n_3 \\ \epsilon_{21} n_1 + \epsilon_{22} n_2 + \epsilon_{23} n_3 \\ \epsilon_{31} n_1 + \epsilon_{32} n_2 + \epsilon_{33} n_3 \end{bmatrix} \quad (2.3.43)$$

alternative interpretation: assume we are interested in strains along a particular material direction, i.e. the stretch of a fiber at $\mathbf{x} \in \mathcal{B}$ characterized through its normal $\mathbf{n}$ with $\|\mathbf{n}\| = 1$

stretch as change of displacement vector $\mathbf{u}$ in the direction of $\mathbf{n}$ given through the Gateaux derivative §??

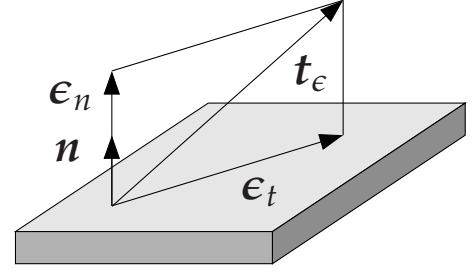
$$\begin{aligned} D\mathbf{u}(\mathbf{x}) \cdot \mathbf{n} &= \frac{d}{d\epsilon} \mathbf{u}(\mathbf{x} + \epsilon \mathbf{n})|_{\epsilon=0} \\ &= \underbrace{\nabla \mathbf{u}(\mathbf{x} + \epsilon \mathbf{n})}_{\text{outer derviative}} \cdot \underbrace{\mathbf{n}}_{\text{inner derivative}}|_{\epsilon=0} = \nabla \mathbf{u}(\mathbf{x}) \cdot \mathbf{n} \end{aligned} \quad (2.3.44)$$

recall that $\nabla \mathbf{u} = \nabla^{\text{sym}} \mathbf{u} + \nabla^{\text{skw}} \mathbf{u} = \epsilon + \omega$ whereby rotation $\omega = \nabla^{\text{skw}} \mathbf{u}$ does not induce strain, thus

$$\mathbf{t}_\epsilon = \nabla^{\text{sym}} \mathbf{u} \cdot \mathbf{n} = \epsilon \cdot \mathbf{n} \quad (2.3.45)$$

2.3.6 Normal–shear decomposition

assume we are interested in strain along a particular fiber characterized through its normal $\mathbf{n}$, stretch of fiber ϵ_n given through normal projection of strain vector $\mathbf{t}_\epsilon$



$$\epsilon_n = \mathbf{t}_\epsilon \cdot \mathbf{n} \quad (2.3.46)$$

alternative interpretation: stretch of a line element can be understood as the projection of change of displacement in the direction of $\mathbf{n}$ as $D\mathbf{u} \cdot \mathbf{n} = \nabla \mathbf{u} \cdot \mathbf{n}$ onto the direction $\mathbf{n}$

$$\epsilon_n = \mathbf{n} \cdot \nabla \mathbf{u} \cdot \mathbf{n} = \mathbf{n} \cdot \boldsymbol{\epsilon} \cdot \mathbf{n} = \boldsymbol{\epsilon} : [\mathbf{n} \otimes \mathbf{n}] \quad (2.3.47)$$

normal-shear (tangential) decomposition of strain vector $\mathbf{t}_\epsilon$

$$\mathbf{t}_\epsilon = \boldsymbol{\epsilon}_n + \boldsymbol{\epsilon}_t \quad (2.3.48)$$

normal strain vector – stretch of fibers in direction of $\mathbf{n}$

$$\boldsymbol{\epsilon}_n = \boldsymbol{\epsilon} : [\mathbf{n} \otimes \mathbf{n}] \mathbf{n} \quad (2.3.49)$$

shear (tangential) strain vector – sliding of fibers parallel to $\mathbf{n}$

$$\boldsymbol{\epsilon}_t = \mathbf{t}_\epsilon - \boldsymbol{\epsilon}_n = \boldsymbol{\epsilon} : [\mathbb{I}^{\text{sym}} \cdot \mathbf{n} - \mathbf{n} \otimes \mathbf{n} \otimes \mathbf{n}] \quad (2.3.50)$$

amount of sliding γ_n

$$\gamma_n = 2 \|\boldsymbol{\epsilon}_t\| = 2 \sqrt{\boldsymbol{\epsilon}_t \cdot \boldsymbol{\epsilon}_t} = 2 \sqrt{\mathbf{t}_\epsilon \cdot \mathbf{t}_\epsilon - \epsilon_n^2} \quad (2.3.51)$$

in general, i.e. for an arbitrary direction $\mathbf{n}$, we have normal and shear contributions to the strain vector, however, three particular directions $\{\mathbf{n}_{\epsilon i}\}_{i=1,2,3}$ can be identified, for which $\mathbf{t}_\epsilon = \boldsymbol{\epsilon}_n$ and thus $\boldsymbol{\epsilon}_t = \mathbf{0}$, the corresponding $\{\mathbf{n}_{\epsilon i}\}_{i=1,2,3}$ are called principal strain directions and $\{\epsilon_{ni}\}_{i=1,2,3} = \{\lambda_{\epsilon i}\}_{i=1,2,3}$ are the principal strains or stretches

2.3.7 Principal strains – stretches

assume strain tensor ϵ to be known at $x \in \mathcal{B}$, principal strains $\{\lambda_{\epsilon i}\}_{i=1,2,3}$ and principal strain directions $\{\mathbf{n}_{\epsilon i}\}_{i=1,2,3}$ can be derived from solution of special eigenvalue problem according to §1.1.3

$$\epsilon \cdot \mathbf{n}_{\epsilon i} = \lambda_{\epsilon i} \mathbf{n}_{\epsilon i} \quad [\epsilon - \lambda_{\epsilon i}] \cdot \mathbf{n}_{\epsilon i} = \mathbf{0} \quad (2.3.52)$$

solution

$$\det(\epsilon - \lambda_{\epsilon} \mathbf{I}) = 0 \quad (2.3.53)$$

or in terms of roots of characteristic equation

$$\lambda_{\epsilon}^3 - I_{\epsilon} \lambda_{\epsilon}^2 + II_{\epsilon} \lambda_{\epsilon} - III_{\epsilon} = 0 \quad (2.3.54)$$

roots of characteristic equations in terms of principal invariants of ϵ

$$\begin{aligned} I_{\epsilon} &= \text{tr}(\epsilon) = \lambda_{\epsilon 1} + \lambda_{\epsilon 2} + \lambda_{\epsilon 3} \\ II_{\epsilon} &= \frac{1}{2}[\text{tr}^2(\epsilon) - \text{tr}(\epsilon^2)] = \lambda_{\epsilon 2} \lambda_{\epsilon 3} + \lambda_{\epsilon 3} \lambda_{\epsilon 1} + \lambda_{\epsilon 1} \lambda_{\epsilon 2} \\ III_{\epsilon} &= \det(\epsilon) = \lambda_{\epsilon 1} \lambda_{\epsilon 2} \lambda_{\epsilon 3} \end{aligned} \quad (2.3.55)$$

spectral representation of ϵ

$$\epsilon = \sum_{i=1}^3 \lambda_{\epsilon i} \mathbf{n}_{\epsilon i} \otimes \mathbf{n}_{\epsilon i} \quad (2.3.56)$$

principal strains (stretches) $\lambda_{\epsilon i}$ are purely normal, no shear deformation (sliding) γ_n in principal directions, i.e. $\mathbf{t}_{\epsilon i} = \epsilon \mathbf{n}_{\epsilon i} = \lambda_{\epsilon i} \mathbf{n}_{\epsilon i}$ and $\epsilon_t = \mathbf{0}$ thus $\gamma_n = 0$

due to symmetry of strains $\epsilon = \epsilon^t$, strain tensor possesses three real eigenvalues $\{\lambda_{\epsilon i}\}_{i=1,2,3}$, corresponding eigendirections $\{\mathbf{n}_{\epsilon i}\}_{i=1,2,3}$ are thus orthogonal $\mathbf{n}_{\epsilon i} \cdot \mathbf{n}_{\epsilon j} = \delta_{ij}$

2.3.8 Compatibility

until now, we have assumed the displacement field $\mathbf{u}(\mathbf{x}, t)$ to be given, such that the strain field $\boldsymbol{\epsilon} = \nabla^{\text{sym}} \mathbf{u}$ could have been derived uniquely as partial derivative of $\mathbf{u}$ with respect to the position $\mathbf{x}$ at fixed time t

assume now, that for a given strain field $\boldsymbol{\epsilon}(\mathbf{x}, t)$, we want to know whether these strains $\boldsymbol{\epsilon}$ are compatible with a continuous single-valued displacement field $\mathbf{u}$

symmetric second order incompatibility tensor

$$\boldsymbol{\eta} = \text{crl}(\text{crl}(\boldsymbol{\epsilon})) \quad (2.3.57)$$

index representation of incompatibility tensor

$$\boldsymbol{\eta} = \eta_{ij} \mathbf{e}_i \otimes \mathbf{e}_j = \overset{3}{e}_{ikm} \epsilon_{kn,ml} \overset{3}{e}_{jln} \mathbf{e}_i \otimes \mathbf{e}_j = \mathbf{0} \quad (2.3.58)$$

coordinate representation of compatibility condition

$$\epsilon_{kl,mn} + \epsilon_{mn,kl} - \epsilon_{ml,kn} - \epsilon_{kn,ml} = 0 \quad (2.3.59)$$

valid $\forall k, l, m, n$, thus 81 equations which are partly redundant, six independent conditions

St. Venant compatibility conditions

$$\begin{aligned} \eta_{11} &= \epsilon_{22,33} + \epsilon_{33,22} - 2\epsilon_{23,32} &= 0 \\ \eta_{22} &= \epsilon_{33,11} + \epsilon_{11,33} - 2\epsilon_{31,13} &= 0 \\ \eta_{33} &= \epsilon_{11,22} + \epsilon_{22,11} - 2\epsilon_{12,21} &= 0 \\ \eta_{12} &= \epsilon_{13,32} + \epsilon_{23,31} - \epsilon_{33,12} - \epsilon_{12,33} &= 0 \\ \eta_{23} &= \epsilon_{21,13} + \epsilon_{31,12} - \epsilon_{11,23} - \epsilon_{23,11} &= 0 \\ \eta_{31} &= \epsilon_{32,21} + \epsilon_{12,23} - \epsilon_{22,31} - \epsilon_{31,22} &= 0 \end{aligned} \quad (2.3.60)$$

incompatible displacement field, e.g. in dislocation theory

2.3.9 Special case of plane strain

dimensional reduction in case of plane strain with vanishing strains $\epsilon_{13} = \epsilon_{23} = \epsilon_{31} = \epsilon_{32} = \epsilon_{33} = 0$ in out of plane direction, e.g. in geomechanics

$$\epsilon = \epsilon_{ij} \mathbf{e}_i \otimes \mathbf{e}_j \quad (2.3.61)$$

matrix representation of coordinates $[\epsilon_{ij}]$

$$[\epsilon_{ij}] = \begin{bmatrix} \epsilon_{11} & \epsilon_{12} & 0 \\ \epsilon_{21} & \epsilon_{22} & 0 \\ 0 & 0 & 0 \end{bmatrix} \quad (2.3.62)$$

2.3.10 Voigt representation of strain

three dimensional second order strain tensor ϵ

$$\epsilon = \epsilon_{ij} \mathbf{e}_i \otimes \mathbf{e}_j \quad (2.3.63)$$

matrix representation of coordinates $[\epsilon_{ij}]$

$$[\epsilon_{ij}] = \begin{bmatrix} \epsilon_{11} & \epsilon_{12} & \epsilon_{13} \\ \epsilon_{21} & \epsilon_{22} & \epsilon_{23} \\ \epsilon_{31} & \epsilon_{23} & \epsilon_{33} \end{bmatrix} \quad (2.3.64)$$

due to symmetry $[\epsilon_{ij}] = [\epsilon_{ji}]$ and thus $\epsilon_{12} = \epsilon_{21}$, $\epsilon_{23} = \epsilon_{32}$, $\epsilon_{31} = \epsilon_{13}$, strain tensor ϵ contains only six independent components $\epsilon_{11}, \epsilon_{22}, \epsilon_{33}, \epsilon_{12}, \epsilon_{23}, \epsilon_{31}$, it proves convenient to represent second order tensor ϵ through a vector $\underline{\epsilon}$

$$\underline{\epsilon} = [\epsilon_{11}, \epsilon_{22}, \epsilon_{33}, 2\epsilon_{12}, 2\epsilon_{23}, 2\epsilon_{31}]^t \quad (2.3.65)$$

vector representation $\underline{\epsilon}$ of strain ϵ in case of plane strain

$$\underline{\epsilon} = [\epsilon_{11}, \epsilon_{22}, 2\epsilon_{12}]^t \quad (2.3.66)$$

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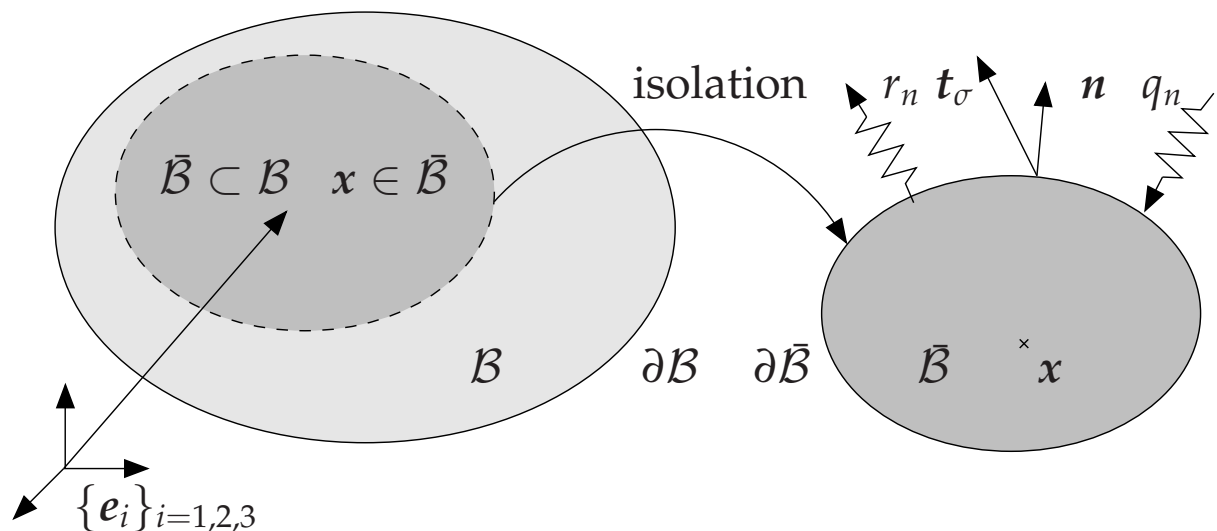
lecture notes 07

tuesday, january 26th, 2010

3 Balance equations

3.1 Basic ideas

- until now:
kinematics, i.e. characterization of deformation of a material body $\mathcal{B}$ without studying its physical cause
- now:
balance equations, i.e. general statements that characterize the cause of cause of the motion of any body $\mathcal{B}$



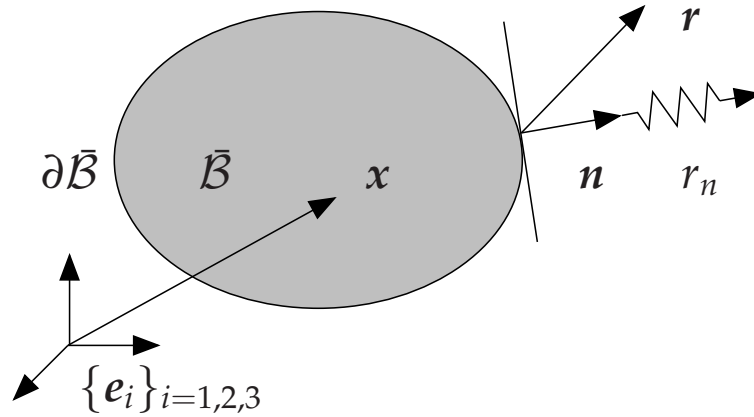
basic strategy

- isolation of an arbitrary subset $\bar{\mathcal{B}}$ of the body $\mathcal{B}$
- characterization of the influence of the remaining body $\mathcal{B} \setminus \bar{\mathcal{B}}$ on $\bar{\mathcal{B}}$ through phenomenological quantities, i.e. the contact mass flux r , the contact stress t_σ , the contact heat flux q
- definition of basic physical quantities, i.e. the mass m , the linear momentum I , the moment of momentum D and the energy E of subset $\bar{\mathcal{B}}$
- postulate of balance of these quantities renders global balance equations for subset $\bar{\mathcal{B}}$
- localization of global balance equations renders local balance equations at point $x \in \bar{\mathcal{B}}$

3.1.1 Concept of mass flux

the contact mass flux r_n at a point x is a scalar of the unit [mass/time/surface area]

the contact mass flux r_n characterizes the transport of matter normal to the tangent plane to an imaginary surface passing through this point with normal vector $\mathbf{n}$



definition of contact heat flux q_n in analogy to Cauchy's postulate, lemma and theorem originally introduced for the momentum flux in §3.1.2

Cauchy's postulate

$$r_n = r_n(\mathbf{x}, \mathbf{n}) \quad (3.1.1)$$

Cauchy's lemma

$$r_n(\mathbf{x}, \mathbf{n}) = -r_n(\mathbf{x}, -\mathbf{n}) \quad (3.1.2)$$

Cauchy's theorem

the contact mass flux r_n can be expressed as linear function of the surface normal $\mathbf{n}$ and the mass flux vector $\mathbf{r}$

$$r_n = \mathbf{r} \cdot \mathbf{n} \quad (3.1.3)$$

Mass flux vector

the vector field $\mathbf{r}$ is called mass flux vector

$$\mathbf{r} = r_i \mathbf{e}_i \quad (3.1.4)$$

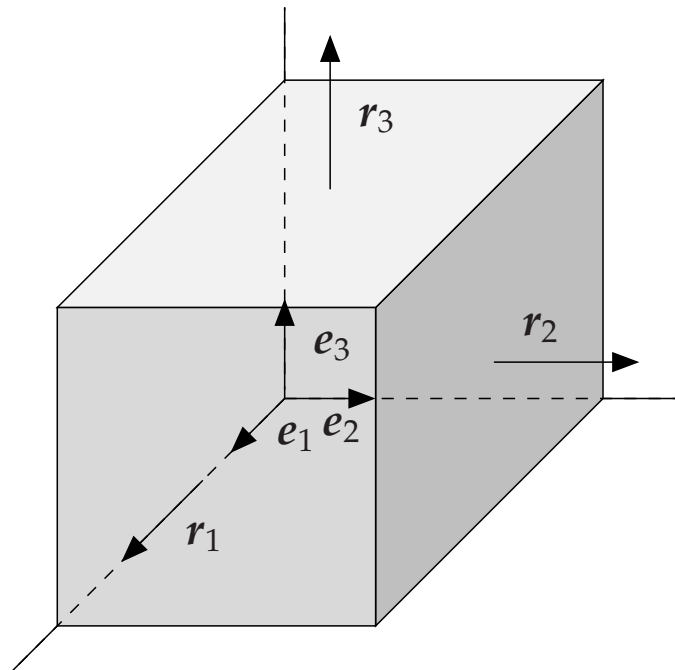
Cauchy's theorem

$$r_n = \mathbf{r} \cdot \mathbf{n} \quad (3.1.5)$$

index representation

$$r_n = (r_i \mathbf{e}_i) \cdot (n_j \mathbf{e}_j) = r_i n_j \delta_{ij} = r_i n_i \quad (3.1.6)$$

geometric interpretation



the coordinates r_i characterize the transport of matter through the planes parallel to the coordinate planes

in classical closed system continuum mechanics (here) the mass flux vector vanishes identically

examples of mass flux: transport of chemical reactants in chemomechanics or cell migration in biomechanics

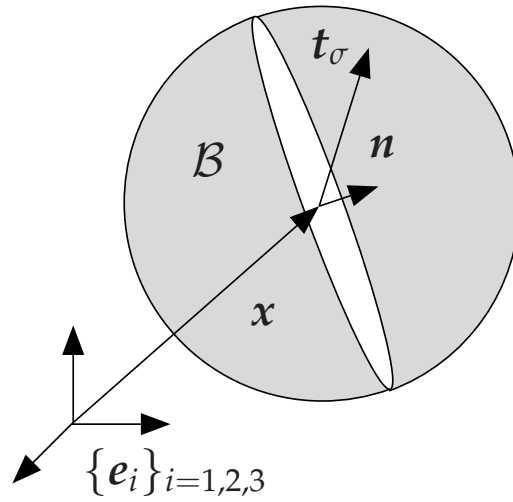
3.1.2 Concept of stress

traction vector

$$t_\sigma = \lim_{\Delta a \rightarrow 0} \frac{\Delta f}{\Delta a} = \frac{df}{da} \quad (3.1.7)$$

interpretation as surface force per unit surface area

Cauchy's postulate



the traction vector t_σ at a point x can be expressed exclusively in terms of the point x and the normal n to the tangent plane to an imaginary surface passing through this point

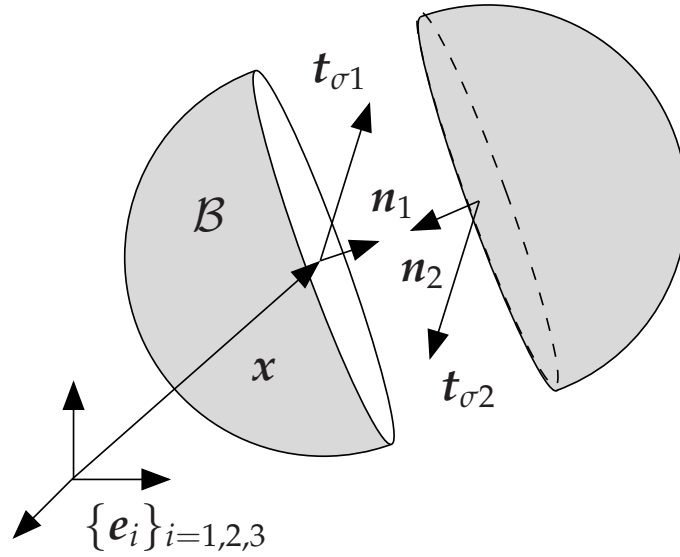
traction vector

$$t_\sigma = t_\sigma(x, n) \quad (3.1.8)$$

Cauchy's lemma

the traction vectors acting on opposite sides of a surface are equal in magnitude and opposite in sign

$$t_{\sigma 1}(x, n_1) = -t_{\sigma 2}(x, n_2) \quad (3.1.9)$$



generalization with $\mathbf{n} = \mathbf{n}_1 = -\mathbf{n}_2$ and $\mathbf{t}_\sigma = \mathbf{t}_{\sigma 1}$

$$\mathbf{t}_\sigma(\mathbf{x}, \mathbf{n}) = -\mathbf{t}_\sigma(\mathbf{x}, -\mathbf{n}) \quad (3.1.10)$$

Cauchy's theorem

the traction vector $\mathbf{t}_\sigma$ can be expressed as a linear map of the surface normal $\mathbf{n}$ mapped via the transposed stress tensor σ^t

$$\mathbf{t}_\sigma = \sigma^t \cdot \mathbf{n} \quad (3.1.11)$$

accordingly with $\mathbf{n} = \mathbf{n}_1 = -\mathbf{n}_2$ and $\mathbf{t}_\sigma = \mathbf{t}_{\sigma 1}$

$$\begin{aligned} \mathbf{t}_{\sigma 1} &= \sigma^t \cdot \mathbf{n}_1 = \sigma^t \cdot \mathbf{n} = \mathbf{t}_\sigma \\ \mathbf{t}_{\sigma 2} &= \sigma^t \cdot \mathbf{n}_2 = -\sigma^t \cdot \mathbf{n} = -\mathbf{t}_\sigma \end{aligned} \quad (3.1.12)$$

Cauchy tetraeder

balance of momentum (pointwise)

$$\mathbf{t}_\sigma(\mathbf{n}) da = -\mathbf{t}_\sigma(\mathbf{n}_i) da_i = \mathbf{t}_\sigma(\mathbf{e}_i) da_i = \mathbf{t}_{\sigma i} da_i \quad (3.1.13)$$

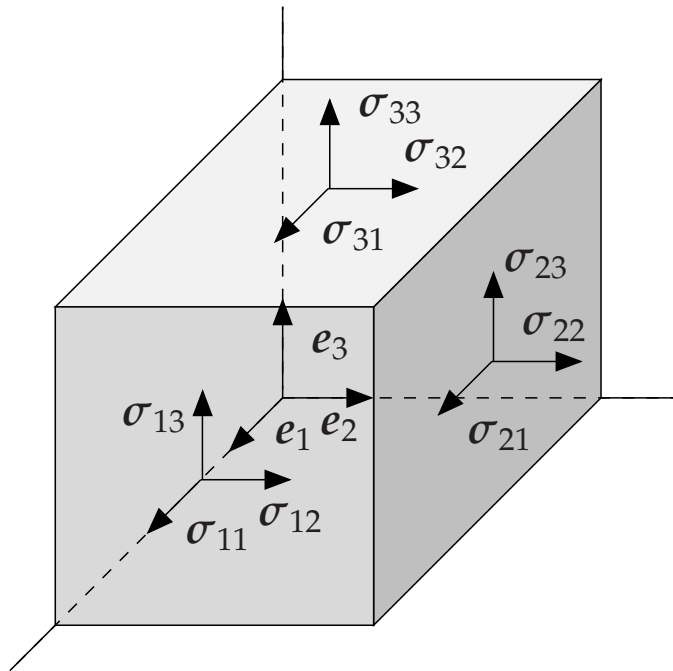
surface theorem, area fractions from Gauss theorem

$$\mathbf{n} da = -\mathbf{n}_i da_i = \mathbf{e}_i da_i \quad \frac{da_i}{da} = \mathbf{e}_i \cdot \mathbf{n} = \cos \angle(\mathbf{e}_i, \mathbf{n}) \quad (3.1.14)$$

matrix representation of tensor coordinates of σ_{ij}

$$[\sigma_{ij}] = \begin{bmatrix} \sigma_{11} & \sigma_{12} & \sigma_{13} \\ \sigma_{21} & \sigma_{22} & \sigma_{23} \\ \sigma_{31} & \sigma_{32} & \sigma_{33} \end{bmatrix} = \begin{bmatrix} \mathbf{t}_{\sigma 1}^t \\ \mathbf{t}_{\sigma 2}^t \\ \mathbf{t}_{\sigma 3}^t \end{bmatrix} \quad (3.1.19)$$

geometric interpretation



with traction vectors on surfaces

$$\begin{aligned} \mathbf{t}_{\sigma 1} &= [\sigma_{11} \quad \sigma_{12} \quad \sigma_{13}]^t \\ \mathbf{t}_{\sigma 2} &= [\sigma_{21} \quad \sigma_{22} \quad \sigma_{23}]^t \\ \mathbf{t}_{\sigma 3} &= [\sigma_{31} \quad \sigma_{32} \quad \sigma_{33}]^t \end{aligned} \quad (3.1.20)$$

first index ... surface normal

second index ... direction (coordinate of traction vector)

diagonal entries ... normal stresses

non-diagonal entries .. shear stresses

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lecture notes 08

thursday, january 28th, 2010

Volumetric–deviatoric decomposition

in analogy to the strain tensor ϵ , the stress tensor σ can be additively decomposed into a volumetric part σ^{vol} and a traceless deviatoric part σ^{dev}

volumetric – deviatoric decomposition of stress tensor σ

$$\sigma = \sigma^{\text{vol}} + \sigma^{\text{dev}} \quad (3.1.21)$$

with volumetric and deviatoric stress tensor σ^{vol} and σ^{dev}

$$\text{tr}(\sigma^{\text{vol}}) = \text{tr}(\sigma) \quad \text{tr}(\sigma^{\text{dev}}) = 0 \quad (3.1.22)$$

- volumetric second order tensor σ^{vol}

$$\sigma^{\text{vol}} = \frac{1}{3}[\sigma : I] I = \mathbb{I}^{\text{vol}} : \sigma \quad (3.1.23)$$

upon double contraction volumetric fourth order unit tensor $\mathbb{I}^{\text{vol}}$ extracts volumetric part σ^{vol} of stress tensor

$$\begin{aligned} \mathbb{I}^{\text{vol}} &= \frac{1}{3} I \otimes I \\ \mathbb{I}^{\text{vol}} &= \frac{1}{3} \delta_{ij} \delta_{kl} \mathbf{e}_i \otimes \mathbf{e}_j \otimes \mathbf{e}_k \otimes \mathbf{e}_l \end{aligned} \quad (3.1.24)$$

- deviatoric second order tensor σ^{dev}

$$\sigma^{\text{dev}} = \sigma - \frac{1}{3}[\sigma : I] I = \mathbb{I}^{\text{dev}} : \sigma \quad (3.1.25)$$

upon double contraction deviatoric fourth order unit tensor $\mathbb{I}^{\text{dev}}$ extracts deviatoric part of stress tensor

$$\begin{aligned} \mathbb{I}^{\text{dev}} &= \mathbb{I}^{\text{sym}} - \mathbb{I}^{\text{vol}} = \mathbb{I}^{\text{sym}} - \frac{1}{3} I \otimes I \\ \mathbb{I}^{\text{dev}} &= \left[\frac{1}{2} \delta_{ik} \delta_{jl} + \frac{1}{2} \delta_{il} \delta_{jk} - \frac{1}{3} \delta_{ij} \delta_{kl} \right] \mathbf{e}_i \otimes \mathbf{e}_j \otimes \mathbf{e}_k \otimes \mathbf{e}_l \end{aligned} \quad (3.1.26)$$

Volumetric stress

volumetric part σ^{vol} of stress tensor σ

$$\sigma^{\text{vol}} = \frac{1}{3} [\sigma : \mathbf{I}] \mathbf{I} = \frac{1}{3} [\mathbf{I} \otimes \mathbf{I}] : \sigma = \mathbb{I}^{\text{vol}} : \sigma \quad (3.1.27)$$

interpretation of trace as hydrostatic pressure

$$p = \frac{1}{3} \text{tr}(\sigma) = \frac{1}{3} \sigma : \mathbf{I} = \frac{1}{3} (\sigma_{11} + \sigma_{22} + \sigma_{33}) \quad (3.1.28)$$

index representation

$$\sigma^{\text{vol}} = \sigma_{ij}^{\text{vol}} \mathbf{e}_i \otimes \mathbf{e}_j \quad (3.1.29)$$

matrix representation of coordinates $[\sigma_{ij}^{\text{vol}}]$

$$[\sigma_{ij}^{\text{vol}}] = p \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad p = \frac{1}{3} \text{tr}(\sigma) \quad (3.1.30)$$

volumetric stress tensor σ^{vol} is a spherical second order tensor as $\sigma^{\text{vol}} = p \mathbf{I}$

volumetric stress tensor σ^{vol} contains the hydrostatic pressure part of the total stress tensor σ

Deviatoric stress

deviatoric stress tensor σ^{dev} preserves the volume and contains the remaining part of the total stress tensor σ

deviatoric part σ^{dev} of the stress tensor σ

$$\sigma^{\text{dev}} = \sigma - \sigma^{\text{vol}} = \sigma - \frac{1}{3} [\sigma : \mathbf{I}] \mathbf{I} = \mathbb{I}^{\text{dev}} : \sigma \quad (3.1.31)$$

index representation

$$\sigma^{\text{dev}} = \sigma_{ij}^{\text{dev}} \mathbf{e}_i \otimes \mathbf{e}_j \quad (3.1.32)$$

matrix representation of coordinates $[\sigma_{ij}^{\text{dev}}]$

$$[\sigma_{ij}^{\text{dev}}] = \frac{1}{3} \begin{bmatrix} 2\sigma_{11} - \sigma_{22} - \sigma_{33} & \sigma_{12} & \sigma_{13} \\ \sigma_{21} & 2\sigma_{22} - \sigma_{11} - \sigma_{33} & \sigma_{23} \\ \sigma_{31} & \sigma_{32} & 2\sigma_{33} - \sigma_{11} - \sigma_{22} \end{bmatrix} \quad (3.1.33)$$

trace of deviatoric stress $\text{tr}(\sigma^{\text{dev}})$

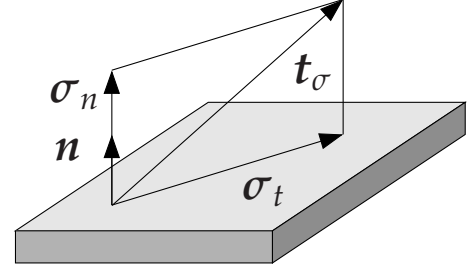
$$\begin{aligned} \text{tr}(\sigma^{\text{dev}}) &= \frac{1}{3} [2\sigma_{11} - \sigma_{22} - \sigma_{33}] \\ &+ \frac{1}{3} [2\sigma_{22} - \sigma_{11} - \sigma_{33}] \\ &+ \frac{1}{3} [2\sigma_{33} - \sigma_{11} - \sigma_{22}] = 0 \end{aligned} \quad (3.1.34)$$

deviatoric stress tensor σ^{dev} is a traceless second order tensor as $\text{tr}(\sigma^{\text{dev}}) = 0$

deviatoric stress tensor σ^{dev} contains the hydrostatic pressure free part of the total stress tensor σ

Normal–shear decomposition

assume we are interested in the stress σ_n normal to a particular plane characterized through its normal $\mathbf{n}$, i.e. the normal projection of the stress vector $\mathbf{t}_\sigma$



$$\sigma_n = \mathbf{t}_\sigma \cdot \mathbf{n} = [\sigma^t \cdot \mathbf{n}] \cdot \mathbf{n} = \sigma^t : [\mathbf{n} \otimes \mathbf{n}] = \sigma^t : \mathbf{N} \quad (3.1.35)$$

normal–shear (tangential) decomposition of stress vector $\mathbf{t}_\sigma$

$$\mathbf{t}_\sigma = \sigma_n + \sigma_t \quad (3.1.36)$$

normal stress vector – stress in direction of $\mathbf{n}$

$$\sigma_n = [\sigma^t : [\mathbf{n} \otimes \mathbf{n}]] \mathbf{n} = \sigma^t : [\mathbf{n} \otimes \mathbf{n} \otimes \mathbf{n}] \quad (3.1.37)$$

shear (tangential) stress vector – stress in the plane

$$\begin{aligned} \sigma_t &= \mathbf{t}_\sigma - \sigma_n = \sigma^t \cdot \mathbf{n} - \sigma^t : [\mathbf{n} \otimes \mathbf{n} \otimes \mathbf{n}] \\ &= \sigma^t : [\mathbb{I}^{\text{sym}} \cdot \mathbf{n} - \mathbf{n} \otimes \mathbf{n} \otimes \mathbf{n}] = \sigma^t : \mathbf{T} \end{aligned} \quad (3.1.38)$$

amount of shear stress τ_n

$$||\tau_n||^2 = (\mathbf{t}_\sigma - \sigma_n) \cdot (\mathbf{t}_\sigma - \sigma_n) = \mathbf{t}_\sigma \cdot \mathbf{t}_\sigma - 2\mathbf{t}_\sigma \cdot \sigma_n + \sigma_n^2 \mathbf{n} \cdot \mathbf{n} \quad (3.1.39)$$

and thus

$$\tau_n = ||\sigma_t|| = \sqrt{\sigma_t \cdot \sigma_t} = \sqrt{\mathbf{t}_\sigma \cdot \mathbf{t}_\sigma - \sigma_n^2} \quad (3.1.40)$$

in general, i.e. for an arbitrary direction $\mathbf{n}$, we have normal and shear contributions to the stress vector, however, three particular directions $\{\mathbf{n}_{\sigma i}\}_{i=1,2,3}$ can be identified, for which $\mathbf{t}_\sigma = \sigma_n$ and thus $\sigma_t = \mathbf{0}$, the corresponding $\{\mathbf{n}_{\sigma i}\}_{i=1,2,3}$ are called principal stress directions and $\{\sigma_{ni}\}_{i=1,2,3} = \{\lambda_{\sigma i}\}_{i=1,2,3}$ are the principal stresses

Principal stresses

assume stress tensor σ^t to be known at $x \in \mathcal{B}$, principal stresses $\{\lambda_{\sigma i}\}_{i=1,2,3}$ and principal stress directions $\{\mathbf{n}_{\sigma i}\}_{i=1,2,3}$ can be derived from solution of special eigenvalue problem according to §1.1.3

$$\sigma^t \cdot \mathbf{n}_{\sigma i} = \lambda_{\sigma i} \mathbf{n}_{\sigma i} \quad [\sigma^t - \lambda_{\sigma i}] \cdot \mathbf{n}_{\sigma i} = \mathbf{0} \quad (3.1.41)$$

solution

$$\det(\sigma^t - \lambda_{\sigma} \mathbf{I}) = 0 \quad (3.1.42)$$

or in terms of roots of characteristic equation

$$\lambda_{\sigma}^3 - I_{\sigma} \lambda_{\sigma}^2 + II_{\sigma} \lambda_{\sigma} - III_{\sigma} = 0 \quad (3.1.43)$$

roots of characteristic equation in terms of principal invariants of σ^t

$$\begin{aligned} I_{\sigma} &= \text{tr}(\sigma^t) &= \lambda_{\sigma 1} + \lambda_{\sigma 2} + \lambda_{\sigma 3} \\ II_{\sigma} &= \frac{1}{2}[\text{tr}^2(\sigma^t) - \text{tr}(\sigma^{t^2})] &= \lambda_{\sigma 2} \lambda_{\sigma 3} + \lambda_{\sigma 3} \lambda_{\sigma 1} + \lambda_{\sigma 1} \lambda_{\sigma 2} \\ III_{\sigma} &= \det(\sigma^t) &= \lambda_{\sigma 1} \lambda_{\sigma 2} \lambda_{\sigma 3} \end{aligned} \quad (3.1.44)$$

spectral representation of σ

$$\sigma^t = \sum_{i=1}^3 \lambda_{\sigma i} \mathbf{n}_{\sigma i} \otimes \mathbf{n}_{\sigma i} \quad (3.1.45)$$

principal stresses $\lambda_{\sigma i}$ are purely normal, no shear stress τ_n in principal directions, i.e. $\mathbf{t}_{\sigma i} = \sigma_n = \lambda_{\sigma i} \mathbf{n}_{\sigma i}$ and $\sigma_t = \mathbf{0}$ thus $\tau_n = 0$

due to symmetry of stresses $\sigma = \sigma^t$, stress tensor possesses three real eigenvalues $\{\lambda_{\sigma i}\}_{i=1,2,3}$, corresponding eigendirections $\{\mathbf{n}_{\sigma i}\}_{i=1,2,3}$ are thus orthogonal $\mathbf{n}_{\sigma i} \cdot \mathbf{n}_{\sigma j} = \delta_{ij}$

Special case of plane stress

dimensional reduction in case of plane stress with vanishing stresses $\sigma_{13} = \sigma_{23} = \sigma_{31} = \sigma_{32} = \sigma_{33} = 0$ in out of plane direction, e.g. for flat sheets

$$\sigma = \sigma_{ij} \mathbf{e}_i \otimes \mathbf{e}_j \quad (3.1.46)$$

matrix representation of coordinates $[\sigma_{ij}]$

$$[\sigma_{ij}] = \begin{bmatrix} \sigma_{11} & \sigma_{12} & 0 \\ \sigma_{21} & \sigma_{22} & 0 \\ 0 & 0 & 0 \end{bmatrix} \quad (3.1.47)$$

Voigt representation of stress

three dimensional second order stress tensor σ

$$\sigma = \sigma_{ij} \mathbf{e}_i \otimes \mathbf{e}_j \quad (3.1.48)$$

matrix representation of coordinates $[\sigma_{ij}]$

$$[\sigma_{ij}] = \begin{bmatrix} \sigma_{11} & \sigma_{12} & \sigma_{13} \\ \sigma_{21} & \sigma_{22} & \sigma_{23} \\ \sigma_{31} & \sigma_{23} & \sigma_{33} \end{bmatrix} \quad (3.1.49)$$

due to symmetry $[\sigma_{ij}] = [\sigma_{ji}]$ and thus $\sigma_{12} = \sigma_{21}$, $\sigma_{23} = \sigma_{32}$, $\sigma_{31} = \sigma_{13}$, stress tensor σ contains only six independent components $\sigma_{11}, \sigma_{22}, \sigma_{33}, \sigma_{12}, \sigma_{23}, \sigma_{31}$, it proves convenient to represent second order tensor σ through a vector $\underline{\sigma}$

$$\underline{\sigma} = [\sigma_{11}, \sigma_{22}, \sigma_{33}, \sigma_{12}, \sigma_{23}, \sigma_{31}]^t \quad (3.1.50)$$

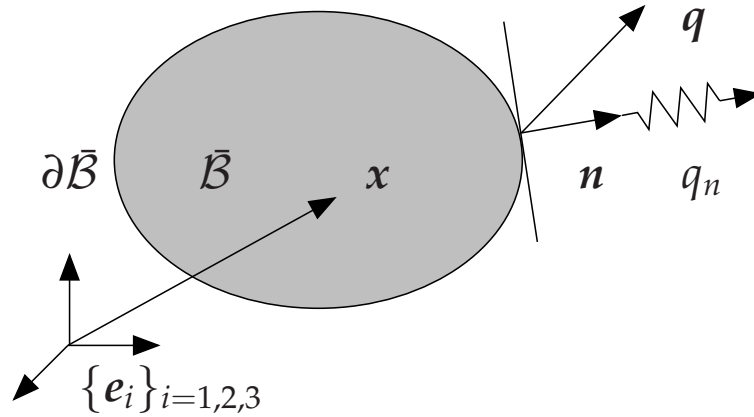
vector representation $\underline{\sigma}$ of stress σ in case of plane stress

$$\underline{\sigma} = [\sigma_{11}, \sigma_{22}, \sigma_{33}, \sigma_{12}]^t \quad (3.1.51)$$

3.1.3 Concept of heat flux

the contact heat flux q_n at a point x is a scalar of the unit [energy/time/surface area]

the contact heat flux q_n characterizes the energy transport normal to the tangent plane to an imaginary surface passing through this point with normal vector n



definition of contact heat flux q_n in analogy to Cauchy's postulate, lemma and theorem originally introduced for the momentum flux in §3.1.2

Cauchy's postulate

$$q_n = q_n(x, n) \quad (3.1.52)$$

Cauchy's lemma

$$q_n(x, n) = -q_n(x, -n) \quad (3.1.53)$$

Cauchy's theorem

the contact heat flux q_n can be expressed as linear function of the surface normal n and the heat flux vector q

$$q_n = q \cdot n \quad (3.1.54)$$

Heat flux vector

the vector field $\mathbf{q}$ is called heat flux vector

$$\mathbf{q} = q_i \mathbf{e}_i \quad (3.1.55)$$

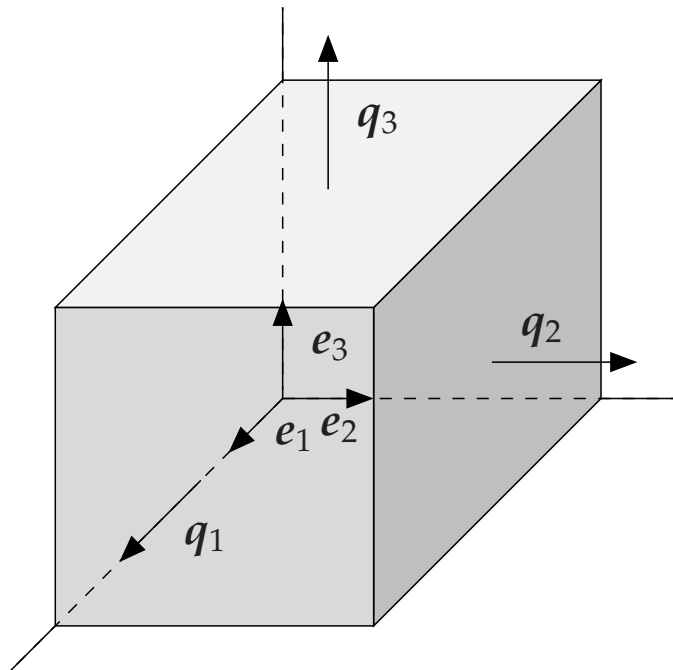
Cauchy's theorem

$$q_n = \mathbf{q} \cdot \mathbf{n} \quad (3.1.56)$$

index representation

$$q_n = (q_i \mathbf{e}_i) \cdot (n_j \mathbf{e}_j) = q_i n_j \delta_{ij} = q_i n_i \quad (3.1.57)$$

geometric interpretation

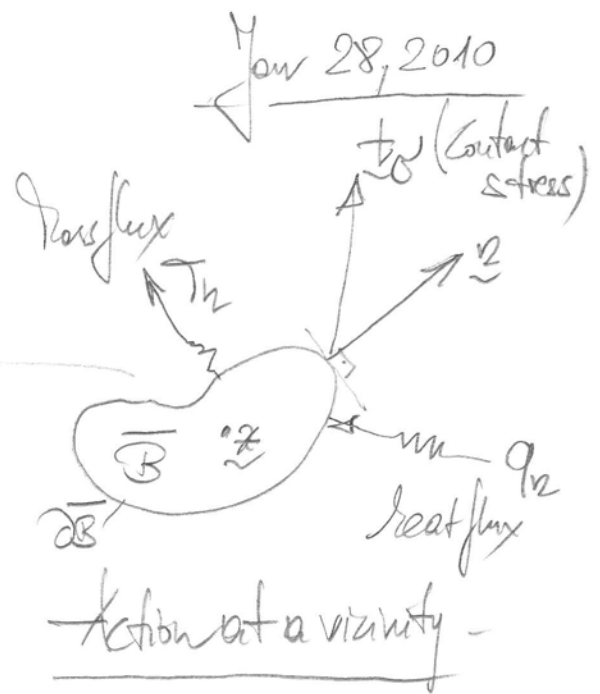
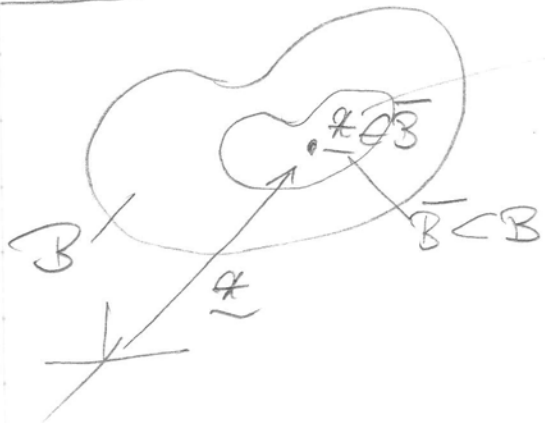


the coordinates q_i characterize the heat energy transport through the planes parallel to the coordinate planes

in continuum mechanics of adiabatic systems the heat flux vector vanishes identically

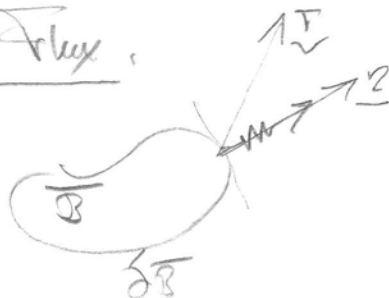
3. BALANCE EQUATIONS

Recall the basic steps -



- Cut out $\bar{B} \subset B$
- Characterize the action at vicinity such as the mass flux m (out), the stress traction vector t_n , and the heat flux q_n
- Postulate the balance of basic physical quantities
 - Mass m
 - Linear Momentum I
 - Moment of Momentum D
 - Energy E
- Solve the global balance equations to obtain local balance equations at point $x \in \bar{B}$.

Mass Flux:



- $T_n = T_n(x, n)$ (Cauchy's postulate)
- $T_n(x, n) = -T_n(x, -n)$ (Cauchy's lemma)
- $T_n = T \cdot n$ (Cauchy's theorem)

Today,

Concept of Stress:



Stress traction vector: $\underline{t}_\sigma := \lim_{\Delta a \rightarrow 0} \frac{\Delta f}{\Delta a} = \frac{df}{da}$

Cauchy's postulate

$$\underline{t}_\sigma = \underline{t}_\sigma(\underline{x}, \underline{n})$$

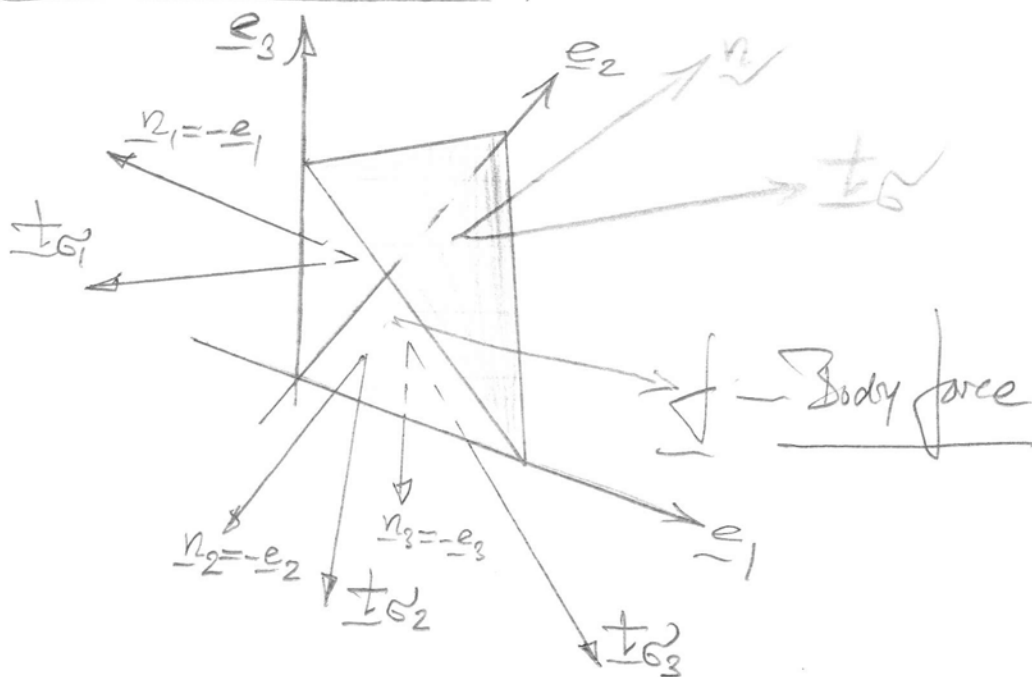
Cauchy's lemma

$$\underline{t}_\sigma(\underline{x}, \underline{n}) = -\underline{t}_\sigma(\underline{x}, -\underline{n})$$

Cauchy's Theorem

$$\underline{t}_\sigma(\underline{x}, \underline{n}) = \underline{\sigma}^t(\underline{x}) \cdot \underline{n}$$

Consider a tetrahedron.



Directional cosines $da_j = da \underline{n} \cdot \underline{e}_j \quad \forall j=1,2,3$

Force Balance: $\underline{t}_\sigma(\underline{n}) da + \sum_{j=1}^3 \underline{t}_{\sigma_j}(\underline{n}_j) da_j + \int dv = \underline{0}$

$$\lim_{\frac{dv}{da} \rightarrow 0} \underline{t}_\sigma(n) da - \sum_j \underline{t}_{\sigma_j}(\underline{e}_j) \cdot \underline{e}_j da + \int dv = 0$$

$$= \underline{t}_\sigma - (\underline{t}_{\sigma_1} \underline{n} \cdot \underline{e}_1 + \underline{t}_{\sigma_2} \underline{n} \cdot \underline{e}_2 + \underline{t}_{\sigma_3} \underline{n} \cdot \underline{e}_3) = 0$$

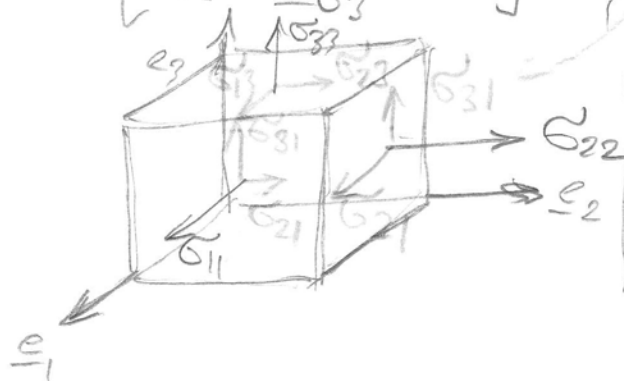
$$\underline{t}_\sigma = [\underline{t}_{\sigma_1} \underline{e}_1 + \underline{t}_{\sigma_2} \underline{e}_2 + \underline{t}_{\sigma_3} \underline{e}_3] \cdot \underline{n}$$

$$\underline{t}_\sigma = \underbrace{[\underline{t}_{\sigma_j} \otimes \underline{e}_j]}_{\underline{\sigma}^t} \cdot \underline{n} =: \underline{\sigma}^t \cdot \underline{n}$$

$$\therefore \underline{\sigma}^t = \underline{t}_{\sigma_j} \otimes \underline{e}_j = t_{\sigma_{ji}} \underline{e}_i \otimes \underline{e}_j = \underline{\sigma}_{ji} \underline{e}_i \otimes \underline{e}_j$$

$$\underline{t}_{\sigma_j} = \underline{\sigma}_{ji} \underline{e}_i$$

$$[\underline{\sigma}_{ij}] = \begin{bmatrix} \leftarrow \underline{t}_{\sigma_1} \rightarrow \\ \leftarrow \underline{t}_{\sigma_2} \rightarrow \\ \leftarrow \underline{t}_{\sigma_3} \rightarrow \end{bmatrix}$$



$$t_i = \underline{\sigma}_{ji} n_j$$

Physical interpretation

Volumetric-Denatanz Decomposition

These hat $\hat{\tau}(\hat{\sigma}_{vol}) = 0$, and $\hat{\sigma}_{vol} = \hat{\tau} \hat{\epsilon}$
 where $\hat{\tau} = \frac{1}{3} \hat{\epsilon} \rightarrow$ negative pressure.

* See the notes for the matrix representation

* EXERCISE: Metal plasticity, incompressible plasticity.

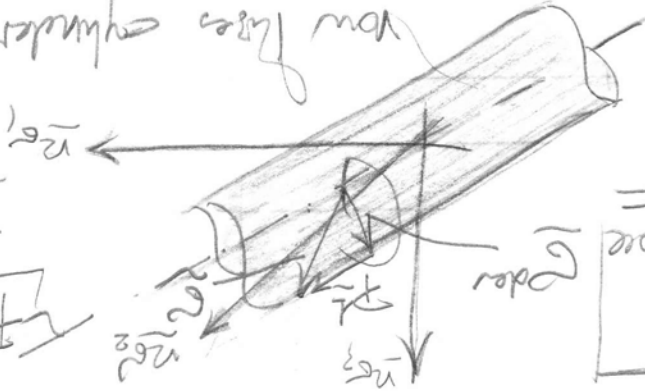
Explain where does the factor $\frac{1}{3}$ come from.

in hydrostatic condition.

$$\phi(\hat{\epsilon}) = \|\hat{\epsilon}_{dev}\| - \sqrt{\frac{2}{3}} \hat{\epsilon}_y \leq 0$$

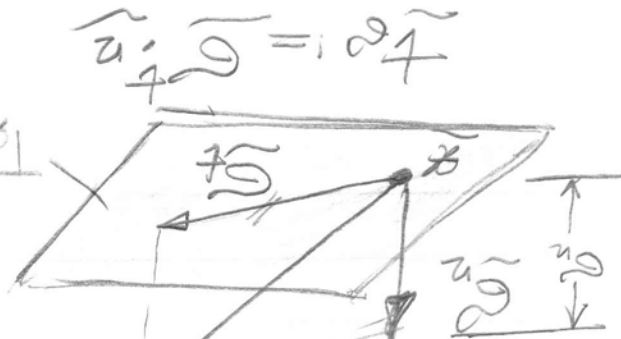
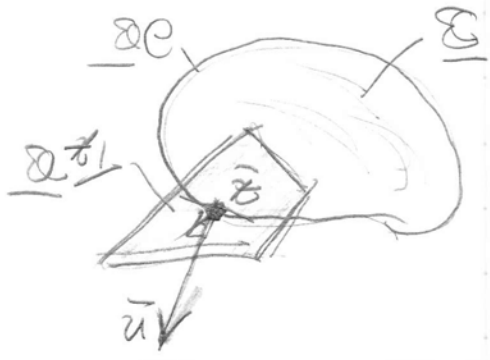
Based on the uniaxial tension test.

Principal-Shear-Space $\hat{\epsilon}_{dev}$
 $\hat{\epsilon} = \hat{\epsilon}_{dev} + \hat{\epsilon}_{vol}$
 $\hat{\epsilon}_{vol} = \frac{1}{3}(\hat{\epsilon}_1 + \hat{\epsilon}_2 + \hat{\epsilon}_3)$
 Hydrostatic axis



von Mises cylinder (yield surface)

Normal-Shear Decomposition:
 $\hat{\epsilon} = \hat{\epsilon}_{dev} + \hat{\epsilon}_{vol}$
 $\hat{\epsilon}_{vol} = \frac{1}{3}(\hat{\epsilon}_1 + \hat{\epsilon}_2 + \hat{\epsilon}_3)$
 $\hat{\epsilon}_{dev} = \hat{\epsilon} - \hat{\epsilon}_{vol}$



Through the Cauchy theorem, we have

$$\underline{\tau} = \underline{\underline{S}}^t \cdot \underline{n} = \underline{S}_n + \underline{S}_t$$

Normal Stress Vector
 Tangential Stress Vector

$$\underline{S}_n = \underline{\tau} \cdot \underline{n} = \underline{\underline{S}}^t : (\underline{n} \otimes \underline{n}) \quad \text{Normal stress}$$

Thus,

$$\underline{S}_n = \underline{S}_n \underline{n} = \left[\underline{\underline{S}}^t : (\underline{n} \otimes \underline{n}) \right] \underline{n}$$

Normal stress Vector

$$= \underline{\underline{S}}^t : [\underline{n} \otimes \underline{n} \otimes \underline{n}]$$

and,

$$\underline{S}_t = \underline{\tau} - \underline{S}_n = \underline{\underline{S}}^t \cdot \underline{n} - \underline{\underline{S}}^t : [\underline{n} \otimes \underline{n} \otimes \underline{n}]$$

$$= \underline{\underline{S}}^t : \left[\underline{\underline{I}}^{\text{sym}} \cdot \underline{n} - \underline{n} \otimes \underline{n} \otimes \underline{n} \right]$$

Shear
Tangential stress Vector

EXERCISE 2: CRYSTAL PLASTICITY (SHEAR STRESS)

SIL PLASTICITY:

Then, the amount of shear stress τ_m is given by

$$\|\tau_m\|^2 = (\underline{\tau} - \underline{S}_n) \cdot (\underline{\tau} - \underline{S}_n) = \underline{S}_t \cdot \underline{S}_t$$

$$\|\tau_m\| = \sqrt{\underline{\tau} \cdot \underline{\tau} - S_n^2}$$

PRINCIPAL STRESSES

Eigenvalue problem $\underline{\underline{\sigma}}^t \cdot \underline{n}_{\sigma_i} = \lambda_{\sigma_i} \cdot \underline{n}_{\sigma_i}$

$$[\underline{\underline{\sigma}}^t - \lambda_{\sigma_i} \underline{\underline{1}}] \cdot \underline{n}_{\sigma_i} = \underline{0}$$

Characteristic Equation

$$\lambda_{\sigma}^3 - I_{\sigma} \lambda_{\sigma}^2 + II_{\sigma} \lambda_{\sigma} - III_{\sigma} = 0$$

where $I_{\sigma} := \text{tr}[\underline{\underline{\sigma}}^t] = \lambda_{\sigma_1} + \lambda_{\sigma_2} + \lambda_{\sigma_3}$

$$II_{\sigma} := \frac{1}{2} [\text{tr}^2(\underline{\underline{\sigma}}^t) - \text{tr}(\underline{\underline{\sigma}}^t \underline{\underline{\sigma}}^t)] = \lambda_{\sigma_1} \lambda_{\sigma_2} + \lambda_{\sigma_2} \lambda_{\sigma_3} + \lambda_{\sigma_1} \lambda_{\sigma_3}$$

$$III_{\sigma} := \det(\underline{\underline{\sigma}}^t) = \lambda_{\sigma_1} \lambda_{\sigma_2} \lambda_{\sigma_3}$$

This leads us to the spectral representation of the stress tensor

$$\underline{\underline{\sigma}}^t = \sum_{i=1}^3 \lambda_{\sigma_i} \underline{n}_{\sigma_i} \otimes \underline{n}_{\sigma_i}$$

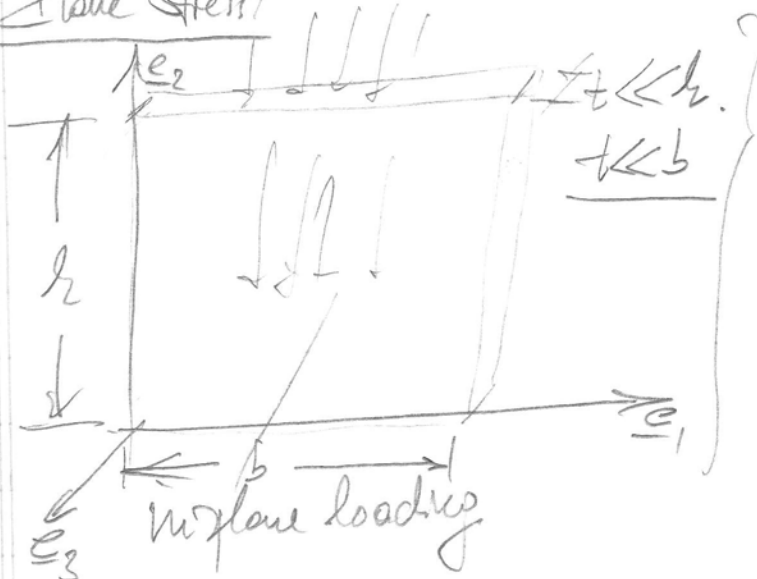
Observe that λ_{σ_i} are purely normal; that is, if we decompose $\underline{\underline{\sigma}}^t$ on a plane with a normal $\underline{n}_{\sigma_i}$; $\underline{\underline{\sigma}} \cdot \underline{n}_{\sigma_i} = \lambda_{\sigma_i} \underline{n}_{\sigma_i}$, $\underline{\underline{\sigma}}^t = \underline{\underline{\sigma}}$

Derivation

$$\begin{aligned} \sum [\underline{\underline{\sigma}}^t \cdot \underline{n}_{\sigma_i} \otimes \underline{n}_{\sigma_i}] &= \sum [\lambda_{\sigma_i} \underline{n}_{\sigma_i}] \otimes \underline{n}_{\sigma_i} \\ \underline{\underline{\sigma}}^t \cdot \sum \underline{n}_{\sigma_i} \otimes \underline{n}_{\sigma_i} &= \sum \lambda_{\sigma_i} \underline{n}_{\sigma_i} \otimes \underline{n}_{\sigma_i} \quad \text{q.e.d.} \end{aligned}$$

1

Plane Stress



$$\sigma_{i3} = \sigma_{3i} = 0 \quad \forall i=1,2,3$$

Plane stress condition

Voirg Notation (Storage): As we will show $\underline{\underline{S}} = \underline{\underline{S}}^t$,

we store the stress tensor

$$\begin{bmatrix} 1 & 4 & 6 \\ & 2 & 5 \\ & & 3 \end{bmatrix}$$

In 3d. $\rightarrow \underline{\underline{S}} = [\sigma_{11} \quad \sigma_{22} \quad \sigma_{33} \quad \sigma_{12} \quad \sigma_{23} \quad \sigma_{13}]^t$

Concept of Heat Flux.



Analogous for:

- Mass flux
- Momentum flux
- Heat flux

Cauchy-Type description

Cauchy's postulate

$$q_n = \hat{q}_n(\underline{x}, \underline{n})$$

Cauchy's lemma

$$q_n(\underline{x}, \underline{n}) = -q_n(\underline{x}, -\underline{n})$$

Cauchy's Theorem

$$q_n = \underline{q} \cdot \underline{n}$$

ME338A
CONTINUUM MECHANICS

lecture notes 09

tuesday, february 02nd, 2010

3.2 Balance of mass

total mass m of a body $\bar{\mathcal{B}}$

$$m := \int_{\bar{\mathcal{B}}} \mathrm{d} m \quad (3.2.1)$$

mass density

$$\rho = \lim_{\Delta V \rightarrow 0} \frac{\Delta m}{\Delta V} = \frac{\mathrm{d} m}{\mathrm{d} v} \quad \mathrm{d} m = \rho \mathrm{d} V \quad (3.2.2)$$

$$m = \int_{\bar{\mathcal{B}}} \rho \mathrm{d} V \quad (3.2.3)$$

mass exchange of body $\bar{\mathcal{B}}$ with environment m^{sur} and m^{vol}

$$m^{\text{sur}} := \int_{\partial \bar{\mathcal{B}}} r_n \mathrm{d} A \quad m^{\text{vol}} := \int_{\bar{\mathcal{B}}} \mathcal{R} \mathrm{d} V \quad (3.2.4)$$

with contact mass flux $r_n = \mathbf{r} \cdot \mathbf{n}$ and mass source $\mathcal{R}$

3.2.1 Global form of balance of mass

"The time rate of change of the total mass m of a body $\bar{\mathcal{B}}$ is balanced with the mass exchange due to contact mass flux m^{sur} and the at-a-distance mass exchange m^{vol} ."

$$\mathrm{D}_t m = m^{\text{sur}} + m^{\text{vol}} \quad (3.2.5)$$

and thus

$$\mathrm{D}_t \int_{\bar{\mathcal{B}}} \rho \mathrm{d} V = \int_{\partial \bar{\mathcal{B}}} r_n \mathrm{d} A + \int_{\bar{\mathcal{B}}} \mathcal{R} \mathrm{d} V \quad (3.2.6)$$

3.2.2 Local form of balance of mass

modification of rate term $D_t m$

$$D_t m = D_t \int_{\bar{B}} \rho \, dV \stackrel{\bar{B}^{\text{fixed}}}{=} \int_{\bar{B}} D_t \rho \, dV \quad (3.2.7)$$

modification of surface term m^{sur}

$$m^{\text{sur}} = \int_{\partial \bar{B}} r_n \, dA \stackrel{\text{Cauchy}}{=} \int_{\partial \bar{B}} \mathbf{r} \cdot \mathbf{n} \, dA \stackrel{\text{Gauss}}{=} \int_{\bar{B}} \text{div}(\mathbf{r}) \, dV \quad (3.2.8)$$

and thus

$$\int_{\bar{B}} D_t \rho \, dV = \int_{\bar{B}} \text{div}(\mathbf{r}) \, dV + \int_{\bar{B}} \mathcal{R} \, dV \quad (3.2.9)$$

for arbitrary bodies $\bar{B} \rightarrow$ local form of mass balance

$$D_t \rho = \text{div}(\mathbf{r}) + \mathcal{R} \quad (3.2.10)$$

Classical continuum mechanics of closed systems

in classical closed system continuum mechanics (here), $\mathbf{r} = \mathbf{0}$ and $\mathcal{R} = 0$, such that the mass density ρ is constant in time

$$D_t \rho(\mathbf{x}, t) = 0 \quad \rho = \rho(\mathbf{x}) = \text{const} \quad (3.2.11)$$

typically $\mathbf{r} \neq \mathbf{0}$ and $\mathcal{R} \neq 0$ only in bio- or chemomechanics

3.3 Balance of linear momentum

total momentum $\mathbf{p}$ of a body $\bar{\mathcal{B}}$

$$\mathbf{p} := \int_{\bar{\mathcal{B}}} \mathbf{D}_t \mathbf{u} \, dm = \int_{\bar{\mathcal{B}}} \mathbf{v} \, dm = \int_{\bar{\mathcal{B}}} \rho \mathbf{v} \, dV \quad (3.3.1)$$

momentum exchange of body $\bar{\mathcal{B}}$ with environment through contact forces $\mathbf{f}^{\text{sur}}$ and at-a-distance forces $\mathbf{f}^{\text{vol}}$

$$\mathbf{f}^{\text{sur}} := \int_{\partial \bar{\mathcal{B}}} \mathbf{t}_\sigma \, dA \quad \mathbf{f}^{\text{vol}} := \int_{\bar{\mathcal{B}}} \mathbf{b} \, dV \quad (3.3.2)$$

with contact/surface force $\mathbf{t}_\sigma = \boldsymbol{\sigma}^t \cdot \mathbf{n}$ and volume force $\mathbf{b}$

3.3.1 Global form of balance of momentum

"The time rate of change of the total momentum $\mathbf{p}$ of a body $\bar{\mathcal{B}}$ is balanced with the momentum exchange due to contact momentum flux / surface force $\mathbf{f}^{\text{sur}}$ and the at-a-distance momentum exchange / volume force $\mathbf{f}^{\text{vol}}$."

$$\mathbf{D}_t \mathbf{p} = \mathbf{f}^{\text{sur}} + \mathbf{f}^{\text{vol}} \quad (3.3.3)$$

and thus

$$\mathbf{D}_t \int_{\bar{\mathcal{B}}} \rho \mathbf{v} \, dV = \int_{\partial \bar{\mathcal{B}}} \mathbf{t}_\sigma \, dA + \int_{\bar{\mathcal{B}}} \mathbf{b} \, dV \quad (3.3.4)$$

3.3.2 Local form of balance of momentum

modification of rate term $\mathbf{D}_t \mathbf{p}$

$$\mathbf{D}_t \mathbf{p} = \mathbf{D}_t \int_{\bar{\mathcal{B}}} \rho \mathbf{v} \, dV \stackrel{\bar{\mathcal{B}}^{\text{fixed}}}{=} \int_{\bar{\mathcal{B}}} \mathbf{D}_t (\rho \mathbf{v}) \, dV \quad (3.3.5)$$

modification of surface term f^{sur}

$$f^{\text{sur}} = \int_{\partial \bar{\mathcal{B}}} \mathbf{t}_\sigma \, dA \stackrel{\text{Cauchy}}{=} \int_{\partial \bar{\mathcal{B}}} \boldsymbol{\sigma}_t \cdot \mathbf{n} \, dA \stackrel{\text{Gauss}}{=} \int_{\bar{\mathcal{B}}} \text{div}(\boldsymbol{\sigma}^t) \, dV \quad (3.3.6)$$

and thus

$$\int_{\bar{\mathcal{B}}} D_t(\rho \mathbf{v}) \, dV = \int_{\bar{\mathcal{B}}} \text{div}(\boldsymbol{\sigma}^t) \, dV + \int_{\bar{\mathcal{B}}} \mathbf{b} \, dV \quad (3.3.7)$$

for arbitrary bodies $\bar{\mathcal{B}} \rightarrow$ local form of momentum balance

$$D_t(\rho \mathbf{v}) = \text{div}(\boldsymbol{\sigma}^t) + \mathbf{b} \quad (3.3.8)$$

3.3.3 Reduction with lower order balance equations

balance equations are typically modified with the help of lower order balance equations

$$D_t(\rho \mathbf{v}) = \mathbf{v} D_t \rho + \rho D_t \mathbf{v} = \text{div}(\boldsymbol{\sigma}^t) + \mathbf{b} \quad (3.3.9)$$

with balance of mass multiplied by velocity $\mathbf{v}$

$$\mathbf{v} D_t \rho = \mathbf{v} \text{div}(\mathbf{r}) + \mathbf{v} \mathcal{R}_0 = \text{div}(\mathbf{v} \otimes \mathbf{r}) - \nabla \mathbf{v} \cdot \mathbf{r} + \mathbf{v} \mathcal{R}_0 \quad (3.3.10)$$

local momentum balance in reduced format

$$\rho D_t \mathbf{v} = \text{div}(\boldsymbol{\sigma}^t - \nabla \mathbf{v} \otimes \mathbf{r}) + \mathbf{b} + \nabla \mathbf{v} \cdot \mathbf{r} - \mathbf{v} \mathcal{R}_0 \quad (3.3.11)$$

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Classical continuum mechanics of closed systems

in classical closed system continuum mechanics (here), $\mathbf{r} = \mathbf{0}$ and $\mathcal{R} = 0$, such that the mass density ρ is constant in time, i.e. $D_t(\rho \mathbf{v}) = \rho D_t \mathbf{v}$ and thus

$$\rho D_t \mathbf{v} = \operatorname{div}(\boldsymbol{\sigma}^t) + \mathbf{b} \quad (3.3.12)$$

the above equation is referred to as 'Cauchy's first equation of motion', Cauchy [1827]

3.4 Balance of angular momentum

total angular momentum $\mathbf{l}$ of a body $\bar{\mathcal{B}}$

$$\mathbf{l} := \int_{\bar{\mathcal{B}}} \mathbf{x} \times D_t \mathbf{u} \, d m = \int_{\bar{\mathcal{B}}} \mathbf{x} \times \mathbf{v} \, d m = \int_{\bar{\mathcal{B}}} \rho \mathbf{x} \times \mathbf{v} \, d V \quad (3.4.1)$$

angular momentum exchange of body $\bar{\mathcal{B}}$ with environment through momentum from contact forces $\mathbf{l}^{\text{sur}}$ and momentum from at-a-distance forces $\mathbf{l}^{\text{vol}}$

$$\mathbf{l}^{\text{sur}} := \int_{\partial \bar{\mathcal{B}}} \mathbf{x} \times \mathbf{t}_\sigma \, d A \quad \mathbf{l}^{\text{vol}} := \int_{\bar{\mathcal{B}}} \mathbf{x} \times \mathbf{b} \, d V \quad (3.4.2)$$

with momentum due to contact/surface forces $\mathbf{x} \times \mathbf{t}_\sigma = \mathbf{x} \times \boldsymbol{\sigma}^t \cdot \mathbf{n}$ and volume forces $\mathbf{x} \times \mathbf{b}$, assumption: no additional external torques

3.4.1 Global form of balance of angular momentum

"The time rate of change of the total angular momentum $\mathbf{l}$ of a body $\bar{\mathcal{B}}$ is balanced with the angular momentum exchange

due to contact momentum flux / surface force $\mathbf{l}^{\text{sur}}$ and due to the at-a-distance momentum exchange / volume force $\mathbf{l}^{\text{vol}}$."

$$\mathbf{D}_t \mathbf{l} = \mathbf{l}^{\text{sur}} + \mathbf{l}^{\text{vol}} \quad (3.4.3)$$

and thus

$$\mathbf{D}_t \int_{\bar{\mathcal{B}}} \rho \mathbf{x} \times \mathbf{v} \, dV = \int_{\partial \bar{\mathcal{B}}} \mathbf{x} \times \mathbf{t}_\sigma \, dA + \int_{\bar{\mathcal{B}}} \mathbf{x} \times \mathbf{b} \, dV \quad (3.4.4)$$

3.4.2 Local form of balance of angular momentum

modification of rate term $\mathbf{D}_t \mathbf{p}$

$$\mathbf{D}_t \mathbf{l} = \mathbf{D}_t \int_{\bar{\mathcal{B}}} \rho \mathbf{x} \times \mathbf{v} \, dV \stackrel{\bar{\mathcal{B}}^{\text{fixed}}}{=} \int_{\bar{\mathcal{B}}} \mathbf{D}_t(\rho \mathbf{x} \times \mathbf{v}) \, dV \quad (3.4.5)$$

modification of surface term $\mathbf{l}^{\text{sur}}$

$$\begin{aligned} \mathbf{l}^{\text{sur}} &= \int_{\partial \bar{\mathcal{B}}} \mathbf{x} \times \mathbf{t}_\sigma \, dA \stackrel{\text{Cauchy}}{=} \int_{\partial \bar{\mathcal{B}}} \mathbf{x} \times \boldsymbol{\sigma}^t \cdot \mathbf{n} \, dA \\ &\stackrel{\text{Gauss}}{=} \int_{\bar{\mathcal{B}}} \text{div}(\mathbf{x} \times \boldsymbol{\sigma}^t) \, dV \\ &= \int_{\bar{\mathcal{B}}} \mathbf{x} \times \text{div}(\boldsymbol{\sigma}^t) \, dV + \int_{\bar{\mathcal{B}}} \nabla \mathbf{x} \times \boldsymbol{\sigma}^t \, dV \end{aligned} \quad (3.4.6)$$

and thus

$$\begin{aligned} \int_{\bar{\mathcal{B}}} \mathbf{D}_t(\rho \mathbf{x} \times \mathbf{v}) \, dV &= \int_{\bar{\mathcal{B}}} \mathbf{x} \times \text{div}(\boldsymbol{\sigma}^t) \, dV \\ &\quad + \int_{\bar{\mathcal{B}}} \mathbf{I} \times \boldsymbol{\sigma}^t \, dV + \int_{\bar{\mathcal{B}}} \mathbf{x} \times \mathbf{b} \, dV \end{aligned} \quad (3.4.7)$$

for arbitrary bodies $\bar{\mathcal{B}} \rightarrow$ local form of ang. mom. balance

$$\mathbf{D}_t(\rho \mathbf{x} \times \mathbf{v}) = \mathbf{x} \times \text{div}(\boldsymbol{\sigma}^t) + \mathbf{I} \times \boldsymbol{\sigma}^t + \mathbf{x} \times \mathbf{b} \quad (3.4.8)$$

3.4.3 Reduction with lower order balance equations

modification with the help of lower order balance equations

$$\begin{aligned} D_t(\rho \mathbf{x} \times \mathbf{v}) &= D_t \mathbf{x} \times (\rho \mathbf{v}) + \mathbf{x} \times D_t(\rho \mathbf{v}) \\ &= \mathbf{x} \times \operatorname{div}(\boldsymbol{\sigma}^t) + \mathbf{I} \times \boldsymbol{\sigma}^t + \mathbf{x} \times \mathbf{b} \end{aligned} \quad (3.4.9)$$

position vector $\mathbf{x}$ fixed, thus $D_t \mathbf{x} = \mathbf{0}$, with balance of linear momentum multiplied by position $\mathbf{x}$

$$\mathbf{x} \times D_t(\rho \mathbf{v}) = \mathbf{x} \times \operatorname{div}(\boldsymbol{\sigma}^t) + \mathbf{x} \times \mathbf{b} \quad (3.4.10)$$

local angular momentum balance in reduced format

$$\mathbf{I} \times \boldsymbol{\sigma}^t \stackrel{3}{=} \boldsymbol{\hat{e}}: \boldsymbol{\sigma} = -2 \operatorname{axl}(\boldsymbol{\sigma}) = \mathbf{0} \quad \boldsymbol{\sigma} = \boldsymbol{\sigma}^t \quad (3.4.11)$$

the above equation is referred to as 'Cauchy's second equation of motion', Cauchy [1827]

in the absense of couple stresses, surface and body couples, the stress tensor is symmetric, $\boldsymbol{\sigma} = \boldsymbol{\sigma}^t$, else: micropolar / Cosserat continua

vector product of second order tensors $\mathbf{A} \times \mathbf{B} \stackrel{3}{=} \boldsymbol{\hat{e}}: [\mathbf{A} \cdot \mathbf{B}^t]$

3.5 Balance of energy

total energy E of a body $\bar{B}$ as the sum of the kinetic energy K and the internal energy I

$$\begin{aligned} E &:= \int_{\bar{B}} e \, dV = \int_{\bar{B}} k + i \, dV = K + I \\ K &:= \int_{\bar{B}} k \, dV = \int_{\bar{B}} \frac{1}{2} \rho \mathbf{v} \cdot \mathbf{v} \, dV \\ I &:= \int_{\bar{B}} i \, dV \end{aligned} \quad (3.5.1)$$

energy exchange of body $\bar{B}$ with environment e^{sur} and e^{vol}

$$\begin{aligned} E^{\text{sur}} &:= \int_{\partial \bar{B}} \mathbf{v} \cdot \mathbf{t}_\sigma \, dA - \int_{\partial \bar{B}} q_n \, dA \\ E^{\text{vol}} &:= \int_{\bar{B}} \mathbf{v} \cdot \mathbf{b} \, dV + \int_{\partial \bar{B}} Q \, dV \end{aligned} \quad (3.5.2)$$

with contact heat flux $q_n = \mathbf{r} \cdot \mathbf{n}$ and heat source Q

3.5.1 Global form of balance of energy

"The time rate of change of the total energy E of a body $\bar{B}$ is balanced with the energy exchange due to contact energy flux E^{sur} and the at-a-distance energy exchange E^{vol} ."

$$D_t E = E^{\text{sur}} + E^{\text{vol}} \quad (3.5.3)$$

and thus

$$D_t \int_{\bar{B}} e \, dV = \int_{\partial \bar{B}} \mathbf{v} \cdot \mathbf{t}_\sigma - q_n \, dA + \int_{\bar{B}} \mathbf{v} \cdot \mathbf{b} + Q \, dV \quad (3.5.4)$$

3.5.2 Local form of balance of energy

modification of rate term $D_t E$

$$D_t E = D_t \int_{\bar{B}} e \, dV \stackrel{\bar{B} \text{ fixed}}{=} \int_{\bar{B}} D_t e \, dV \quad (3.5.5)$$

modification of surface term E^{sur}

$$\begin{aligned} E^{\text{sur}} &= \int_{\partial \bar{B}} \mathbf{v} \cdot \mathbf{t}_\sigma - q_n \, dA \\ &\stackrel{\text{Cauchy}}{=} \int_{\partial \bar{B}} \mathbf{v} \cdot \boldsymbol{\sigma}^t \cdot \mathbf{n} - \mathbf{q} \cdot \mathbf{n} \, dA \\ &\stackrel{\text{Gauss}}{=} \int_{\bar{B}} \operatorname{div}(\mathbf{v} \cdot \boldsymbol{\sigma}^t) + \operatorname{div}(-\mathbf{q}) \, dV \end{aligned} \quad (3.5.6)$$

and thus

$$\int_{\bar{B}} D_t e \, dV = \int_{\bar{B}} \operatorname{div}(\mathbf{v} \cdot \boldsymbol{\sigma}^t - \mathbf{q}) \, dV + \int_{\bar{B}} \mathbf{v} \cdot \mathbf{b} + \mathcal{Q} \, dV \quad (3.5.7)$$

for arbitrary bodies $\bar{B} \rightarrow$ local form of energy balance

$$D_t e = \operatorname{div}(\mathbf{v} \cdot \boldsymbol{\sigma}^t - \mathbf{q}) + \mathbf{v} \cdot \mathbf{b} + \mathcal{Q} \quad (3.5.8)$$

3.5.3 Reduction with lower order balance equations

modification with the help of lower order balance equations with

$$\begin{aligned} D_t e &= D_t(k + i) = v D_t(\rho v) + D_t i \\ &= \operatorname{div}(\mathbf{v} \cdot \boldsymbol{\sigma}^t - \mathbf{q}) + \mathbf{v} \cdot \mathbf{b} + \mathcal{Q} \end{aligned} \quad (3.5.9)$$

with

$$D_t k = D_t\left(\frac{1}{2}\rho \mathbf{v} \cdot \mathbf{v}\right) = \frac{1}{2}D_t(\rho \mathbf{v}) \cdot \mathbf{v} + \frac{1}{2}\mathbf{v} \cdot D_t(\rho \mathbf{v}) = \mathbf{v} \cdot D_t(\rho \mathbf{v})$$

$$(3.5.10)$$

with balance of momentum multiplied by velocity $\boldsymbol{v}$

$$\boldsymbol{v} D_t (\rho \boldsymbol{v}) = \boldsymbol{v} \operatorname{div} (\boldsymbol{\sigma}^t) + \boldsymbol{v} \boldsymbol{b} = \operatorname{div} (\boldsymbol{v} \cdot \boldsymbol{\sigma}^t) - \nabla \boldsymbol{v} : \boldsymbol{\sigma}^t + \boldsymbol{v} \cdot \boldsymbol{b} \quad (3.5.11)$$

with stress power

$$\begin{aligned} \nabla \boldsymbol{v} : \boldsymbol{\sigma}^t &= \boldsymbol{\sigma}^t : \nabla \boldsymbol{v} = \boldsymbol{\sigma}^t : \nabla D_t \boldsymbol{\varphi} \\ &= \boldsymbol{\sigma}^t : D_t \nabla \boldsymbol{\varphi} = \boldsymbol{\sigma}^t : D_t \nabla^{\operatorname{sym}} \boldsymbol{\varphi} = \boldsymbol{\sigma}^t : D_t \boldsymbol{\epsilon} \end{aligned} \quad (3.5.12)$$

local energy balance in reduced format/balance of int. energy

$$D_t i = \boldsymbol{\sigma}^t : D_t \boldsymbol{\epsilon} - \operatorname{div} (\boldsymbol{q}) + \mathcal{Q} \quad (3.5.13)$$

3.5.4 First law of thermodynamics

alternative definitions

$\mathfrak{p}^{\operatorname{ext}} := \operatorname{div} (\boldsymbol{v} \cdot \boldsymbol{\sigma}^t) + \boldsymbol{v} \cdot \boldsymbol{b}$... external mechanical power

$\mathfrak{p}^{\operatorname{int}} := \boldsymbol{\sigma}^t : \nabla \boldsymbol{v} = \boldsymbol{\sigma}^t : D_t \boldsymbol{\epsilon}$... internal mechanical power

$\mathfrak{q}^{\operatorname{ext}} := \operatorname{div} (-\boldsymbol{q}) + \mathcal{Q}$... external thermal power

$$(3.5.14)$$

Balance of total energy / first law of thermodynamics

the rate of change of the total energy e , i.e. the sum of the kinetic energy k and the potential energy i is in equilibrium with the external mechanical power $\mathfrak{p}^{\operatorname{ext}}$ and the external thermal power $\mathfrak{q}^{\operatorname{ext}}$

$$D_t e = D_t k + D_t i = p^{\text{ext}} + q^{\text{ext}} \quad (3.5.15)$$

the above equation is typically referred to as "principle of interconvertibility of heat and mechanical work" , Carnot [1832], Joule [1843], Duhem [1892]

first law of thermodynamics does not provide any information about the direction of a thermodynamic process

balance of kinetic energy

$$D_t k = p^{\text{ext}} - p^{\text{int}} \quad (3.5.16)$$

the balance of kinetic energy is an alternative statement of the balance of linear momentum

balance of internal energy

$$D_t i = q^{\text{ext}} + p^{\text{int}} \quad (3.5.17)$$

kinetic energy k and internal energy i are no conservation properties, they exchange the internal mechanical energy p^{int}

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3.6 Balance of entropy

total entropy H of a body $\bar{\mathcal{B}}$

$$H := \int_{\bar{\mathcal{B}}} h \, dV \quad (3.6.1)$$

entropy exchange of body $\bar{\mathcal{B}}$ with environment through H^{sur} and H^{vol}

$$H^{\text{sur}} := - \int_{\partial\bar{\mathcal{B}}} h_n \, dA \quad H^{\text{vol}} := \int_{\bar{\mathcal{B}}} \mathcal{H} \, dV \quad (3.6.2)$$

with contact entropy flux $h_n = \mathbf{h} \cdot \mathbf{n}$ and entropy source $\mathcal{H}$

internal entropy production of body $\bar{\mathcal{B}}$ as H^{pro}

$$H^{\text{pro}} := \int_{\bar{\mathcal{B}}} h^{\text{pro}} \, dV \geq 0 \quad (3.6.3)$$

whereby H^{pro} is strictly non-negative

3.6.1 Global form of balance of entropy

“The time rate of change of the total entropy H of a body $\bar{\mathcal{B}}$ is balanced with the energy exchange due to contact energy flux H^{sur} , the at-a-distance entropy exchange H^{vol} and the non-negative entropy production H^{pro} with $H^{\text{pro}} \geq 0$ ”.

$$D_t H = H^{\text{sur}} + H^{\text{vol}} + H^{\text{pro}} \quad (3.6.4)$$

and thus

$$D_t \int_{\bar{\mathcal{B}}} h \, dV = - \int_{\partial\bar{\mathcal{B}}} h_n \, dA + \int_{\bar{\mathcal{B}}} \mathcal{H} \, dV + \int_{\bar{\mathcal{B}}} h^{\text{pro}} \, dV \quad (3.6.5)$$

3.6.2 Local form of balance of entropy

modification of rate term $D_t H$

$$D_t H = D_t \int_{\bar{B}} h \, dV \stackrel{\bar{B}^{\text{fixed}}}{=} \int_{\bar{B}} D_t h \, dV \quad (3.6.6)$$

modification of surface term H^{sur}

$$H^{\text{sur}} = - \int_{\partial \bar{B}} h_n \, dA \stackrel{\text{Cauchy}}{=} - \int_{\partial \bar{B}} \mathbf{h} \cdot \mathbf{n} \, dA \stackrel{\text{Gauss}}{=} - \int_{\bar{B}} \text{div}(\mathbf{h}) \, dV \quad (3.6.7)$$

and thus

$$\int_{\bar{B}} D_t h \, dV = - \int_{\bar{B}} \text{div}(\mathbf{h}) \, dV + \int_{\bar{B}} \mathcal{H} \, dV + \int_{\bar{B}} h^{\text{pro}} \, dV \quad (3.6.8)$$

for arbitrary bodies $\bar{B} \rightarrow$ local form of entropy balance

$$D_t h = -\text{div}(\mathbf{h}) + \mathcal{H} + h^{\text{pro}} \quad (3.6.9)$$

3.6.3 Reduction with lower order balance equations

modification with the help of lower order balance equations

assumption of existence of absolute temperature $\theta \geq 0$
which renders the following constitutive assumption

$$\mathbf{h} = \frac{1}{\theta} \mathbf{q} \quad \mathcal{H} = \frac{1}{\theta} \mathcal{Q} \quad (3.6.10)$$

such that

$$\begin{aligned} D_t h &= - \text{div} \left(\frac{1}{\theta} \mathbf{q} \right) + \frac{1}{\theta} \mathcal{Q} + h^{\text{pro}} \\ \theta D_t h &= - \theta \frac{1}{\theta} \text{div}(\mathbf{q}) - \mathbf{q} \cdot \nabla \left(\frac{1}{\theta} \right) + \mathcal{Q} + \theta h^{\text{pro}} \\ D_t(\theta h) &= - \theta \frac{1}{\theta} \text{div}(\mathbf{q}) - \mathbf{q} \cdot \nabla \ln(\theta) + \mathcal{Q} + \theta h^{\text{pro}} - h D_t \theta \end{aligned}$$

$$(3.6.11)$$

with balance of internal energy

$$D_t i = \mathbf{q}^{\text{ext}} + \mathbf{p}^{\text{int}} = -\text{div}(\mathbf{q}) + \mathcal{Q} + \boldsymbol{\sigma}^t : D_t \boldsymbol{\epsilon} \quad (3.6.12)$$

combination of balance of entropy and internal energy

$$D_t (i - \theta h) = \boldsymbol{\sigma}^t : D_t \boldsymbol{\epsilon} - h D_t \theta - \mathbf{q} \cdot \nabla \ln(\theta) - \theta h^{\text{pro}} \quad (3.6.13)$$

Legendre transform: free Helmholtz energy

$$\psi = i - \theta h \quad (3.6.14)$$

and thus

$$D_t \psi = \boldsymbol{\sigma}^t : D_t \boldsymbol{\epsilon} - h D_t \theta - \mathbf{q} \cdot \nabla \ln(\theta) - \theta h^{\text{pro}} \quad (3.6.15)$$

3.6.4 Second law of thermodynamics

“The production of entropy h^{pro} is non-negative.”

$$h^{\text{pro}} \geq 0 \quad (3.6.16)$$

entropy is a measure of microscopic disorder of a system,
positive entropy production gives a preferred direction to
thermodynamic processes

Clausius: ‘heat never flows from a colder to a warmer system’

introduction of dissipation (-rate) $\mathcal{D}$

$$\mathcal{D} := \theta h^{\text{pro}} \geq 0 \quad (3.6.17)$$

dissipation inequality

$$\mathcal{D} = \boldsymbol{\sigma}^t : D_t \boldsymbol{\epsilon} - h D_t \theta - D_t \psi - \mathbf{q} \cdot \nabla \ln(\theta) \geq 0 \quad (3.6.18)$$

the above equation is referred to as 'Clausius–Duhem inequality', Clausius [1822-1888]

$$\begin{aligned}\mathcal{D} &= 0 \quad \dots \text{ reversible process} \\ \mathcal{D} &> 0 \quad \dots \text{ irreversible process}\end{aligned}\tag{3.6.19}$$

conjugate pairs

$$\begin{aligned}\text{stress } \sigma &\text{ vs. strain } \epsilon \\ \text{entropy } h &\text{ vs. temperature } \theta\end{aligned}\tag{3.6.20}$$

decomposition into local and conductive part

$$\mathcal{D} = \mathcal{D}^{\text{loc}} + \mathcal{D}^{\text{con}} \geq 0\tag{3.6.21}$$

local part / Clausius–Planck inequality

$$\mathcal{D}^{\text{loc}} = \sigma^t : D_t \epsilon - h D_t \theta - D_t \psi \geq 0\tag{3.6.22}$$

conductive part / Fourier inequality

$$\mathcal{D}^{\text{con}} = -q \cdot \nabla \ln(\theta) \geq 0\tag{3.6.23}$$

example: Fourier law of heat conduction

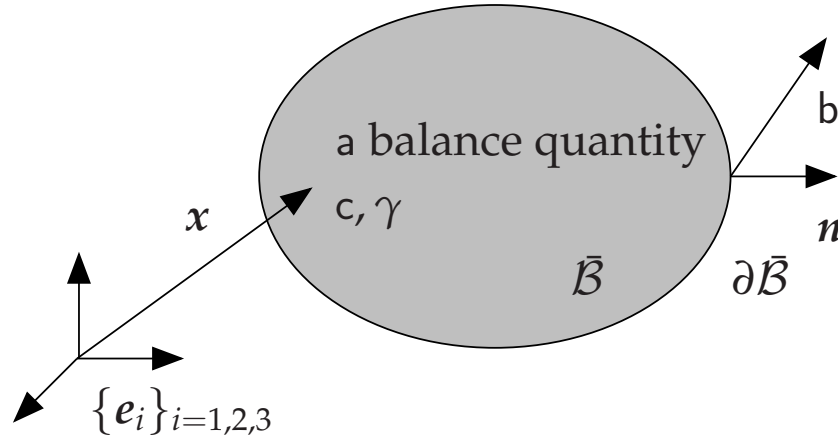
$$q = -\kappa \cdot \nabla \theta \quad \text{isotropic} \quad \kappa = \kappa I\tag{3.6.24}$$

Fourier inequality a priori satisfied by construction

$$\nabla \theta \cdot \kappa \cdot \nabla \theta \geq 0\tag{3.6.25}$$

3.7 Generic balance equation

any balance law can be expressed a general format



3.7.1 Global / integral format

$$D_t A = B + C + \Gamma \quad (3.7.1)$$

or alternatively

$$D_t \int_{\bar{\mathcal{B}}} a \, dV = \int_{\partial\bar{\mathcal{B}}} b \cdot n \, dA + \int_{\bar{\mathcal{B}}} c \, dV + \int_{\bar{\mathcal{B}}} \gamma \, dV \quad (3.7.2)$$

whereby

A ... balance quantity

B ... surface transport through $\partial\bar{\mathcal{B}}$

C ... volume source in $\bar{\mathcal{B}}$

Γ ... production in $\bar{\mathcal{B}}$

(3.7.3)

3.7.2 Local / differential format

local format of balance law follows from

- application of Gauss' theorem
- localization theorem, i.e. $\bar{\mathcal{B}}$ arbitrarily small

$$D_t a = \operatorname{div} (b) + c + \gamma \quad (3.7.4)$$

	quantity a	flux b	source c	production γ
mass	ρ	$\boldsymbol{r}$	$\mathcal{R}$	–
lin. mom.	$\rho \boldsymbol{v}$	$\boldsymbol{\sigma}^t$	$\boldsymbol{b}$	–
ang. mom.	$\boldsymbol{x} \times \rho \boldsymbol{v}$	$\boldsymbol{x} \times \boldsymbol{\sigma}^t$	$\boldsymbol{x} \times \boldsymbol{b}$	–
energy	e	$\boldsymbol{v} \cdot \boldsymbol{\sigma}^t - q$	$\boldsymbol{v} \cdot \boldsymbol{b} + \mathcal{Q}$	–
kin. energy	k	$\boldsymbol{v} \cdot \boldsymbol{\sigma}^t$	$\boldsymbol{v} \cdot \boldsymbol{b}$	$-\boldsymbol{\sigma}^t : D_t \boldsymbol{\epsilon}$
int. energy	i	$-q$	$\mathcal{Q}$	$\boldsymbol{\sigma}^t : D_t \boldsymbol{\epsilon}$
entropy	h	$-\boldsymbol{h}$	$\mathcal{H}$	h^{pro}

Table 3.1: generic balance law

mass, linear momentum, angular momentum and total energy are conservation properties, while kinetic energy, internal energy and entropy are not, they possess a production term

3.8 Thermodynamic potentials

internal energy

$$i = i(\epsilon, h, \dots) \rightarrow \sigma = D_\epsilon i \quad \text{and} \quad \theta = D_h i \quad (3.8.1)$$

Legendre-Fenchel transform $h \rightarrow \theta$

$$\psi(\epsilon, \theta) = \inf_h (i(\epsilon, h) - \theta h) = i(\epsilon, h(\theta)) - \theta h(\theta) \quad (3.8.2)$$

Helmholtz free energy

$$\psi = \psi(\epsilon, \theta, \dots) \rightarrow \sigma = D_\epsilon \psi \quad \text{and} \quad h = -D_\theta \psi \quad (3.8.3)$$

Legendre-Fenchel transform $\epsilon \rightarrow \sigma$

$$g(\sigma, \theta) = \sup_\epsilon (\sigma : \epsilon - \psi(\epsilon, \theta)) = \sigma : \epsilon(\sigma) - \psi(\epsilon(\sigma), \theta) \quad (3.8.4)$$

Gibbs free energy

$$g = g(\sigma, \theta, \dots) \rightarrow \epsilon = D_\sigma g \quad \text{and} \quad h = D_\theta g \quad (3.8.5)$$

Legendre-Fenchel transform $\theta \rightarrow h$

$$\eta(\sigma, h) = \inf_\theta (g(\sigma, \theta) - h \theta) = g(\sigma, \theta(h)) - h \theta(h) \quad (3.8.6)$$

enthalpy

$$\eta = \eta(\epsilon, h, \dots) \rightarrow \epsilon = D_\sigma \eta \quad \text{and} \quad \theta = -D_h \eta \quad (3.8.7)$$

Legendre-Fenchel transform $\sigma \rightarrow \epsilon$

$$i(\epsilon, h) = \sup_\sigma (\epsilon : \sigma - \eta(\sigma, h)) = \epsilon : \sigma(\epsilon) - \eta(\sigma(\epsilon), h) \quad (3.8.8)$$

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4 Constitutive equations

motivation:

unknowns

• density ρ	1
• displacement $\boldsymbol{u}$	3
• temperature θ	1
• mass flux $\boldsymbol{r}$	3
• stress $\boldsymbol{\sigma}$	9
• heat flux $\boldsymbol{q}$	3

in total	<u><u>20</u></u>
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equations

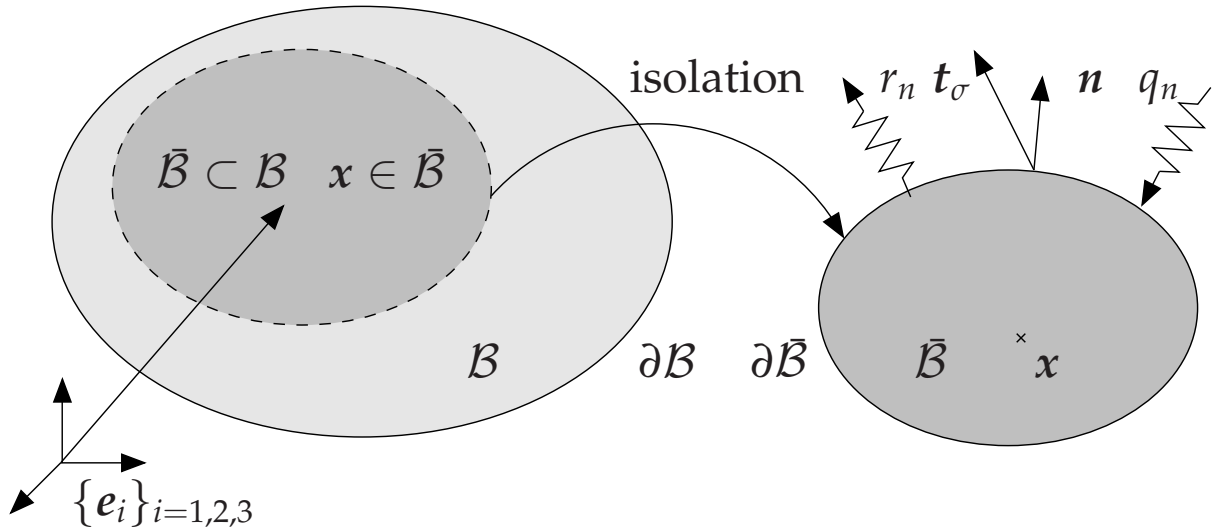
• balance of mass	1
• balance of momentum	3
• balance of angular momentum	3
• balance of energy	1

in total	<u><u>8</u></u>
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balance of unknowns vs. number of equations: $20-8=12$ equations are missing, introduction of 12 material specific constitutive equations, i.e. equations for the mass flux $\boldsymbol{r}$ (3 eqns.), the symmetric stress $\boldsymbol{\sigma} = \boldsymbol{\sigma}^t$ (6 eqns.) and the heat flux $\boldsymbol{q}$ (3 eqns.)

4.1 Linear constitutive equations

- for the mass flux $\mathbf{r}$ with $r_n = \mathbf{r} \cdot \mathbf{n}$
- for the momentum flux / stress $\boldsymbol{\sigma}^t = \boldsymbol{\sigma}$ with $\mathbf{t}_n = \boldsymbol{\sigma}^t \cdot \mathbf{n}$
- for the heat flux $\mathbf{q}$ with $q_n = \mathbf{q} \cdot \mathbf{n}$



in the simplest case, we could introduce ad hoc definitions of the mass flux, the momentum flux and the heat flux in terms of the spatial gradients of the density, the deformation and the temperature

4.1.1 Mass flux – Fick's law

linear relation between mass flux $\mathbf{r}$ (vector) and density gradient $\nabla \rho$ (vector) in terms of mass conduction coefficient $\mathbf{R}$ (second order tensor)

$$\mathbf{r} = \mathbf{R} \cdot \nabla \rho \quad (4.1.1)$$

index representation

$$r_i \mathbf{e}_i = [R_{ij} \mathbf{e}_i \otimes \mathbf{e}_j] \cdot [\rho_{,k} \mathbf{e}_k] = R_{ij} \mathbf{e}_i \delta_{jk} \rho_{,k} = R_{ij} \rho_{,j} \mathbf{e}_i \quad (4.1.2)$$

matrix representation of coordinates

$$[r_i] = \begin{bmatrix} r_1 \\ r_2 \\ r_3 \end{bmatrix} = \begin{bmatrix} R_{11} \rho_{,1} + R_{12} \rho_{,2} + R_{13} \rho_{,3} \\ R_{21} \rho_{,1} + R_{22} \rho_{,2} + R_{23} \rho_{,3} \\ R_{31} \rho_{,1} + R_{32} \rho_{,2} + R_{33} \rho_{,3} \end{bmatrix} \quad (4.1.3)$$

special case of isotropie

$$\mathbf{R} = R \mathbf{I} \quad \mathbf{r} = R \nabla \rho \quad [r_i] = \begin{bmatrix} r_1 \\ r_2 \\ r_3 \end{bmatrix} = R \begin{bmatrix} \rho_{,1} \\ \rho_{,2} \\ \rho_{,3} \end{bmatrix} \quad (4.1.4)$$

a linear relation between the flux of matter $\mathbf{r}$ and the gradient of concentrations $\nabla \rho$ is referred to as Fick's law

4.1.2 Momentum flux – Hook's law

linear relation between momentum flux σ (second order tensor) and displacement gradient $\nabla \mathbf{u}$ or rather $\epsilon = \nabla^{\text{sym}} \mathbf{u}$ (second order tensor) in terms of elasticity tensor $\mathbb{E}$ (fourth order tensor)

$$\sigma = \mathbb{E} : \nabla^{\text{sym}} \mathbf{u} = \mathbb{E} : \epsilon \quad (4.1.5)$$

index representation

$$\begin{aligned} \sigma_{ij} \mathbf{e}_i \otimes \mathbf{e}_j &= [E_{ijkl} \mathbf{e}_i \otimes \mathbf{e}_j \otimes \mathbf{e}_k \otimes \mathbf{e}_l] \cdot [\epsilon_{mn} \mathbf{e}_m \otimes \mathbf{e}_n] \\ &= E_{ijkl} \mathbf{e}_i \otimes \mathbf{e}_j \delta_{km} \delta_{ln} \epsilon_{mn} = E_{ijkl} \epsilon_{kl} \mathbf{e}_i \otimes \mathbf{e}_j \end{aligned} \quad (4.1.6)$$

special case of isotropie

i.e. identical Eigenbasis of stress & strain

$$\sigma = \sum_{i_1}^3 \lambda_{\sigma i} \mathbf{n}_{\sigma i} \otimes \mathbf{n}_{\sigma i} \quad \epsilon = \sum_{i_1}^3 \lambda_{\epsilon i} \mathbf{n}_{\epsilon i} \otimes \mathbf{n}_{\epsilon i} \quad (4.1.7)$$

representation theorem for isotropic tensor-valued tensor-functions

$$\sigma(\epsilon) = f_1 \mathbf{I} + f_2 \epsilon + f_3 \epsilon \cdot \epsilon \quad (4.1.8)$$

with $f_i = f_i(I_\epsilon, II_\epsilon, III_\epsilon)$ function of strain invariants

$$\begin{aligned} I_\epsilon &= \text{tr}(\epsilon) = \lambda_{\epsilon 1} + \lambda_{\epsilon 2} + \lambda_{\epsilon 3} \\ II_\epsilon &= \frac{1}{2} [\text{tr}^2(\epsilon) - \text{tr}(\epsilon^2)] = \lambda_{\epsilon 2} \lambda_{\epsilon 3} + \lambda_{\epsilon 3} \lambda_{\epsilon 1} + \lambda_{\epsilon 1} \lambda_{\epsilon 2} \\ III_\epsilon &= \det(\epsilon) = \lambda_{\epsilon 1} \lambda_{\epsilon 2} \lambda_{\epsilon 3} \end{aligned} \quad (4.1.9)$$

a linear relation between the momentum flux σ and the strains ϵ represents the generalized form of Hook's law

Hypoelasticity / Cauchy Elasticity

a hypoelastic / Cauchy elastic constitutive law can be represented in the following form

$$\sigma = \sigma(\epsilon) \quad (4.1.10)$$

- invertible relation between stress σ and strain ϵ (rates)
- possible dissipation of energy in closed strain circles

$$\oint \mathcal{D}^{\text{loc}} dt = \oint \sigma : \epsilon dt - \oint D_t \psi dt = \oint \sigma : \epsilon dt \quad (4.1.11)$$

homogeneous strain path from ϵ_{t_1} to ϵ_{t_2}

$$\epsilon(\alpha) = [1 - \alpha] \epsilon_{t_1} + \alpha \epsilon_{t_2} \quad d\epsilon = [\epsilon_{t_2} - \epsilon_{t_1}] d\alpha \quad (4.1.12)$$

stress work for linear elastic material

$$\begin{aligned} \int_{t_1}^{t_2} \sigma : d\epsilon &= \int_0^1 \epsilon(\alpha) : \mathbb{E} : [\epsilon_{t_2} - \epsilon_{t_1}] d\alpha \\ &= \frac{1}{2} [\epsilon_{t_2} + \epsilon_{t_1}] : \mathbb{E} : [\epsilon_{t_2} - \epsilon_{t_1}] \end{aligned} \quad (4.1.13)$$

dissipation in isothermal closed strain cycle $t_1 \rightarrow t_2 \rightarrow t_3 \rightarrow t_1$

$$\oint \mathcal{D}^{\text{loc}} dt = \epsilon_{t_1} : \mathbb{E}^{\text{skw}} : \epsilon_{t_2} + \epsilon_{t_2} : \mathbb{E}^{\text{skw}} : \epsilon_{t_3} + \epsilon_{t_3} : \mathbb{E}^{\text{skw}} : \epsilon_{t_1} \neq 0 \quad (4.1.14)$$

4.1.3 Heat flux – Fourier's law

linear relation between heat flux $\mathbf{q}$ (vector) and temperature gradient $\nabla\theta$ (vector) in terms of heat conduction coefficient κ (second order tensor)

$$\mathbf{q} = \kappa \cdot \nabla\theta \quad (4.1.15)$$

index representation

$$q_i \mathbf{e}_i = [\kappa_{ij} \mathbf{e}_i \otimes \mathbf{e}_j] \cdot [\theta_{,k} \mathbf{e}_k] = \kappa_{ij} \mathbf{e}_i \delta_{jk} \theta_{,k} = \kappa_{ij} \theta_{,j} \mathbf{e}_i \quad (4.1.16)$$

matrix representation of coordinates

$$[q_i] = \begin{bmatrix} q_1 \\ q_2 \\ q_3 \end{bmatrix} = \begin{bmatrix} \kappa_{11} \theta_{,1} + \kappa_{12} \theta_{,2} + \kappa_{13} \theta_{,3} \\ \kappa_{21} \theta_{,1} + \kappa_{22} \theta_{,2} + \kappa_{23} \theta_{,3} \\ \kappa_{31} \theta_{,1} + \kappa_{32} \theta_{,2} + \kappa_{33} \theta_{,3} \end{bmatrix} \quad (4.1.17)$$

special case of isotropie

$$\kappa = \kappa \mathbf{I} \quad \mathbf{q} = \kappa \nabla\theta \quad [q_i] = \begin{bmatrix} q_1 \\ q_2 \\ q_3 \end{bmatrix} = \kappa \begin{bmatrix} \theta_{,1} \\ \theta_{,2} \\ \theta_{,3} \end{bmatrix} \quad (4.1.18)$$

a linear relation between the heat flux vector $\mathbf{q}$ and the temperature gradient $\nabla\theta$ is referred to as Fourier's law which goes back to Fourier [1822]

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4.2 Hyperelasticity

4.2.1 Specific stored energy

a hyperelastic / Green elastic constitutive law can be represented in the following form

$$\sigma = \sigma(\epsilon) \quad \text{and} \quad \mathcal{D}^{\text{loc}} = \mathcal{W} - D_t \psi = 0 \quad (4.2.1)$$

- invertible relation between stress σ and strain ϵ (and stress and strain rates $D_t \sigma$ and $D_t \epsilon$) based on a potential
- potential corresponds to elastically stored specific energy
- by construction no dissipation of energy in closed strain circles

stress power $\mathcal{W}$

$$\mathcal{W} = \sigma : D_t \epsilon \doteq D_t \psi \quad (4.2.2)$$

ensuring $\mathcal{D}^{\text{loc}} = \mathcal{W} - D_t \psi = 0$ by construction, thus

$$\psi = \psi(\epsilon) \quad \text{and} \quad D_t \psi = D_\epsilon \psi : D_t \epsilon \quad (4.2.3)$$

specific stored energy W as path independent integral of stress power $\mathcal{W}$

$$W(\epsilon) = \psi(\epsilon) \quad (4.2.4)$$

with

$$W(\epsilon_{t2}) - W(\epsilon_{t1}) = \int_{t_1}^{t_2} D_t W dt = \int_{t_1}^{t_2} \mathcal{W} dt = \int_{t_1}^{t_2} \sigma : d\epsilon \quad (4.2.5)$$

generic hyperelastic / Green elastic constitutive law

$$\sigma = D_{\epsilon} W \quad \text{with} \quad W = W(\epsilon) \quad (4.2.6)$$

- path independent $W(\epsilon_{t2}) - W(\epsilon_{t1}) = \int_{t_1}^{t_2} dW$
- no dissipation $\oint dW = 0$
- symmetric $\frac{D^2 W}{D\epsilon \otimes D\epsilon}$

relation between stress rates and strain rates defines continuum tangent stiffness (fourth order tensor) $\mathbb{E}^{\text{tan}}$

$$D_t \sigma = \mathbb{E}^{\text{tan}} : D_t \epsilon \quad (4.2.7)$$

fourth order tangent stiffness / elastic material tangent

$$\mathbb{E}^{\text{tan}} = \frac{D^2 W}{D\epsilon \otimes D\epsilon} = E_{ijkl} \mathbf{e}_i \otimes \mathbf{e}_j \otimes \mathbf{e}_k \otimes \mathbf{e}_l \quad (4.2.8)$$

minor and major symmetries: reduction from $3^4 = 81$ to $6^2 = 36$ to 21 coefficients

$$E_{ijkl} = E_{jikl} = E_{jilk} = E_{ijlk} \quad \text{and} \quad E_{ijkl} = E_{klij} \quad (4.2.9)$$

4.2.2 Specific complementary energy

specific stored energy

$$\sigma = D_{\epsilon} W \quad \text{with} \quad W = W(\epsilon) \quad (4.2.10)$$

Legendre-Fenchel transform $\sigma \rightarrow \epsilon$

$$W^*(\sigma) = \sup_{\epsilon} (\sigma : \epsilon - W(\epsilon)) \quad (4.2.11)$$

specific complementary stored energy

$$W^* = W^*(\sigma) = \sigma : \epsilon(\sigma) - W(\epsilon(\sigma)) \quad (4.2.12)$$

general hyperelastic constitutive law

$$\epsilon = D_{\sigma} W^* \quad \text{with} \quad W^* = W^*(\sigma) \quad (4.2.13)$$

relation between strain rates and stress rates defines continuum tangent compliance (fourth order tensor) $\mathbb{C}^{\text{tan}}$

$$D_t \epsilon = \mathbb{C}^{\text{tan}} : D_t \sigma \quad \text{with} \quad \mathbb{C}^{\text{tan}} = \mathbb{E}^{\text{tan}^{-1}} \quad (4.2.14)$$

fourth order tangent compliance

$$\mathbb{C}^{\text{tan}} = \frac{D^2 W}{D\sigma \otimes D\sigma} = C_{ijkl} \mathbf{e}_i \otimes \mathbf{e}_j \otimes \mathbf{e}_k \otimes \mathbf{e}_l \quad (4.2.15)$$

minor and major symmetries: reduction from $3^4 = 81$ to $6^2 = 36$ to 21 coefficients

$$C_{ijkl} = C_{jikl} = C_{jilk} = C_{ijlk} \quad \text{and} \quad C_{ijkl} = C_{klij} \quad (4.2.16)$$

4.3 Isotropic hyperelasticity

4.3.1 Specific stored energy

isotropy: identical eigenbasis of stress and strain

$$\boldsymbol{\sigma} = \sum_{i_1}^3 \lambda_{\sigma i} \mathbf{n}_{\sigma i} \otimes \mathbf{n}_{\sigma i} \quad \boldsymbol{\epsilon} = \sum_{i_1}^3 \lambda_{\epsilon i} \mathbf{n}_{\epsilon i} \otimes \mathbf{n}_{\epsilon i} \quad (4.3.1)$$

representation theorem for isotropic tensor-valued tensor functions

$$\boldsymbol{\sigma}(\boldsymbol{\epsilon}) = f_1 \mathbf{I} + f_2 \boldsymbol{\epsilon} + f_3 \boldsymbol{\epsilon} \cdot \boldsymbol{\epsilon} \quad (4.3.2)$$

with $f_i = f_i(I_\epsilon, II_\epsilon, III_\epsilon)$ function of strain invariants

$$\begin{aligned} I_\epsilon &= \text{tr}(\boldsymbol{\epsilon}) \\ II_\epsilon &= \frac{1}{2}[\text{tr}^2(\boldsymbol{\epsilon}) - \text{tr}(\boldsymbol{\epsilon}^2)] \\ III_\epsilon &= \det(\boldsymbol{\epsilon}) \end{aligned} \quad (4.3.3)$$

representation theorem for isotropic scalar-valued tensor functions

$$W(\boldsymbol{\epsilon}) = W(I_\epsilon, II_\epsilon, III_\epsilon) \quad (4.3.4)$$

with $W(\boldsymbol{\epsilon}) = W(\mathbf{Q} \cdot \boldsymbol{\epsilon} \cdot \mathbf{Q}^t) \forall \mathbf{Q} \in SO(3)$

stress for hyperelastic material

$$\boldsymbol{\sigma} = \mathbf{D}_\epsilon W = \frac{\mathbf{D}W}{\mathbf{D}I_\epsilon} \frac{\mathbf{D}I_\epsilon}{\mathbf{D}\boldsymbol{\epsilon}} + \frac{\mathbf{D}W}{\mathbf{D}II_\epsilon} \frac{\mathbf{D}II_\epsilon}{\mathbf{D}\boldsymbol{\epsilon}} + \frac{\mathbf{D}W}{\mathbf{D}III_\epsilon} \frac{\mathbf{D}III_\epsilon}{\mathbf{D}\boldsymbol{\epsilon}} \quad (4.3.5)$$

with derivatives of invariants $I_\epsilon, II_\epsilon, III_\epsilon$ with respect to second order tensor ϵ

$$\begin{aligned} D_\epsilon I_\epsilon &= I \\ D_\epsilon II_\epsilon &= -\epsilon + I_\epsilon I \\ D_\epsilon III_\epsilon &= III_\epsilon \epsilon^{-t} = \epsilon^2 - I_\epsilon \epsilon + II_\epsilon I \end{aligned} \quad (4.3.6)$$

general representation of stress

$$\sigma = D_{I_\epsilon} W I + D_{II_\epsilon} W [-\epsilon + I_\epsilon I] + D_{III_\epsilon} W [\epsilon^2 - I_\epsilon \epsilon + II_\epsilon I] \quad (4.3.7)$$

comparison of coefficients

$$\begin{aligned} f_1 &= D_{I_\epsilon} W + I_\epsilon D_{II_\epsilon} W + II_\epsilon D_{III_\epsilon} W \\ f_2 &= -D_{II_\epsilon} W - I_\epsilon D_{III_\epsilon} W \\ f_3 &= D_{III_\epsilon} W \end{aligned} \quad (4.3.8)$$

assumption of linearity (quadratic term vanishes), two Lamé constants λ and μ

$$f_1 = I_\epsilon \lambda = [\epsilon : I] \lambda \quad f_2 = 2\mu \quad f_3 = 0 \quad (4.3.9)$$

specific stored energy (quadratic in strains)

$$W = \frac{1}{2} \epsilon : \mathbb{E} : \epsilon = \frac{1}{2} \lambda [\epsilon : I]^2 + \mu [\epsilon^2 : I] \quad (4.3.10)$$

stress tensor (linear in strains)

$$\sigma = D_\epsilon W = \mathbb{E} : \epsilon = f_1 I + f_2 \epsilon = \lambda [\epsilon : I] I + 2\mu \epsilon \quad (4.3.11)$$

matrix representation of coordinates

$$[\sigma_{ij}] = \begin{bmatrix} \lambda I_\epsilon + 2\mu\epsilon_{11} & 2\mu\epsilon_{12} & 2\mu\epsilon_{13} \\ 2\mu\epsilon_{21} & \lambda I_\epsilon + 2\mu\epsilon_{22} & 2\mu\epsilon_{23} \\ 2\mu\epsilon_{31} & 2\mu\epsilon_{32} & \lambda I_\epsilon + 2\mu\epsilon_{33} \end{bmatrix} \quad (4.3.12)$$

linear elastic continuum tangent stiffness (constant in strains)

$$\mathbb{E}^{\text{tan}} = \lambda \mathbf{I} \otimes \mathbf{I} + 2\mu \mathbb{I}^{\text{sym}} \quad D_t \sigma = \mathbb{E}^{\text{tan}} : D_t \epsilon \quad (4.3.13)$$

linear elastic continuum secant stiffness

$$\mathbb{E} = \lambda \mathbf{I} \otimes \mathbf{I} + 2\mu \mathbb{I}^{\text{sym}} \quad \sigma = \mathbb{E} : \epsilon \quad (4.3.14)$$

Voigt representation of stiffness tensor

$$\mathbb{E} = \begin{bmatrix} \lambda + 2\mu & \lambda & \lambda & 0 & 0 & 0 \\ \lambda & \lambda + 2\mu & \lambda & 0 & 0 & 0 \\ \lambda & \lambda & \lambda + 2\mu & 0 & 0 & 0 \\ 0 & 0 & 0 & \mu & 0 & 0 \\ 0 & 0 & 0 & 0 & \mu & 0 \\ 0 & 0 & 0 & 0 & 0 & \mu \end{bmatrix} \quad (4.3.15)$$

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volumetric - deviatoric decomposition

specific stored energy (quadratic in volumetric and deviatoric strains)

$$\begin{aligned} W &= W^{\text{vol}}(\boldsymbol{\epsilon}^{\text{vol}}) + W^{\text{dev}}(\boldsymbol{\epsilon}^{\text{dev}}) \\ &= \frac{1}{2} \kappa [\boldsymbol{\epsilon}^{\text{vol}} : \mathbf{I}]^2 + \mu [\boldsymbol{\epsilon}^{\text{dev}2} : \mathbf{I}] \end{aligned} \quad (4.3.16)$$

with

$$\begin{aligned} \boldsymbol{\epsilon}^{\text{vol}} &= \frac{1}{3} [\boldsymbol{\epsilon} : \mathbf{I}] \mathbf{I} = \mathbb{I}^{\text{vol}} : \boldsymbol{\epsilon} \\ \boldsymbol{\epsilon}^{\text{dev}} &= \boldsymbol{\epsilon} - \frac{1}{3} [\boldsymbol{\epsilon} : \mathbf{I}] \mathbf{I} = \mathbb{I}^{\text{dev}} : \boldsymbol{\epsilon} \end{aligned} \quad (4.3.17)$$

stress tensor

$$\boldsymbol{\sigma} = \mathbf{D}_{\boldsymbol{\epsilon}} W = \mathbb{E} : \boldsymbol{\epsilon} = 3 \kappa \boldsymbol{\epsilon}^{\text{vol}} + 2 \mu \boldsymbol{\epsilon}^{\text{dev}} \quad (4.3.18)$$

matrix representation of coordinates

$$[\sigma_{ij}] = \begin{bmatrix} \kappa I_{\epsilon} + 2 \mu \epsilon_{11}^{\text{dev}} & 2 \mu \epsilon_{12}^{\text{dev}} & 2 \mu \epsilon_{13}^{\text{dev}} \\ 2 \mu \epsilon_{21}^{\text{dev}} & \kappa I_{\epsilon} + 2 \mu \epsilon_{22}^{\text{dev}} & 2 \mu \epsilon_{23}^{\text{dev}} \\ 2 \mu \epsilon_{31}^{\text{dev}} & 2 \mu \epsilon_{32}^{\text{dev}} & \kappa I_{\epsilon} + 2 \mu \epsilon_{33}^{\text{dev}} \end{bmatrix} \quad (4.3.19)$$

linear elastic continuum tangent stiffness

$$\mathbb{E}^{\text{tan}} = 3 \kappa \mathbb{I}^{\text{vol}} + 2 \mu \mathbb{I}^{\text{dev}} \quad \mathbf{D}_t \boldsymbol{\sigma} = \mathbb{E}^{\text{tan}} : \mathbf{D}_t \boldsymbol{\epsilon} \quad (4.3.20)$$

linear elastic continuum secant stiffness

$$\mathbb{E} = 3 \kappa \mathbb{I}^{\text{vol}} + 2 \mu \mathbb{I}^{\text{dev}} \quad \sigma = \mathbb{E} : \epsilon \quad (4.3.21)$$

Voigt representation of stiffness tensor

$$\mathbb{E} = \begin{bmatrix} \kappa + \frac{4}{3}\mu & \kappa - \frac{2}{3}\mu & \kappa - \frac{2}{3}\mu & 0 & 0 & 0 \\ \kappa - \frac{2}{3}\mu & \kappa + \frac{4}{3}\mu & \kappa - \frac{2}{3}\mu & 0 & 0 & 0 \\ \kappa - \frac{2}{3}\mu & \kappa - \frac{2}{3}\mu & \kappa + \frac{4}{3}\mu & 0 & 0 & 0 \\ 0 & 0 & 0 & \mu & 0 & 0 \\ 0 & 0 & 0 & 0 & \mu & 0 \\ 0 & 0 & 0 & 0 & 0 & \mu \end{bmatrix} \quad (4.3.22)$$

transformation with fourth order unit tensors

$$\begin{aligned} \mathbb{I}^{\text{vol}} &= \frac{1}{3} \mathbf{I} \otimes \mathbf{I} \\ \mathbb{I}^{\text{dev}} &= \mathbb{I}^{\text{sym}} - \mathbb{I}^{\text{vol}} = \mathbb{I}^{\text{sym}} - \frac{1}{3} \mathbf{I} \otimes \mathbf{I} \end{aligned} \quad (4.3.23)$$

thus

$$\begin{aligned} \mathbb{E} = \mathbb{E}^{\text{tan}} &= 3 \kappa \mathbb{I}^{\text{vol}} + 2 \mu \mathbb{I}^{\text{dev}} \\ &= 3 \left[\kappa - \frac{2}{3}\mu \right] \mathbb{I}^{\text{vol}} + 2 \mu \mathbb{I}^{\text{sym}} \\ &= 3 \lambda \mathbb{I}^{\text{vol}} + 2 \mu \mathbb{I}^{\text{sym}} \\ &= \lambda \mathbf{I} \otimes \mathbf{I} + 2 \mu \mathbb{I}^{\text{sym}} \end{aligned} \quad (4.3.24)$$

bulk modulus and shear modulus

$$\kappa = \lambda + \frac{2}{3} \mu \quad \text{and} \quad \mu \quad (4.3.25)$$

comparison of coordinates E_{1111} and E_{1122}

$$\begin{aligned}\kappa + \frac{4}{3}\mu &= \lambda + \frac{2}{3}\mu + \frac{4}{3}\mu = \lambda + 2\mu \\ \kappa - \frac{2}{3}\mu &= \lambda + \frac{2}{3}\mu - \frac{2}{3}\mu = \lambda\end{aligned}\tag{4.3.26}$$

restrictions to elastic constants from positive stored energy,
i.e. positive definite elastic stiffness

$$\boldsymbol{\epsilon} : \mathbb{E} : \boldsymbol{\epsilon} > 0 \quad \forall \boldsymbol{\epsilon} \neq \mathbf{0} \quad \rightarrow \quad \kappa, \mu > 0, \lambda > -\frac{2}{3}\mu \tag{4.3.27}$$

4.3.2 Specific complementary energy

fourth order elasticity tensor

$$\mathbb{E} = 2\mu\mathbb{I}^{\text{sym}} + \lambda\mathbf{I} \otimes \mathbf{I} \quad \sigma = \mathbb{E} : \epsilon \quad (4.3.28)$$

inversion by making use of Sherman-Morrison-Woodbury theorem

$$\mathbb{A} = \mathbb{B} + \alpha\mathbf{C} \otimes \mathbf{D} \quad (4.3.29)$$

inverse of rank 1 modified fourth order tensor

$$\mathbb{A}^{-1} = \mathbb{B}^{-1} - \alpha \frac{\mathbb{B}^{-1} : \mathbf{C} \otimes \mathbf{D} : \mathbb{B}^{-1}}{1 + \alpha \mathbf{D} : \mathbb{B}^{-1} : \mathbf{C}} \quad (4.3.30)$$

with $\mathbb{A} = \mathbb{E}$, $\mathbb{A}^{-1} = \mathbb{C}$, $\mathbb{B} = 2\mu\mathbb{I}^{\text{sym}}$, $\alpha = \lambda$, $\mathbf{C} = \mathbf{I}$ and $\mathbf{D} = \mathbf{I}$ we obtain the fourth order compliance tensor

$$\mathbb{C} = \frac{1}{2\mu}\mathbb{I}^{\text{sym}} - \frac{\lambda}{2\mu[2\mu + 3\lambda]}\mathbf{I} \otimes \mathbf{I} \quad \epsilon = \mathbb{C} : \sigma \quad (4.3.31)$$

or rather

$$\mathbb{C} = \gamma\mathbf{I} \otimes \mathbf{I} + \frac{1}{2\mu}\mathbb{I}^{\text{sym}} \quad \text{with} \quad \gamma = -\frac{\lambda}{2\mu[2\mu + 3\lambda]} \quad (4.3.32)$$

check

with $\mathbb{I}^{\text{sym}} : \mathbb{I}^{\text{sym}} = \mathbb{I}^{\text{sym}}$, $\mathbb{I}^{\text{sym}} : [\mathbf{I} \otimes \mathbf{I}] = \mathbf{I} \otimes \mathbf{I}$ and $[\mathbf{I} \otimes \mathbf{I}] : [\mathbf{I} \otimes \mathbf{I}] = 3\mathbf{I} \otimes \mathbf{I}$

$$\begin{aligned} \mathbb{E} : \mathbb{E}^{-1} &= [2\mu\mathbb{I}^{\text{sym}} + \lambda\mathbf{I} \otimes \mathbf{I}] : \left[\frac{1}{2\mu}\mathbb{I}^{\text{sym}} - \frac{\lambda}{2\mu[2\mu + 3\lambda]}\mathbf{I} \otimes \mathbf{I} \right] \\ &= \mathbb{I}^{\text{sym}} + \left[\frac{\lambda}{2\mu} - \frac{2\mu\lambda}{2\mu[2\mu + 3\lambda]} - \frac{3\lambda^2}{2\mu[2\mu + 3\lambda]} \right] \mathbf{I} \otimes \mathbf{I} = \mathbb{I}^{\text{sym}} \end{aligned} \quad (4.3.33)$$

Voigt representation of compliance tensor

$$\mathfrak{C} = \begin{bmatrix} \gamma + \frac{1}{2\mu} & \gamma & \gamma & 0 & 0 & 0 \\ \gamma & \gamma + \frac{1}{2\mu} & \gamma & 0 & 0 & 0 \\ \gamma & \gamma & \gamma + \frac{1}{2\mu} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{1}{4\mu} & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{1}{4\mu} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{1}{4\mu} \end{bmatrix} \quad (4.3.34)$$

strain tensor (linear in stresses)

$$\epsilon = D_{\sigma} W^* = \mathfrak{C} : \sigma = \gamma [\sigma : I] I + \frac{1}{2\mu} \sigma \quad (4.3.35)$$

matrix representation of coordinates

$$[\epsilon_{ij}] = \begin{bmatrix} \gamma I_{\sigma} + \frac{1}{2\mu} \sigma_{11} & \frac{1}{2\mu} \sigma_{12} & \frac{1}{2\mu} \sigma_{13} \\ \frac{1}{2\mu} \sigma_{21} & \gamma I_{\sigma} + \frac{1}{2\mu} \sigma_{22} & \frac{1}{2\mu} \sigma_{23} \\ \frac{1}{2\mu} \sigma_{31} & \frac{1}{2\mu} \sigma_{32} & \gamma I_{\sigma} + \frac{1}{2\mu} \sigma_{33} \end{bmatrix} \quad (4.3.36)$$

specific complementary energy (quadratic in stresses)

$$W^* = \frac{1}{2} \sigma : \mathfrak{C} : \sigma = \frac{1}{2} \gamma [\sigma : I]^2 + \frac{1}{4\mu} [\sigma^2 : I] \quad (4.3.37)$$

volumetric - deviatoric decomposition

fourth order elasticity tensor

$$\mathbb{E} = 2\mu\mathbb{I}^{\text{dev}} + 3\kappa\mathbb{I}^{\text{vol}} \quad \sigma = \mathbb{E} : \epsilon \quad (4.3.38)$$

inversion by making use of orthogonality of $\mathbb{I}^{\text{vol}}$ and $\mathbb{I}^{\text{dev}}$ yields fourth order compliance tensor

$$\mathbb{C} = \frac{1}{2\mu}\mathbb{I}^{\text{dev}} + \frac{1}{3\kappa}\mathbb{I}^{\text{vol}} \quad \epsilon = \mathbb{C} : \sigma \quad (4.3.39)$$

check

with $\mathbb{I}^{\text{dev}} : \mathbb{I}^{\text{dev}} = \mathbb{I}^{\text{dev}}$, $\mathbb{I}^{\text{vol}} : \mathbb{I}^{\text{vol}} = \mathbb{I}^{\text{vol}}$ and $\mathbb{I}^{\text{dev}} : \mathbb{I}^{\text{vol}} = \mathbf{0}$

$$\begin{aligned} \mathbb{E} : \mathbb{E}^{-1} &= [2\mu\mathbb{I}^{\text{dev}} + 3\kappa\mathbb{I}^{\text{vol}}] : [\frac{1}{2\mu}\mathbb{I}^{\text{dev}} + \frac{1}{3\kappa}\mathbb{I}^{\text{vol}}] \\ &= \mathbb{I}^{\text{dev}} + \mathbb{I}^{\text{vol}} = \mathbb{I} \end{aligned} \quad (4.3.40)$$

Voigt representation of compliance

$$\mathbb{C} = \begin{bmatrix} \frac{1}{9\kappa} + \frac{2}{6\mu} & \frac{1}{9\kappa} - \frac{1}{6\mu} & \frac{1}{9\kappa} - \frac{1}{6\mu} & 0 & 0 & 0 \\ \frac{1}{9\kappa} - \frac{1}{6\mu} & \frac{1}{9\kappa} + \frac{2}{6\mu} & \frac{1}{9\kappa} - \frac{1}{6\mu} & 0 & 0 & 0 \\ \frac{1}{9\kappa} - \frac{1}{6\mu} & \frac{1}{9\kappa} - \frac{1}{6\mu} & \frac{1}{9\kappa} + \frac{2}{6\mu} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{1}{4\mu} & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{1}{4\mu} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{1}{4\mu} \end{bmatrix} \quad (4.3.41)$$

comparison of coordinates C_{1111} and C_{1122}

$$\begin{aligned}
 \frac{1}{9\kappa} + \frac{2}{6\mu} &= \frac{1}{9\lambda + 6\mu} + \frac{2}{6\mu} = \frac{1}{3[3\lambda + 2\mu]} - \frac{1}{3[2\mu]} + \frac{3}{3[2\mu]} \\
 &= \frac{2\mu - 3\lambda - 2\mu}{3[2\mu][3\lambda + 2\mu]} + \frac{1}{2\mu} \\
 &= \frac{-\lambda}{2\mu[2\mu + 3\lambda]} + \frac{1}{2\mu} = \gamma + \frac{1}{2\mu} \\
 \frac{1}{9\kappa} - \frac{1}{6\mu} &= \frac{1}{9\lambda + 6\mu} - \frac{1}{6\mu} = \frac{1}{3[3\lambda + 2\mu]} - \frac{1}{3[2\mu]} \\
 &= \frac{2\mu - 3\lambda - 2\mu}{3[2\mu][3\lambda + 2\mu]} = -\frac{\lambda}{2\mu[2\mu + 3\lambda]} = \gamma
 \end{aligned} \tag{4.3.42}$$

strain tensor (linear in stresses)

$$\epsilon = D_{\sigma} W^* = \mathfrak{C} : \sigma = \kappa [\sigma : \mathbf{I}] \mathbf{I} + \frac{1}{4\mu} \sigma^{\text{dev}} \tag{4.3.43}$$

matrix representation of coordinates

$$[\epsilon_{ij}] = \begin{bmatrix} \kappa I_{\epsilon} + 2\mu \epsilon_{11}^{\text{dev}} & 2\mu \epsilon_{12}^{\text{dev}} & 2\mu \epsilon_{13}^{\text{dev}} \\ 2\mu \epsilon_{21}^{\text{dev}} & \kappa I_{\epsilon} + 2\mu \epsilon_{22}^{\text{dev}} & 2\mu \epsilon_{23}^{\text{dev}} \\ 2\mu \epsilon_{31}^{\text{dev}} & 2\mu \epsilon_{32}^{\text{dev}} & \kappa I_{\epsilon} + 2\mu \epsilon_{33}^{\text{dev}} \end{bmatrix} \tag{4.3.44}$$

specific stored energy (quadratic in volumetric and deviatoric stresses)

$$\begin{aligned}
 W^* &= W^{*\text{vol}}(\sigma^{\text{vol}}) + W^{*\text{dev}}(\sigma^{\text{dev}}) \\
 &= \frac{1}{2} \kappa \epsilon^{\text{vol}2} + \mu [\epsilon^{\text{dev}2} : \mathbf{I}]
 \end{aligned} \tag{4.3.45}$$

4.3.3 Elastic constants

isotropic linear elasticity can be characterized by only two elastic constants

	E, ν	E, μ	λ, μ	κ, μ
E	E	E	$\frac{\mu [3\lambda + 2\mu]}{\lambda + \mu}$	$\frac{9\kappa\mu}{3\kappa + \mu}$
ν	ν	$\frac{E - 2\mu}{2\mu}$	$\frac{\lambda}{2[\lambda + \mu]}$	$\frac{3\kappa - 2\mu}{6\kappa + 2\mu}$
μ	$\frac{E}{2[1 + \nu]}$	μ	μ	μ
λ	$\frac{E\nu}{[1 + \nu][1 - 2\nu]}$	$\frac{\mu[E - 2\mu]}{3\mu - E}$	λ	$\kappa - \frac{2}{3}\mu$
κ	$\frac{E}{3[1 - 2\nu]}$	$\frac{E\mu}{3[3\mu - E]}$	$\lambda + \frac{2}{3}\mu$	κ

Table 4.1: relations between elastic constants

4.4 Transversely isotropic hyperelasticity

fiber direction $\mathbf{n}$ and structural tensor $\mathbf{N}$

$$\mathbf{N} = \mathbf{n} \otimes \mathbf{n} \quad \text{with} \quad |\mathbf{n}| = 1 \quad (4.4.1)$$

specific stored energy:

isotropic tensor function with two arguments

$$\begin{aligned} W &= W(\boldsymbol{\epsilon}, \mathbf{n}) = W(\boldsymbol{\epsilon}, -\mathbf{n}) = W(\boldsymbol{\epsilon}, \mathbf{N}) \\ &= W(\mathbf{Q} \cdot \boldsymbol{\epsilon} \cdot \mathbf{Q}^t, \mathbf{Q} \cdot \mathbf{N} \cdot \mathbf{Q}^t) \quad \forall \mathbf{Q} \in SO(3) \end{aligned} \quad (4.4.2)$$

representation theorem for isotropic tensor functions with two arguments

$$W = W(\boldsymbol{\epsilon}, \mathbf{N}) = W(i_{\boldsymbol{\epsilon}}, i_{\mathbf{N}}, i_{\boldsymbol{\epsilon}\mathbf{N}}) \quad (4.4.3)$$

irreducible set of ten invariants, integrity basis

$$\begin{aligned} \bar{i}_{\boldsymbol{\epsilon}} &= \{\boldsymbol{\epsilon} : \mathbf{I}, \boldsymbol{\epsilon}^2 : \mathbf{I}, \boldsymbol{\epsilon}^3 : \mathbf{I}\} \\ i_{\mathbf{N}} &= \{\mathbf{N} : \mathbf{I}, \mathbf{N}^2 : \mathbf{I}, \mathbf{N}^3 : \mathbf{I}\} \\ i_{\boldsymbol{\epsilon}, \mathbf{N}} &= \{\boldsymbol{\epsilon} : \mathbf{N}, \boldsymbol{\epsilon} : \mathbf{N}^2, \boldsymbol{\epsilon}^2 : \mathbf{N}, \boldsymbol{\epsilon}^2 : \mathbf{N}^2\} \end{aligned} \quad (4.4.4)$$

recall different representation of set of three invariants

$$\begin{aligned} \bar{i}_{\boldsymbol{\epsilon}} &= \{\boldsymbol{\epsilon} : \mathbf{I}, \boldsymbol{\epsilon}^2 : \mathbf{I}, \boldsymbol{\epsilon}^3 : \mathbf{I}\} && \text{basic invariants} \\ i_{\boldsymbol{\epsilon}} &= \{\text{tr}(\boldsymbol{\epsilon}), \frac{1}{2}[\text{tr}^2(\boldsymbol{\epsilon}) - \text{tr}(\boldsymbol{\epsilon}^2)], \det(\boldsymbol{\epsilon})\} && \text{principal invariants} \\ i_{\boldsymbol{\epsilon}} &= \{(\lambda_{\epsilon 1} + \lambda_{\epsilon 2} + \lambda_{\epsilon 3}), (\lambda_{\epsilon 2}\lambda_{\epsilon 3} + \lambda_{\epsilon 3}\lambda_{\epsilon 1} + \lambda_{\epsilon 1}\lambda_{\epsilon 3}), (\lambda_{\epsilon 1}\lambda_{\epsilon 2}\lambda_{\epsilon 3})\} \\ &&& \text{eigenvalue representation of principal invariants} \end{aligned} \quad (4.4.5)$$

until now: principal invariants $i_{\boldsymbol{\epsilon}}$, now: basic invariants $\bar{i}_{\boldsymbol{\epsilon}}$

with properties of structural tensor, idempotence & normalization

$$\begin{aligned} N^2 &= N \cdot N = [\mathbf{n} \otimes \mathbf{n}] \cdot [\mathbf{n} \otimes \mathbf{n}] = \mathbf{n} \otimes \mathbf{n} = N \\ N : I &= \mathbf{n} \cdot \mathbf{n} = |\mathbf{n}|^2 = 1 \end{aligned} \quad (4.4.6)$$

reduced representation with five invariants

$$W = W(\boldsymbol{\epsilon}, N) = W(\bar{I}_\epsilon, \bar{II}_\epsilon, I\bar{II}_\epsilon, I\bar{V}_\epsilon, \bar{V}_\epsilon) \quad (4.4.7)$$

representation theorem for isotropic tensor functions with two arguments

$$\boldsymbol{\sigma}(\boldsymbol{\epsilon}, N) = f_1 I + f_2 \boldsymbol{\epsilon} + f_3 \boldsymbol{\epsilon} \cdot \boldsymbol{\epsilon} + f_4 N + f_5 [\boldsymbol{\epsilon} \cdot N + N \cdot \boldsymbol{\epsilon}] \quad (4.4.8)$$

stress for transversely isotropic hyperelastic material

$$\begin{aligned} \boldsymbol{\sigma} = D_{\boldsymbol{\epsilon}} W &= \frac{DW}{D\bar{I}_\epsilon} \frac{D\bar{I}_\epsilon}{D\boldsymbol{\epsilon}} + \frac{DW}{D\bar{II}_\epsilon} \frac{D\bar{II}_\epsilon}{D\boldsymbol{\epsilon}} + \frac{DW}{DI\bar{II}_\epsilon} \frac{DI\bar{II}_\epsilon}{D\boldsymbol{\epsilon}} \\ &= \frac{DW}{DI\bar{V}_\epsilon} \frac{DI\bar{V}_\epsilon}{D\boldsymbol{\epsilon}} + \frac{DW}{D\bar{V}_\epsilon} \frac{D\bar{V}_\epsilon}{D\boldsymbol{\epsilon}} \end{aligned} \quad (4.4.9)$$

with derivatives of invariants $\bar{I}_\epsilon, \bar{II}_\epsilon, I\bar{II}_\epsilon, I\bar{V}_\epsilon, \bar{V}_\epsilon$ with respect to second order tensor $\boldsymbol{\epsilon}$

$$\begin{aligned} \bar{I}_\epsilon &= \boldsymbol{\epsilon} : I \quad \text{linear} \quad D_{\boldsymbol{\epsilon}} \bar{I}_\epsilon = I \\ \bar{II}_\epsilon &= \boldsymbol{\epsilon}^2 : I \quad \text{quadr.} \quad D_{\boldsymbol{\epsilon}} \bar{II}_\epsilon = 2\boldsymbol{\epsilon} \\ I\bar{II}_\epsilon &= \boldsymbol{\epsilon}^3 : I \quad \text{cubic} \quad D_{\boldsymbol{\epsilon}} I\bar{II}_\epsilon = 3\boldsymbol{\epsilon}^2 \\ I\bar{V}_\epsilon &= \boldsymbol{\epsilon} : N \quad \text{linear} \quad D_{\boldsymbol{\epsilon}} I\bar{V}_\epsilon = N \\ \bar{V}_\epsilon &= \boldsymbol{\epsilon}^2 : N \quad \text{quadr.} \quad D_{\boldsymbol{\epsilon}} \bar{V}_\epsilon = \boldsymbol{\epsilon} \cdot N + N \cdot \boldsymbol{\epsilon} \end{aligned} \quad (4.4.10)$$

with $\bar{V}_\epsilon = \boldsymbol{\epsilon}^2 : N = [\boldsymbol{\epsilon} \cdot \boldsymbol{\epsilon}] : N = \boldsymbol{\epsilon} : [\boldsymbol{\epsilon} \cdot N] = \boldsymbol{\epsilon} : [N \cdot \boldsymbol{\epsilon}]$

alternatively

$$I\bar{V}_\epsilon = \epsilon : N = n \cdot \epsilon \cdot n \quad \text{and} \quad \bar{V}_\epsilon = \epsilon^2 : N = n \cdot \epsilon^2 \cdot n$$

general representation of stress

$$\begin{aligned} \sigma = D_\epsilon W &= D_{\bar{I}_\epsilon} W \mathbf{I} + 2 D_{\bar{I}I_\epsilon} W \epsilon + 3 D_{\bar{I}I\bar{I}_\epsilon} W \epsilon^2 \\ &= D_{I\bar{V}_\epsilon} W \mathbf{N} + D_{\bar{V}_\epsilon} W [\epsilon \cdot \mathbf{N} + \mathbf{N} \cdot \epsilon] \end{aligned} \quad (4.4.11)$$

comparison of coefficients

assumption of linearity (quadratic term vanishes), six material parameters $f_{11}, f_{14}, f_2, f_{41}, f_{44}, f_5$

$$\begin{aligned} f_1 &= D_{\bar{I}_\epsilon} W = f_{11} \bar{I}_\epsilon + f_{14} I\bar{V}_\epsilon \\ f_2 &= 2 D_{\bar{I}I_\epsilon} W = \text{const.} \\ f_3 &= 3 D_{\bar{I}I\bar{I}_\epsilon} W = 0 \\ f_4 &= D_{I\bar{V}_\epsilon} W = f_{41} \bar{I}_\epsilon + f_{44} I\bar{V}_\epsilon \\ f_5 &= D_{\bar{V}_\epsilon} W = \text{const.} \end{aligned} \quad (4.4.12)$$

specific stored energy (quadratic in strains)

$$\begin{aligned} W = \frac{1}{2} \sigma : \epsilon &= \frac{1}{2} f_{11} \bar{I}_\epsilon^2 + \frac{1}{2} f_{14} I\bar{V}_\epsilon \bar{I}_\epsilon + \frac{1}{2} f_2 \bar{I}I_\epsilon \\ &+ \frac{1}{2} f_{41} \bar{I}_\epsilon I\bar{V}_\epsilon + \frac{1}{2} f_{44} I\bar{V}_\epsilon^2 + f_5 \bar{V}_\epsilon^2 \end{aligned} \quad (4.4.13)$$

stress tensor (linear in strains)

$$\begin{aligned} \sigma = D_\epsilon W &= [f_{11} \bar{I}_\epsilon + f_{14} I\bar{V}_\epsilon] \mathbf{I} + f_2 \epsilon \\ &+ [f_{41} \bar{I}_\epsilon + f_{44} I\bar{V}_\epsilon] \mathbf{N} + f_5 [\epsilon \cdot \mathbf{N} + \mathbf{N} \cdot \epsilon] \end{aligned} \quad (4.4.14)$$

linear elastic continuum tangent stiffness (constant in strains)

$$\mathbb{E}^{\text{tan}} = \mathbf{D}\boldsymbol{\epsilon}\boldsymbol{\sigma}$$

$$\mathbb{E}^{\text{tan}} = f_{11}\mathbf{I} \otimes \mathbf{I} + f_{14}\mathbf{I} \otimes \mathbf{N} + f_{41}\mathbf{N} \otimes \mathbf{I} + f_{44}\mathbf{N} \otimes \mathbf{N} + f_2\mathbb{I}^{\text{sym}} + f_5/\Lambda \quad (4.4.15)$$

whereby $\Lambda = 2\mathbf{D}\boldsymbol{\epsilon}[\boldsymbol{\epsilon} \cdot \mathbf{N} + \mathbf{N} \cdot \boldsymbol{\epsilon}]$

due to symmetry reduction to five material parameters $f_{14} = f_{41}$

$$\mathbb{E}^{\text{tan}} = f_{11}\mathbf{I} \otimes \mathbf{I} + f_{14}[\mathbf{I} \otimes \mathbf{N} + \mathbf{N} \otimes \mathbf{I}] + f_{44}\mathbf{N} \otimes \mathbf{N} + f_2\mathbb{I}^{\text{sym}} + f_5/\Lambda \quad (4.4.16)$$

linear elastic continuum secant stiffness

$$\mathbb{E} = f_{11}\mathbf{I} \otimes \mathbf{I} + f_{14}[\mathbf{I} \otimes \mathbf{N} + \mathbf{N} \otimes \mathbf{I}] + f_{44}\mathbf{N} \otimes \mathbf{N} + f_2\mathbb{I}^{\text{sym}} + f_5/\Lambda \quad (4.4.17)$$

physical interpretation of parameters

interpretation of f_{11} , $f_{14} = f_{41}$, f_{44} , f_2 and f_5 Spencer [1984]

$$W = \frac{1}{2}\lambda \bar{I}_\epsilon^2 + \alpha \bar{I}_\epsilon \bar{I} \bar{V}_\epsilon + \frac{1}{2}\beta \bar{I} \bar{V}_\epsilon^2 + \mu_\perp \bar{I} \bar{I}_\epsilon + 2[\mu_\parallel - \mu_\perp] \bar{V}_\epsilon^2 \quad (4.4.18)$$

stress tensor (linear in strains)

$$\begin{aligned}\boldsymbol{\sigma} = \mathbf{D}_\epsilon W = & [\lambda \bar{I}_\epsilon + \alpha I\bar{V}_\epsilon] \mathbf{I} + 2\mu_\perp \boldsymbol{\epsilon} \\ & + [\alpha \bar{I}_\epsilon + \beta I\bar{V}_\epsilon] \mathbf{N} + 2[\mu_\parallel - \mu_\perp] [\boldsymbol{\epsilon} \cdot \mathbf{N} + \mathbf{N} \cdot \boldsymbol{\epsilon}] \end{aligned} \quad (4.4.19)$$

Voigt representation of stiffness tensor for $\mathbf{n} = [1, 0, 0]^t$, i.e. transverse isotropy with respect to the $\mathbf{e}_1$ axis

$$\mathbb{E} = \begin{bmatrix} \lambda + 2\alpha + 4\mu_\parallel & \lambda + \alpha & \lambda + \alpha & 0 & 0 & 0 \\ -2\mu_\perp + \beta & \lambda + \alpha & \lambda + \alpha & 0 & 0 & 0 \\ \lambda + \alpha & \lambda + 2\mu_\perp & \lambda & 0 & 0 & 0 \\ \lambda + \alpha & \lambda & \lambda + 2\mu_\perp & 0 & 0 & 0 \\ 0 & 0 & 0 & \mu_\parallel & 0 & 0 \\ 0 & 0 & 0 & 0 & \mu_\perp & 0 \\ 0 & 0 & 0 & 0 & 0 & \mu_\parallel \end{bmatrix} \quad (4.4.20)$$

comparison of coefficients

$$f_{11} = \lambda \quad f_2 = 2\mu_\perp \quad f_{14} = \alpha = f_{41} \quad f_{44} = \beta \quad f_5 = 2[\mu_\parallel - \mu_\perp] \quad (4.4.21)$$

with shear moduli $\mu_\parallel$ for shear in fiber direction and $\mu_\perp$ for shear parallel to the fiber direction

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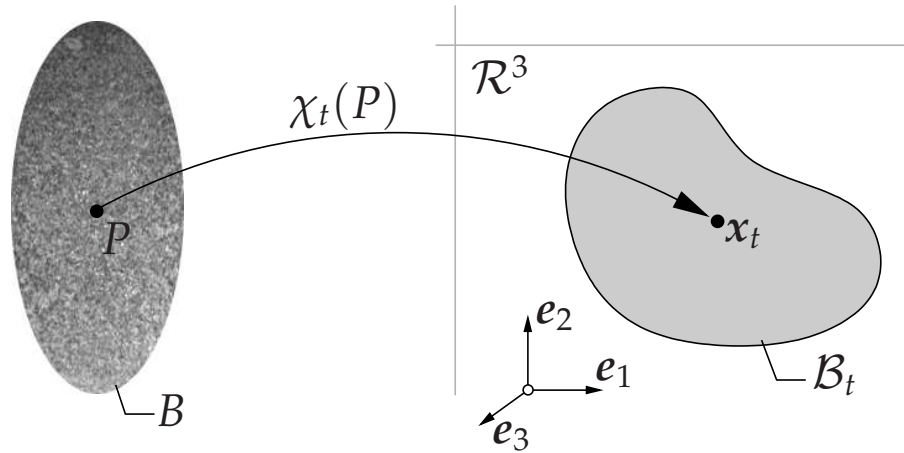
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5 Introduction to Nonlinear Continuum Mechanics

5.1 Kinematics

5.1.1 Motion and Configurations

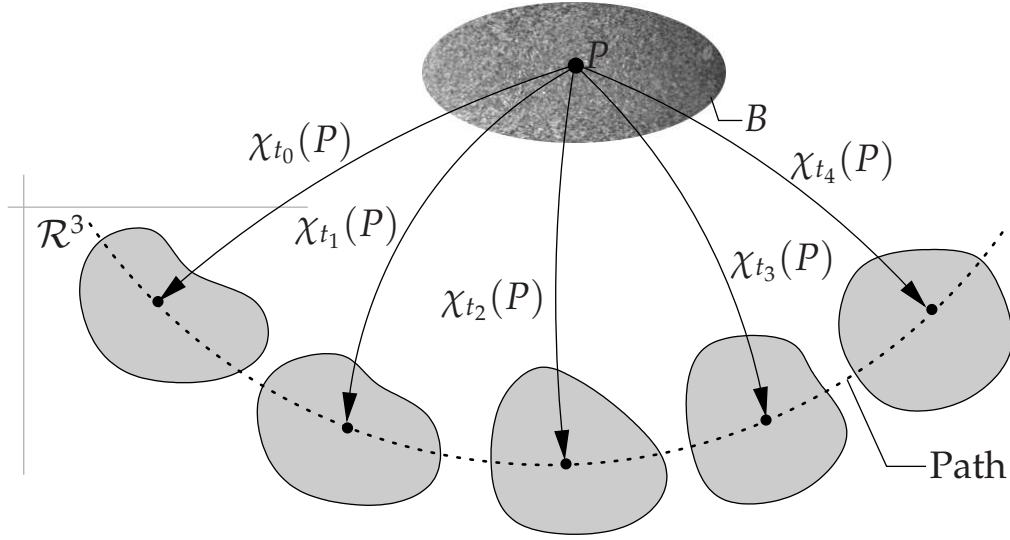
Consider a *material body* B composed of infinitely many *material points* $P \in B$ that are identified with geometrical points in the three-dimensional Euclidean space $\mathcal{R}^3$.



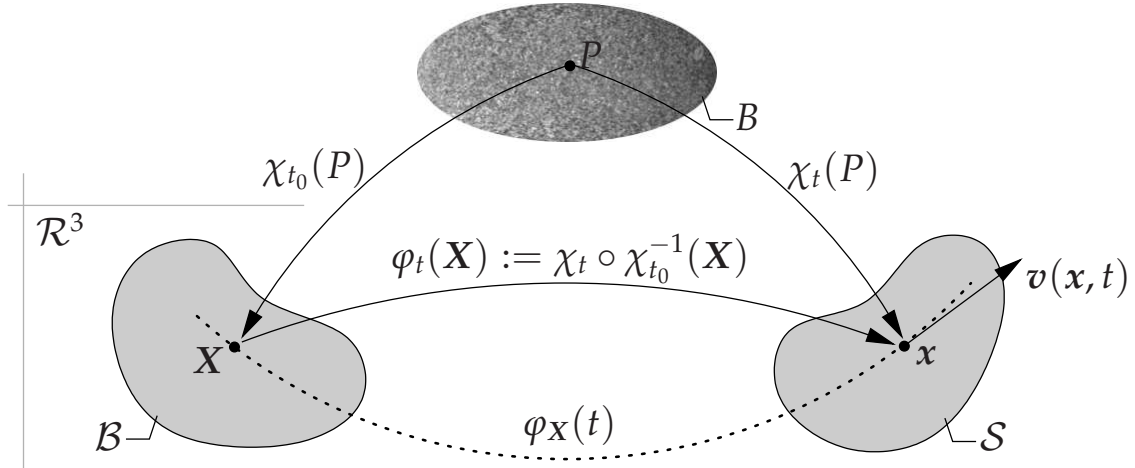
The *configuration* of the body B in $\mathcal{R}^3$ at time t is described by the one-to-one relation

$$\chi_t : \begin{cases} B \rightarrow \mathcal{B}_t \in \mathcal{R}^3, \\ P \in B \mapsto \mathbf{x}_t = \chi_t(P) \in \mathcal{B}_t. \end{cases} \quad (5.1.1)$$

Having this definition at hand, we go on to describe the *motion* of the body as a family of configurations parameterized



by time in the Euclidean space $\mathcal{R}^3$. Throughout the motion, the material point P occupies a sequence of places in space, which forms the *path (trajectory)* of the material point.



While describing the motion of a solid body, it is common practice to name its placement at time t_0 as the *reference configuration* that generally possesses an undistorted stress-free state and is henceforth denoted as $\mathcal{B} \equiv \chi_{t_0}(B)$. Likewise, the configuration of the body at current time t is hereafter denoted as $\mathcal{S} \equiv \chi_t(B)$. The reference and the spatial positions occupied by a material point P within the Euclidean space

are labeled by the *reference* coordinates $\mathbf{X} := \chi_{t_0}(P) \in \mathcal{B}$ and the *spatial* coordinates $\mathbf{x} := \chi_t(P) \in \mathcal{S}$, respectively. In order to describe the *motion* of the body in the Euclidean space, we introduce a non-linear *deformation map* $\varphi_t(\mathbf{X})$ between $\chi_{t_0}(P)$ and $\chi_t(P)$

$$\varphi_t(\mathbf{X}) : \begin{cases} \mathcal{B} \rightarrow \mathcal{S} , \\ \mathbf{X} \mapsto \mathbf{x} = \varphi_t(\mathbf{X}) := \chi_t \circ \chi_0^{-1}(\mathbf{X}) \end{cases} \quad (5.1.2)$$

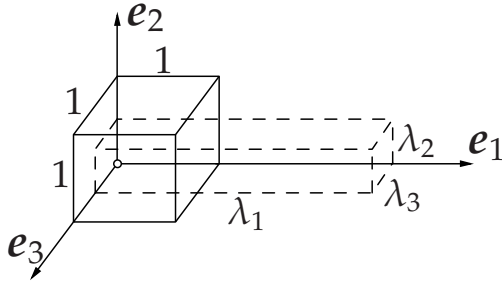
that maps the material points $\mathbf{X} \in \mathcal{B}$ onto their deformed *spatial* positions $\mathbf{x} = \varphi_t(\mathbf{X}) \in \mathcal{S}$ at time $t \in \mathcal{R}_+$. Since the deformation map $\varphi_t(\mathbf{X})$ is bijective, it can be inverted uniquely to obtain the *inverse deformation map*

$$\varphi_t^{-1}(\mathbf{x}) : \begin{cases} \mathcal{S} \rightarrow \mathcal{B} , \\ \mathbf{x} \mapsto \mathbf{X} = \varphi_t^{-1}(\mathbf{x}) := \chi_0 \circ \chi_t^{-1}(\mathbf{x}) . \end{cases} \quad (5.1.3)$$

In indicial notation, we have

$$x_a = \varphi_a(X_1, X_2, X_3, t) \quad \text{and} \quad X_A = \varphi_A^{-1}(x_1, x_2, x_3, t) . \quad (5.1.4)$$

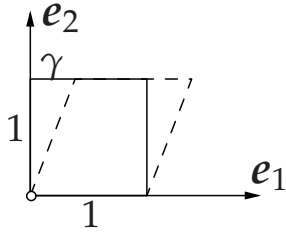
Although we use a common coordinate system to describe the motion of the body, it will be convenient to assign the lower-case indices to tensorial fields belonging to the current configuration and the upper-case indices for referential fields.

Examples:*Simple extension of cube*

$$x_1 = \lambda_1(t) X_1$$

$$x_2 = \lambda_2(t) X_2$$

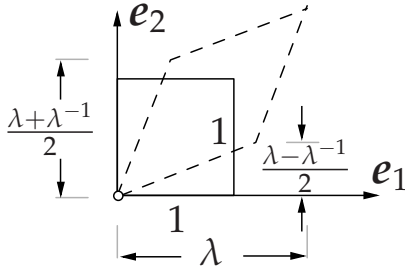
$$x_3 = \lambda_3(t) X_3$$

Simple shear

$$x_1 = X_1 + \gamma(t) X_2$$

$$x_2 = X_2$$

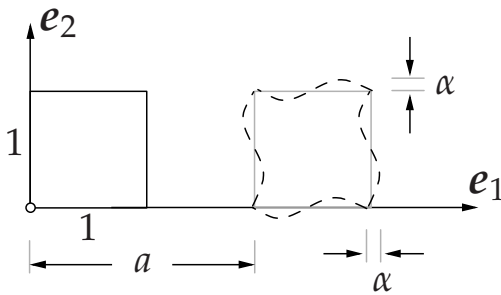
$$x_3 = X_3$$

Pure shear

$$x_1 = \frac{\lambda + \lambda^{-1}}{2} X_1 + \frac{\lambda - \lambda^{-1}}{2} X_2$$

$$x_2 = \frac{\lambda - \lambda^{-1}}{2} X_1 + \frac{\lambda + \lambda^{-1}}{2} X_2$$

$$x_3 = X_3$$

Periodic deformation

$$x_1 = X_1 + \alpha(t) \sin(2\pi X_2) + a$$

$$x_2 = X_2 - \alpha(t) \sin(2\pi X_1)$$

$$x_3 = X_3$$

5.1.2 Velocity and Acceleration

Recall the definition of motion in (5.1.2),

$$\mathbf{x} := \varphi(\mathbf{X}, t) = \varphi_t(\mathbf{X}) = \varphi_{\mathbf{X}}(t) \quad (5.1.5)$$

where $\varphi_t(\mathbf{X})$ denotes the configuration of $\mathcal{B}$ at time t and $\varphi_{\mathbf{X}}(t)$ is the path of the particle P .

5.1.2.1 Material Velocity and Acceleration

We can now define the *material velocity*

$$\mathbf{V}_t(\mathbf{X}) := \partial_t \varphi(\mathbf{X}, t) = \frac{d}{dt} \varphi_{\mathbf{X}}(t) , \quad (5.1.6)$$

and the *material acceleration*

$$\mathbf{A}_t(\mathbf{X}) := \partial_t \mathbf{V}(\mathbf{X}, t) = \frac{d}{dt} \mathbf{V}_{\mathbf{X}}(t) \quad (5.1.7)$$

of the motion. It is important to notice that both the material velocity $\mathbf{V}(\mathbf{X}, t)$ and the material acceleration $\mathbf{A}(\mathbf{X}, t)$ belong to the current configuration $\mathcal{S}$ but are functions of material coordinates $\mathbf{X}$ and time t .

5.1.2.2 Spatial Velocity and Acceleration

The spatial counterparts of the rates of material description of the motion can be obtained by reparameterizing the functional dependency of these fields in terms of spatial coordinates $\mathbf{x}$. The *spatial velocity* and the *spatial acceleration* are then defined as

$$\mathbf{v}(\mathbf{x}, t) := \mathbf{V}(\varphi_t^{-1}(\mathbf{x}), t) = \mathbf{V}_t(\mathbf{X}) \circ \varphi_t^{-1}(\mathbf{x}) , \quad (5.1.8)$$

and

$$\mathbf{a}(\mathbf{x}, t) := \mathbf{A}(\varphi_t^{-1}(\mathbf{x}), t) = \mathbf{A}_t(\mathbf{X}) \circ \varphi_t^{-1}(\mathbf{x}) , \quad (5.1.9)$$

respectively. The *path* $\varphi_{\mathbf{X}}(t)$ is called the *integral curve* of $\mathbf{v}$.

5.1.2.3 Material Time Derivative of Spatial Fields

Consider a spatial field

$$f(\boldsymbol{x}, t) : \varphi_t(\mathcal{B}) \times \mathcal{R}_+ \rightarrow \mathcal{R} .$$

The *material time derivative* of $f(\boldsymbol{x}, t)$ is defined as

$$\frac{D}{Dt}f(\boldsymbol{x}, t) = \dot{f}(\boldsymbol{x}, t) := \frac{\partial}{\partial t}f(\boldsymbol{x}, t) + \nabla_x f \cdot \boldsymbol{v} \quad (5.1.10)$$

For $F(\boldsymbol{X}, t) := f(\boldsymbol{x}, t) \circ \varphi_t(\boldsymbol{X})$, the following equality

$$\frac{\partial}{\partial t}F(\boldsymbol{X}, t) = \dot{f}(\boldsymbol{x}, t) \quad (5.1.11)$$

holds. An immediate example is the spatial acceleration $\boldsymbol{a}(\boldsymbol{x}, t)$ that is none other than the *material time derivative* of the spatial velocity

$$\boldsymbol{a}(\boldsymbol{x}, t) := \boldsymbol{A}_t(\boldsymbol{X}) \circ \varphi_t^{-1}(\boldsymbol{x}) = \partial_t \boldsymbol{v} + \nabla_x \boldsymbol{v} \cdot \boldsymbol{v} . \quad (5.1.12)$$

where $\nabla_x \boldsymbol{v}$ stands for the *spatial gradient of velocity*.

Example: Rigid Body Motion

Chadwick (1976), p.52

The rigid body motion of a body is described by

$$\mathbf{x} = \varphi(\mathbf{X}, t) = \mathbf{c}(t) + \mathbf{Q}(t) \cdot \mathbf{X},$$

where $\mathbf{Q}$ is proper orthogonal, i.e. $\mathbf{Q} \cdot \mathbf{Q}^T = \mathbf{1}$ and $\det \mathbf{Q} = +1$. In *rigid* motion the distance between two points does not change. That is, for $\mathbf{T} := \mathbf{X}_B - \mathbf{X}_A$ and $\mathbf{t} := \mathbf{x}_B - \mathbf{x}_A = \mathbf{Q} \cdot \mathbf{T}$ with $\mathbf{x}_{\{A,B\}} = \varphi(\mathbf{X}_{\{A,B\}}, t)$, we end up with the identity

$$\|\mathbf{t}\| := \sqrt{\mathbf{t} \cdot \mathbf{t}} = \sqrt{(\mathbf{Q} \cdot \mathbf{T}) \cdot (\mathbf{Q} \cdot \mathbf{T})} = \sqrt{\mathbf{T} \cdot (\mathbf{Q}^T \cdot \mathbf{Q} \cdot \mathbf{T})} = \|\mathbf{T}\|.$$

The motion can be then inverted to obtain

$$\mathbf{X} = \varphi_t^{-1}(\mathbf{x}) = \mathbf{Q}^T \cdot (\mathbf{x} - \mathbf{c}).$$

The material velocity and material acceleration

$$\mathbf{V}_t(\mathbf{X}) := \partial_t \varphi(\mathbf{X}, t) = \dot{\mathbf{c}} + \dot{\mathbf{Q}} \cdot \mathbf{X}$$

$$\mathbf{A}_t(\mathbf{X}) := \partial_t \mathbf{V}(\mathbf{X}, t) = \ddot{\mathbf{c}} + \ddot{\mathbf{Q}} \cdot \mathbf{X}$$

The spatial velocity and spatial acceleration

$$\mathbf{v}(\mathbf{x}, t) := \mathbf{V}(\varphi_t^{-1}(\mathbf{x}), t) = \dot{\mathbf{c}} + \dot{\mathbf{Q}} \cdot \mathbf{Q}^T \cdot (\mathbf{x} - \mathbf{c})$$

$$\mathbf{a}(\mathbf{x}, t) := \mathbf{A}(\varphi_t^{-1}(\mathbf{x}), t) = \ddot{\mathbf{c}} + \ddot{\mathbf{Q}} \cdot \mathbf{Q}^T \cdot (\mathbf{x} - \mathbf{c})$$

Since $\mathbf{Q} \cdot \mathbf{Q}^T = \mathbf{1}$, we have

$$\dot{\mathbf{Q}} \cdot \mathbf{Q}^T = -\mathbf{Q} \cdot \dot{\mathbf{Q}}^T = -(\dot{\mathbf{Q}} \cdot \mathbf{Q}^T)^T$$

Therefore,

$$\boldsymbol{\Omega} := \dot{\mathbf{Q}} \cdot \mathbf{Q}^T = -\boldsymbol{\Omega}^T$$

is a skew-symmetric tensor, which has an associated *axial vector* $\omega \in \mathcal{R}^3$ defined by

$$\Omega \cdot \beta = \omega \times \beta \quad \forall \beta \in \mathcal{R}^3.$$

Furthermore, it can be shown that

$$\ddot{\mathbf{Q}} \cdot \mathbf{Q}^T = \dot{\Omega} - \dot{\mathbf{Q}} \cdot \dot{\mathbf{Q}}^T = \dot{\Omega} + \Omega^2,$$

$$\ddot{\mathbf{Q}} \cdot \mathbf{Q}^T \cdot \beta = \dot{\Omega} \cdot \beta + \Omega^2 \cdot \beta = \dot{\omega} \times \beta + \omega \times (\omega \times \beta) \quad \forall \beta \in \mathcal{R}^3.$$

Incorporating these results in the spatial velocity and spatial acceleration expressions for $\beta \equiv (\mathbf{x} - \mathbf{c})$, we end up with

$$\mathbf{v}(\mathbf{x}, t) = \dot{\mathbf{c}} + \omega \times (\mathbf{x} - \mathbf{c})$$

$$\mathbf{a}(\mathbf{x}, t) = \ddot{\mathbf{c}} + \dot{\omega} \times (\mathbf{x} - \mathbf{c}) + \omega \times [\omega \times (\mathbf{x} - \mathbf{c})]$$

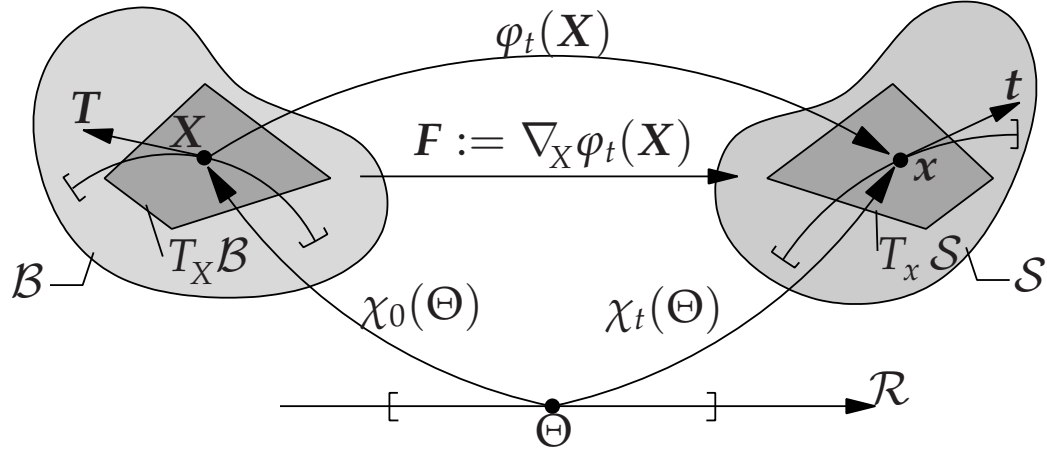
The latter can also be obtained by taking the material time derivative of the spatial velocity field as described in (5.1.12).

$$\begin{aligned} \mathbf{a}(\mathbf{x}, t) &= \partial_t \mathbf{v} + \nabla_x \mathbf{v} \cdot \mathbf{v} \\ &= \partial_t [\dot{\mathbf{c}} + \omega \times (\mathbf{x} - \mathbf{c})] \\ &\quad + \nabla_x [\dot{\mathbf{c}} + \omega \times (\mathbf{x} - \mathbf{c})] \cdot [\dot{\mathbf{c}} + \omega \times (\mathbf{x} - \mathbf{c})] \\ &= \ddot{\mathbf{c}} + \dot{\omega} \times (\mathbf{x} - \mathbf{c}) - \omega \times \dot{\mathbf{c}} \\ &\quad + \omega \times [\dot{\mathbf{c}} + \omega \times (\mathbf{x} - \mathbf{c})] \\ &= \ddot{\mathbf{c}} + \dot{\omega} \times (\mathbf{x} - \mathbf{c}) + \omega \times [\omega \times (\mathbf{x} - \mathbf{c})] \end{aligned}$$

5.1.3 Fundamental Geometric Maps

This section is devoted to the geometric mapping of basic *tangential*, *normal* and *volumetric* objects.

5.1.3.1 Deformation Gradient. Tangent Map



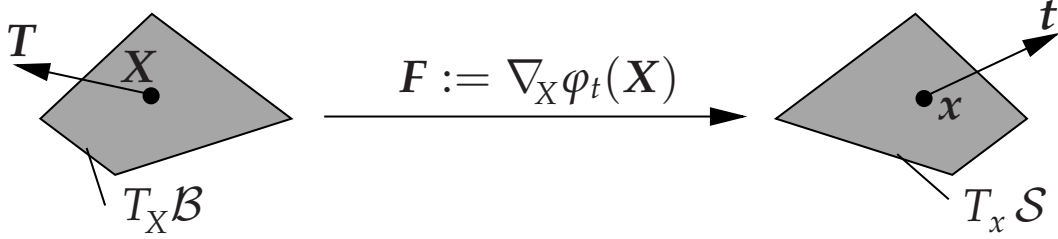
Probably the most fundamental deformation measure used in kinematics of finite deformation is the *deformation gradient*. It can be considered as a linear map of the referential *tangent vectors* onto the spatial counterparts. To this end, let $\chi_0(\Theta)$ and $\chi_t(\Theta)$ be the material and spatial curves parameterized by a common variable $\Theta \in \mathcal{R}$ on $\mathcal{B}$ and $\mathcal{S}$, respectively. It is important to observe that the spatial curve is related to the reference curve by the non-linear deformation map $\chi_t(\Theta) = \varphi_t(\chi_0(\Theta))$. Tangents of the curves belonging to the respective tangent spaces are defined as

$$T := \frac{d\chi_0(\Theta)}{d\Theta} \in T_X \mathcal{B} \quad \text{and} \quad t := \frac{d\chi_t(\Theta)}{d\Theta} \in T_x \mathcal{S} \quad (5.1.13)$$

Through the chain rule, we obtain

$$t = \frac{d\chi_t(\Theta)}{d\Theta} = \nabla_X \varphi_t(X) \cdot \frac{d\chi_0(\Theta)}{d\Theta} = F \cdot T, \quad (5.1.14)$$

where we introduced the *deformation gradient* $F := \nabla_X \varphi_t(\mathbf{X})$ as the Fréchet derivative of the deformation.



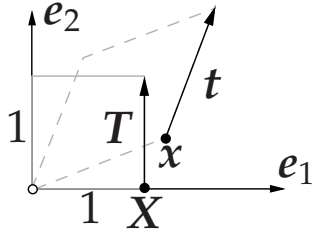
The deformation gradient is also called the *tangent map*

$$F(\mathbf{X}, t) : \begin{cases} T_X \mathcal{B} \rightarrow T_x \mathcal{S}, \\ T \mapsto t = F \cdot T, \end{cases} \quad (5.1.15)$$

between the tangent spaces $T_X \mathcal{B}$ and $T_x \mathcal{S}$ of the configurations (manifolds) $\mathcal{B}$ and $\mathcal{S}$, respectively. Note that the deformation gradient is a *two-point* tensor that involves points from the both configurations; that is, in index notation

$$F_{aA} := \partial \varphi_a / \partial X_A \quad \text{and} \quad t_a = F_{aA} T_A.$$

The determinant of the deformation gradient $\det(F)$ is restricted to non-zero ($\det(F) \neq 0$), positive ($\det(F) > 0$) values due to the facts that $\varphi_t(\mathbf{X})$ is one-to-one and we rule out interpenetration of the material throughout the motion.

Examples:*Pure shear (Homogeneous)*

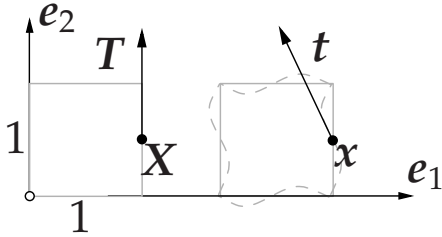
$$x_1 = \frac{\lambda + \lambda^{-1}}{2} X_1 + \frac{\lambda - \lambda^{-1}}{2} X_2$$

$$x_2 = \frac{\lambda - \lambda^{-1}}{2} X_1 + \frac{\lambda + \lambda^{-1}}{2} X_2$$

$$x_3 = X_3$$

$$F = \frac{1}{2} \begin{bmatrix} \lambda + \lambda^{-1} & \lambda - \lambda^{-1} & 0 \\ \lambda - \lambda^{-1} & \lambda + \lambda^{-1} & 0 \\ 0 & 0 & 2 \end{bmatrix}$$

$$T = e_2, \quad t = F \cdot T = \frac{1}{2} [(\lambda - \lambda^{-1})e_1 + (\lambda + \lambda^{-1})e_2]$$

Periodic deformation (Inhomogeneous)

$$x_1 = X_1 + \alpha(t) \sin(2\pi X_2) + a$$

$$x_2 = X_2 - \alpha(t) \sin(2\pi X_1)$$

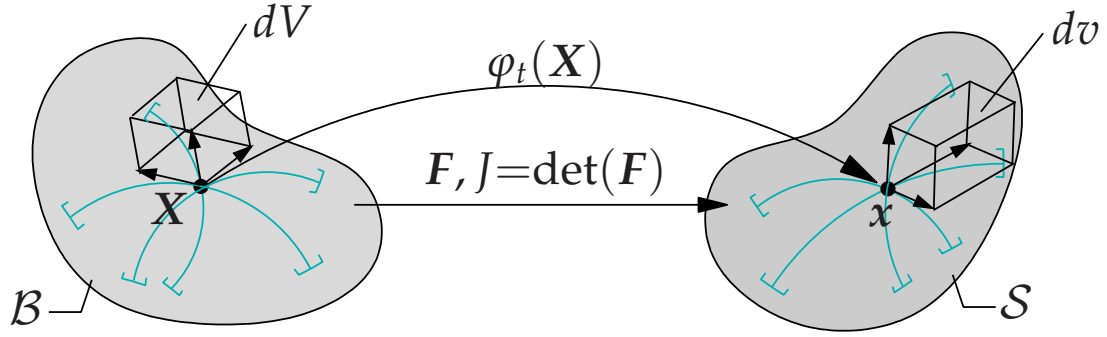
$$x_3 = X_3$$

$$F = \begin{bmatrix} 1 & 2\pi\alpha \cos(2\pi X_2) & 0 \\ -2\pi\alpha \cos(2\pi X_1) & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}_{X=(1,0.5,0)}$$

$$= \begin{bmatrix} 1 & -2\pi\alpha & 0 \\ -2\pi\alpha & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$T = e_2, \quad t = F \cdot T = -2\pi\alpha e_1 + e_2$$

5.1.3.2 Volume Transformation. Jacobi Map



Let dV and dv denote the infinitesimal volumes of parallelepipeds

$$dV := d\mathbf{X}_1 \cdot (d\mathbf{X}_2 \times d\mathbf{X}_3) \quad \text{and} \quad dv := d\mathbf{x}_1 \cdot (d\mathbf{x}_2 \times d\mathbf{x}_3) \quad (5.1.16)$$

defined as the scalar triple product of vectors $d\mathbf{X}_{i=1,2,3} \in T_{\mathbf{X}}\mathcal{B}$ and $d\mathbf{x}_{i=1,2,3} \in T_{\mathbf{x}}\mathcal{S}$, respectively. Each spatial tangent vector $d\mathbf{x}_i$ is defined as a tangential map of its material counterpart, i.e. $d\mathbf{x}_i := \mathbf{F} \cdot d\mathbf{X}_i$ for $i = 1, 2, 3$. We define the *volume map*

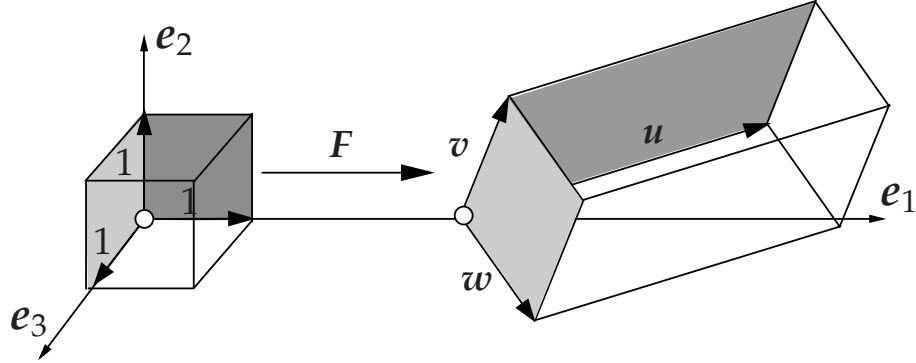
$$\begin{aligned} dv &= (\mathbf{F} \cdot d\mathbf{X}_1) \cdot [(\mathbf{F} \cdot d\mathbf{X}_2) \times (\mathbf{F} \cdot d\mathbf{X}_3)] \\ &= \det(\mathbf{F}) d\mathbf{X}_1 \cdot (d\mathbf{X}_2 \times d\mathbf{X}_3) = \det \mathbf{F} dV =: J dV \end{aligned} \quad (5.1.17)$$

through the conventional coordinate-free definition of the determinant of a second order tensor. Recall that the value of the Jacobian J is restricted to positive real numbers $\mathcal{R}_+$ in order to ensure the one-to-one relation between $\mathbf{x}$ and $\mathbf{X}$ and the impenetrability of a material. Then, we say that the volume map, $\det \mathbf{F}$, maps the reference volume elements onto

their spatial counterparts

$$J = \det \mathbf{F} := \begin{cases} \mathcal{R}_+ \rightarrow \mathcal{R}_+, \\ dV \mapsto dv = \det \mathbf{F} dV. \end{cases} \quad (5.1.18)$$

Example:



Assume that vectors u, v, w are defined as

$$u := F \cdot e_1, \quad v := F \cdot e_2 \quad \text{and} \quad w := F \cdot e_3.$$

Components of the deformation gradient are then

$$\begin{aligned} u_i &= F_{i1}, \\ v_i &= F_{i2}, \\ w_i &= F_{i3}, \end{aligned} \quad \rightsquigarrow \mathbf{F} = \begin{bmatrix} u_1 & v_1 & w_1 \\ u_2 & v_2 & w_2 \\ u_3 & v_3 & w_3 \end{bmatrix}$$

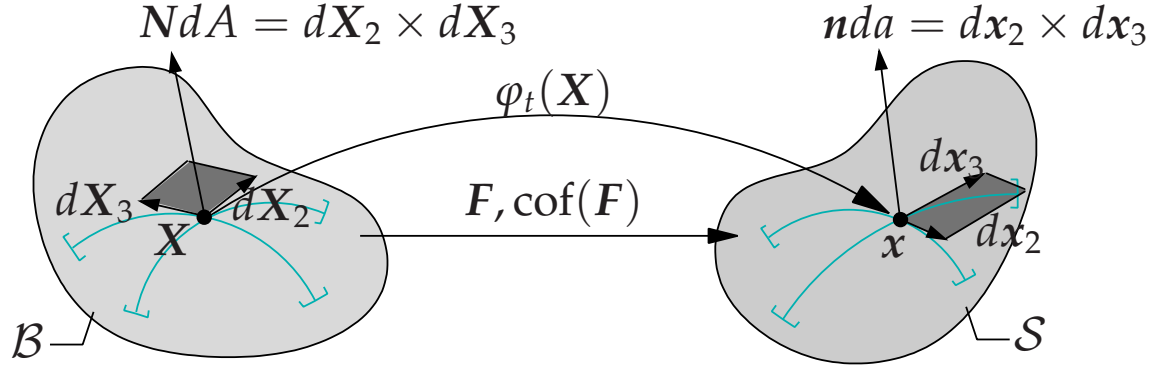
Volume of the deformed parallelepiped is identical to $\det(\mathbf{F})$:

$$\begin{aligned} dv &:= w \cdot (u \times v) = (F \cdot e_1) \cdot [(F \cdot e_2) \times (F \cdot e_3)] \\ &= \det(\mathbf{F}) e_1 \cdot [e_2 \times e_3] \\ &= \det(\mathbf{F}) \end{aligned}$$

5.1.3.3 Adjoint Transformation. Normal Map

The reference and spatial *area normals* are defined as

$$NdA := d\mathbf{X}_2 \times d\mathbf{X}_3 \quad \text{and} \quad nda := d\mathbf{x}_2 \times d\mathbf{x}_3 .$$



With these definitions at hand, we can recast (5.1.17) into the following form

$$d\mathbf{x}_1 \cdot nda = J d\mathbf{X}_1 \cdot NdA . \quad (5.1.19)$$

If we incorporate the identity $d\mathbf{x}_1 = F d\mathbf{X}_1$ in (5.1.19)

$$(F \cdot d\mathbf{X}_1) \cdot nda = d\mathbf{X}_1 \cdot (F^T \cdot nda) = J d\mathbf{X}_1 \cdot NdA ,$$

and solve this equality for nda for an arbitrary tangent vector $d\mathbf{X}_1$, we end up with the *area (normal) map*

$$nda = \text{cof}(F) \cdot NdA \quad \text{with} \quad \text{cof}(F) := J F^{-T} . \quad (5.1.20)$$

It transforms the normals of material surfaces onto the normal vectors of spatial surfaces. Furthermore, we observe that the tensorial quantity carrying out the mapping operation in (5.1.20) is none other than F^{-T} . Thus, we consider F^{-T} as the *normal map* transforming the reference normals N onto the spatial normals (co-vectors) $\mathbf{n}$ belonging to the

respective *co-tangent* spaces $T_X^*\mathcal{B}$ and $T_x^*\mathcal{S}$. The *normal map* is then defined as

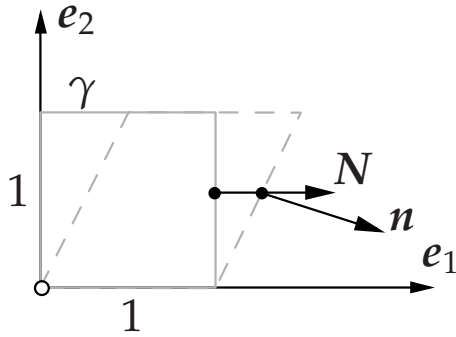
$$F^{-T} := \begin{cases} T_X^*\mathcal{B} \rightarrow T_x^*\mathcal{S}, \\ N \mapsto \boldsymbol{n} = F^{-T} \cdot N. \end{cases} \quad (5.1.21)$$

The *co-factor* of the deformation gradient $\text{cof}(F)$ is also defined as the derivative of the volume map $J := \det F$ with respect to the deformation gradient F

$$\text{cof } F := \partial_F \det F = \det(F) F^{-T}. \quad (5.1.22)$$

Example:

Simple shear



$$\begin{aligned} x_1 &= X_1 + \gamma(t)X_2 \\ x_2 &= X_2 \\ x_3 &= X_3 \end{aligned}$$

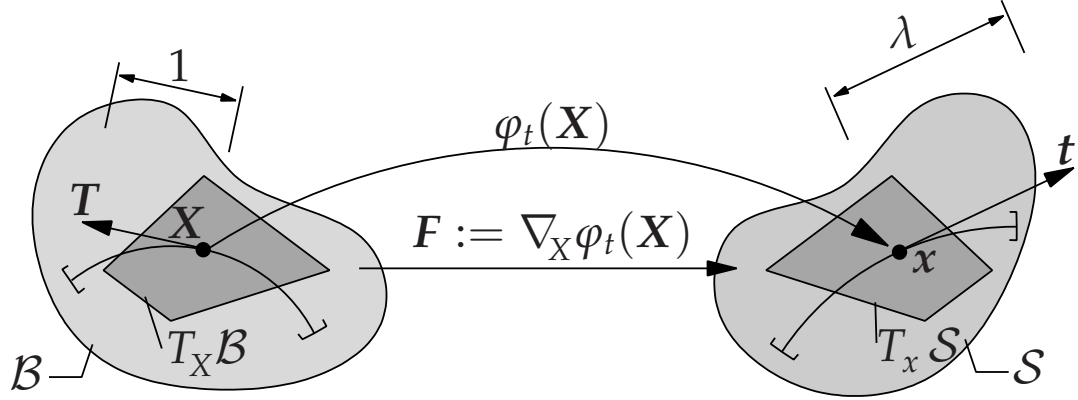
$$F = \begin{bmatrix} 1 & \gamma & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad F^{-T} = \begin{bmatrix} 1 & 0 & 0 \\ -\gamma & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad \begin{aligned} N &= e_1 \\ n &= F^{-T} \cdot N \\ &= e_1 - \gamma e_2 \end{aligned}$$

Observe that the area change of the side surface is given by $\|\boldsymbol{n}\| = \sqrt{1 + \gamma^2}$.

5.1.4 Deformation and Strain Tensors

This section outlines the fundamental deformation and strain tensors that often enter the strain energy functions.

5.1.4.1 Stretch Vector and Stretch



Let $T \in T_X \mathcal{B}$ be a material tangent vector emanating from $X \in \mathcal{B}$. We define the associated *stretch vector* $t \in T_x \mathcal{S}$ as the Gateaux derivative of the deformation $\varphi_t(X)$ in the direction T

$$\begin{aligned} t &:= \left. \frac{d}{d\epsilon} \right|_{\epsilon=0} \varphi_t(X + \epsilon T) \\ &= \left. \frac{d}{d\epsilon} \right|_{\epsilon=0} [\varphi_t(X) + \epsilon F \cdot T + O(\epsilon^2)] \\ &= F \cdot T. \end{aligned} \quad (5.1.23)$$

The ratio of the length of the spatial tangent vector t to the length of the corresponding reference tangent vector T is called the *stretch*

$$\lambda := \frac{\|t\|}{\|T\|} = \frac{\sqrt{t \cdot t}}{\sqrt{T \cdot T}} > 0. \quad (5.1.24)$$

By choosing $\|T\| = 1$ as the *reference* value, the stretch can

be expressed as

$$\begin{aligned}\lambda &= \sqrt{\mathbf{t} \cdot \mathbf{t}} = \sqrt{(\mathbf{F} \cdot \mathbf{T}) \cdot (\mathbf{F} \cdot \mathbf{T})} \\ &= \sqrt{\mathbf{T} \cdot (\mathbf{F}^T \cdot \mathbf{F}) \cdot \mathbf{T}} = \sqrt{\mathbf{T} \cdot \mathbf{C} \cdot \mathbf{T}} =: \|\mathbf{T}\|_C ,\end{aligned}\quad (5.1.25)$$

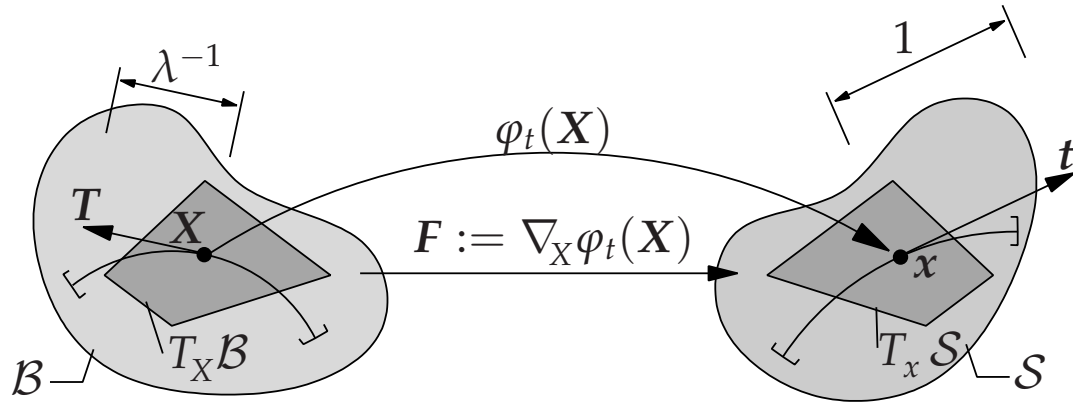
where we already introduced the *right Cauchy-Green tensor*

$$\mathbf{C} := \mathbf{F}^T \cdot \mathbf{F} , \quad C_{AB} = F_{aA} F_{aB} . \quad (5.1.26)$$

It is important to observe that the right Cauchy-Green tensor is symmetric and positive-definite; that is,

$$\mathbf{C} = \mathbf{C}^T = (\mathbf{F}^T \cdot \mathbf{F})^T = \mathbf{F}^T \cdot \mathbf{F} \quad \text{and} \quad \mathbf{T} \cdot \mathbf{C} \cdot \mathbf{T} \geq 0 \quad \forall \mathbf{T} \in \mathcal{R}^3 . \quad (5.1.27)$$

Thus, its eigenvalues $\lambda_{\alpha=1,2,3}^2$ are positive and real numbers. The stretch formulae $\lambda = \|\mathbf{t}\|_1$ and $\lambda = \|\mathbf{T}\|_C$ are often referred to as the Eulerian and Lagrangean descriptions, respectively. Observe that the latter allows us to compute the stretch in terms of material quantities.



Now, let us consider a dual Eulerian (spatial) approach by setting $\|\mathbf{t}\|_1 = 1$ as a reference value. We then express the inverse stretch

$$\begin{aligned}\lambda^{-1} &= \sqrt{\mathbf{T} \cdot \mathbf{T}} = \sqrt{(\mathbf{F}^{-1} \cdot \mathbf{t}) \cdot (\mathbf{F}^{-1} \cdot \mathbf{t})} \\ &= \sqrt{\mathbf{t} \cdot (\mathbf{F}^{-T} \cdot \mathbf{F}^{-1}) \cdot \mathbf{t}} = \sqrt{\mathbf{t} \cdot \mathbf{b}^{-1} \mathbf{t}} =: ||\mathbf{t}||_{\mathbf{b}^{-1}}\end{aligned}\quad (5.1.28)$$

in terms of the *inverse left Cauchy-Green tensor* $\mathbf{b}^{-1}$

$$\mathbf{b}^{-1} := \mathbf{F}^{-T} \cdot \mathbf{F}^{-1}, \quad b_{ab}^{-1} = F_{Aa}^{-1} F_{Ab}^{-1}. \quad (5.1.29)$$

The *left Cauchy-Green tensor* $\mathbf{b}$ is then defined as

$$\mathbf{b} := \mathbf{F} \cdot \mathbf{F}^T, \quad b_{ab} = F_{aA} F_{bA}. \quad (5.1.30)$$

From this definition, we readily note that the left Cauchy-Green (Finger) tensor is also symmetric and positive-definite, i.e.

$$\mathbf{b} = \mathbf{b}^T = (\mathbf{F} \cdot \mathbf{F}^T)^T = \mathbf{F} \cdot \mathbf{F}^T \quad \text{and} \quad \mathbf{t} \cdot \mathbf{b} \cdot \mathbf{t} \geq 0 \quad \forall \mathbf{t} \in \mathcal{R}^3. \quad (5.1.31)$$

In (5.1.25) and (5.1.28), we observe that $\mathbf{C}$ and $\mathbf{b}^{-1}$ act as metric tensors in the respective Lagrangean and Eulerian description of the length deformation.

5.1.4.2 Strain Tensors

Having the stretch defined in (5.1.25), we are now in a position to define the *Green-Lagrange strain tensor*. The Green-Lagrange strain measure ε_{GL} , Lagrangean strain measure, compares the squared lengths of the spatial vector $||\mathbf{t}||_1^2 = ||\mathbf{T}||_C^2$ and the reference vector $||\mathbf{T}||_1^2 = 1$ in an *additive* manner

$$\varepsilon_{GL} := \frac{1}{2} [\lambda^2 - 1]. \quad (5.1.32)$$

Insertion of the above definitions yields

$$\begin{aligned}
 \varepsilon_{GL} &:= \frac{1}{2} [||\mathbf{T}||_C^2 - ||\mathbf{T}||_1^2] \\
 &= \frac{1}{2} [\mathbf{T} \cdot \mathbf{C} \cdot \mathbf{T} - \mathbf{T} \cdot \mathbf{1} \cdot \mathbf{T}] \\
 &= \mathbf{T} \cdot \frac{1}{2} [\mathbf{C} - \mathbf{1}] \cdot \mathbf{T} =: \mathbf{T} \cdot \mathbf{E} \cdot \mathbf{T} .
 \end{aligned} \tag{5.1.33}$$

The Green-Lagrange strain tensor $\mathbf{E}$ is then defined as

$$\mathbf{E} := \frac{1}{2} [\mathbf{C} - \mathbf{1}] , \quad E_{AB} = \frac{1}{2} [C_{AB} - \delta_{AB}] . \tag{5.1.34}$$

Analogous to the dual approach we considered for the inverse stretch, the Eulerian strain measure, the so-called *Almansi strain* measure is defined as

$$\varepsilon_A := \frac{1}{2} [1 - \lambda^{-2}] , \tag{5.1.35}$$

where $||\mathbf{t}||_1^2 = 1$ and $||\mathbf{t}||_{b^{-1}}^2 = \lambda^{-2}$. Incorporating these definitions, we have

$$\begin{aligned}
 \varepsilon_A &:= \frac{1}{2} [||\mathbf{t}||_1^2 - ||\mathbf{t}||_{b^{-1}}^2] \\
 &= \frac{1}{2} [\mathbf{t} \cdot \mathbf{1} \cdot \mathbf{t} - \mathbf{t} \cdot \mathbf{b}^{-1} \cdot \mathbf{t}] \\
 &= \mathbf{t} \cdot \frac{1}{2} [\mathbf{1} - \mathbf{b}^{-1}] \cdot \mathbf{t} =: \mathbf{t} \cdot \mathbf{e} \cdot \mathbf{t} .
 \end{aligned} \tag{5.1.36}$$

with

$$\mathbf{e} := \frac{1}{2} [\mathbf{1} - \mathbf{b}^{-1}] , \quad e_{ab} = \frac{1}{2} [\delta_{ab} - b_{ab}^{-1}] . \tag{5.1.37}$$

denoting the Eulerian *Almansi strain tensor*.

It is important to note that linearization of the both strain tensors, $\mathbf{E}$ and $\mathbf{e}$ about the undeformed state leads to the strain tensor $\varepsilon = \text{sym}(\nabla_x \mathbf{u})$ in the geometrically linear theory.

$$\varepsilon = \text{sym}(\nabla_x \mathbf{u}) = \text{Lin}|_{F=1} \mathbf{E} = \text{Lin}|_{F=1} \mathbf{e} \tag{5.1.38}$$

5.1.5 Material and Spatial Velocity Gradients

Recall that the stretch vector $\mathbf{t} = \mathbf{F} \cdot \mathbf{T} \in T_x \mathcal{S}$ was defined as a tangent map of the material tangent $\mathbf{T} \in T_X \mathcal{B}$. The total time derivative of $\mathbf{t}$ is then given by

$$\dot{\mathbf{t}} = \dot{\mathbf{F}} \cdot \mathbf{T} =: \mathbf{L} \cdot \mathbf{T}, \quad (5.1.39)$$

where the time derivative of the deformation gradient can be expressed as

$$\mathbf{L} := \dot{\mathbf{F}} = \frac{\partial}{\partial t} (\nabla_X \varphi(\mathbf{X}, t)) = \nabla_X \left(\frac{\partial}{\partial t} \varphi(\mathbf{X}, t) \right) = \nabla_X \mathbf{V}(\mathbf{X}, t) \quad (5.1.40)$$

the gradient of the material velocity with respect to the material coordinates. Therefore, we call $\mathbf{L}$ the *material velocity gradient* that maps the material tangent vector $\mathbf{T}$ on the total time derivative of its spatial counter part as shown in (5.1.39).

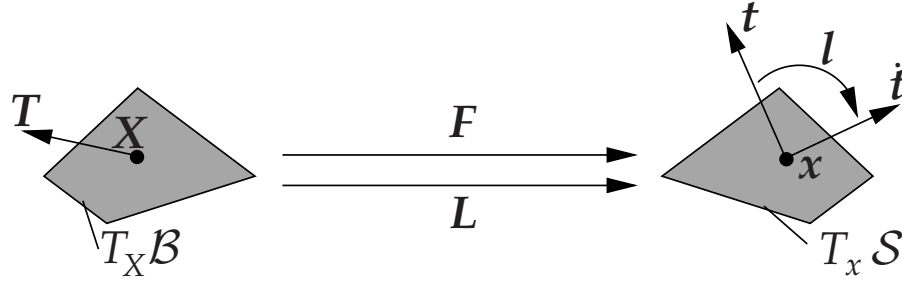
The material vector $\mathbf{T}$ in (5.1.39) can be expressed as $\mathbf{T} = \mathbf{F}^{-1} \cdot \mathbf{t}$. Here, the inverse of the deformation gradient can be obtained through $\mathbf{F}^{-1} = \nabla_x \mathbf{X}$. Then, the rate of the stretch vector assumes the form

$$\dot{\mathbf{t}} = (\dot{\mathbf{F}} \cdot \mathbf{F}^{-1}) \cdot \mathbf{t} =: \mathbf{l} \cdot \mathbf{t} \quad (5.1.41)$$

where we introduced the *spatial velocity gradient*

$$\mathbf{l} := \mathbf{L} \cdot \mathbf{F}^{-1} = \nabla_X \mathbf{V}(\mathbf{X}, t) \cdot \nabla_x \mathbf{X} = \nabla_x \mathbf{v}(\mathbf{x}, t) \quad (5.1.42)$$

by using the relation $\mathbf{v}(\mathbf{x}, t) = \mathbf{V}(\mathbf{X}, t) \circ \varphi_t^{-1}(\mathbf{X})$.



Furthermore, we note that the material and spatial velocity gradients can be conceived as the maps

$$L(\mathbf{X}, t) : \begin{cases} T_{\mathbf{X}}\mathcal{B} \rightarrow T_x\mathcal{S} , \\ T \mapsto \dot{\mathbf{t}} = L \cdot T , \end{cases} \quad (5.1.43)$$

and

$$l(\mathbf{x}, t) : \begin{cases} T_x\mathcal{S} \rightarrow T_x\mathcal{S} , \\ \mathbf{t} \mapsto \dot{\mathbf{t}} = l \cdot \mathbf{t} . \end{cases} \quad (5.1.44)$$

5.1.6 Rate of Deformation and Spin Tensors

Having the spatial velocity gradient l defined, we are now in a position to define the *rate of deformation tensor* d and the *spin tensor* w by additively splitting

$$l = d + w \quad (5.1.45)$$

into its symmetric and skew-symmetric parts. That is, the *rate of deformation tensor* is defined by

$$d := \text{sym}(l) = \frac{1}{2} (l + l^T) , \quad (5.1.46)$$

and

$$w := \text{skew}(l) = \frac{1}{2} (l - l^T) , \quad (5.1.47)$$

denotes the *spin tensor*. Therefore the map (5.1.41) can also be additively decomposed into

$$\dot{\mathbf{t}} = \mathbf{l} \cdot \mathbf{t} = \mathbf{d} \cdot \mathbf{t} + \mathbf{w} \cdot \mathbf{t} , \quad (5.1.48)$$

in which the part $\mathbf{d} \cdot \mathbf{t}$ related with the rate of stretching and the part $\mathbf{w} \cdot \mathbf{t}$ associated with rotation.

In order to interpret the geometrical meaning of the rate of deformation tensor, we consider the time rate of square of the stretch by following its material description in (5.1.25)

$$\frac{d}{dt}(\lambda^2) = 2\lambda\dot{\lambda} = \mathbf{T} \cdot \dot{\mathbf{C}} \cdot \mathbf{T} . \quad (5.1.49)$$

Since $\mathbf{C} = \mathbf{F}^T \cdot \mathbf{F}$ and $\mathbf{t} = \mathbf{F} \cdot \mathbf{T}$, we have $\dot{\mathbf{C}} = \dot{\mathbf{F}}^T \cdot \mathbf{F} + \mathbf{F}^T \cdot \dot{\mathbf{F}}$ and $\mathbf{T} = \mathbf{F}^{-1} \cdot \mathbf{t}$. Insertion of these results leads us to

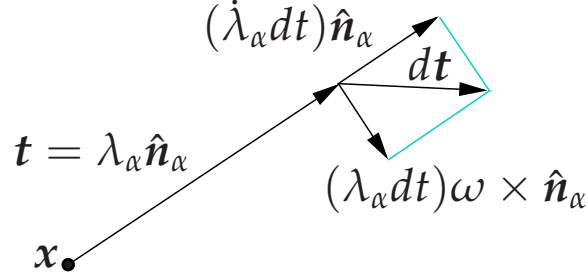
$$\begin{aligned} 2\lambda\dot{\lambda} &= \mathbf{T} \cdot \dot{\mathbf{C}} \cdot \mathbf{T} \\ &= (\mathbf{F}^{-1} \cdot \mathbf{t}) \cdot (\dot{\mathbf{F}}^T \cdot \mathbf{F} + \mathbf{F}^T \cdot \dot{\mathbf{F}}) \cdot (\mathbf{F}^{-1} \cdot \mathbf{t}) \\ &= \mathbf{t} \cdot (\mathbf{F}^{-T} \cdot \dot{\mathbf{F}}^T + \dot{\mathbf{F}} \cdot \mathbf{F}^{-1}) \cdot \mathbf{t} \\ &= \mathbf{t} \cdot (\mathbf{l}^T + \mathbf{l}) \cdot \mathbf{t} = \mathbf{t} \cdot 2\mathbf{d} \cdot \mathbf{t} \end{aligned} \quad (5.1.50)$$

Introducing a unit vector $\mathbf{g}$, i.e. $\|\mathbf{g}\|=1$, parallel to $\mathbf{t} = \lambda\mathbf{g}$, we end up with the time derivative of the *logarithmic strain* $\varepsilon_{\ln} := \ln(\lambda)$ in the direction $\mathbf{g}$

$$\dot{\varepsilon}_{\ln} = \overline{\ln \lambda} = \dot{\lambda}/\lambda = \mathbf{d} : (\mathbf{g} \otimes \mathbf{g}) = \mathbf{g} \cdot \mathbf{d} \cdot \mathbf{g}$$

identical to the component $d_{gg} = \mathbf{g} \cdot \mathbf{d} \cdot \mathbf{g}$ of the rate of deformation tensor. This means, if we had $\mathbf{g} = \hat{\mathbf{n}}_\alpha$ with $\hat{\mathbf{n}}_\alpha$ de-

noting an eigenvector of $\mathbf{d} = \sum_{\alpha=1}^3 d_{\alpha} \hat{\mathbf{n}}_{\alpha} \otimes \hat{\mathbf{n}}_{\alpha}$ then the corresponding eigenvalue d_{α} would be identical to rate of the logarithmic strain in that principal direction, i.e. $d_{\alpha} = \dot{\lambda}_{\alpha} / \lambda_{\alpha}$.



We can now exploit this result to elucidate the geometrical meaning of the spin tensor $\mathbf{w}$. For this purpose, we consider a spatial vector $\mathbf{t} = \lambda_{\alpha} \hat{\mathbf{n}}_{\alpha}$ chosen to be parallel to a principal (eigen) vector of $\mathbf{d}$. Its time derivative can then be written as

$$\dot{\mathbf{t}} = \dot{\lambda}_{\alpha} \hat{\mathbf{n}}_{\alpha} + \lambda_{\alpha} \dot{\hat{\mathbf{n}}}_{\alpha}. \quad (5.1.51)$$

From (5.1.48), we also know that

$$\dot{\mathbf{t}} = \mathbf{l} \cdot \mathbf{t} = \lambda_{\alpha} \mathbf{d} \cdot \hat{\mathbf{n}}_{\alpha} + \lambda_{\alpha} \mathbf{w} \cdot \hat{\mathbf{n}}_{\alpha} = \dot{\lambda}_{\alpha} \hat{\mathbf{n}}_{\alpha} + \lambda_{\alpha} \mathbf{w} \cdot \hat{\mathbf{n}}_{\alpha} \quad (5.1.52)$$

where we incorporated the result $\mathbf{d} \cdot \hat{\mathbf{n}}_{\alpha} = d_{\alpha} \hat{\mathbf{n}}_{\alpha} = \dot{\lambda}_{\alpha} / \lambda_{\alpha} \hat{\mathbf{n}}_{\alpha}$. Comparison of the expressions in (5.1.51) and (5.1.52) yields the identity

$$\dot{\hat{\mathbf{n}}}_{\alpha} = \mathbf{w} \cdot \hat{\mathbf{n}}_{\alpha} = \boldsymbol{\omega} \times \hat{\mathbf{n}}_{\alpha} \quad (5.1.53)$$

where $\boldsymbol{\omega}$ stands for the axial vector of $\mathbf{w}$. From this result, it can be concluded that the spin tensor $\mathbf{w}$ governs the infinitesimal rotation of the eigenvectors of the rate of deformation tensor. Combining the both results for this particular vector $\mathbf{t} = \lambda_{\alpha} \hat{\mathbf{n}}_{\alpha}$, we end up with

$$\dot{\mathbf{t}} = \dot{\lambda}_{\alpha} \hat{\mathbf{n}}_{\alpha} + \lambda_{\alpha} \boldsymbol{\omega} \times \hat{\mathbf{n}}_{\alpha}. \quad (5.1.54)$$

5.1.7 Push-Forward and Pull-Back Relations

Lagrangian and Eulerian fields are related through the *push-forward* and *pull-back* operations. For the rate of deformation tensors, for example, we observe in (5.1.50) that

$$\mathbf{d} = \mathbf{F}^{-T} \cdot \frac{1}{2} \dot{\mathbf{C}} \cdot \mathbf{F}^{-1} \quad (5.1.55)$$

holds. We then say that the rate of deformation tensor $\mathbf{d}$ is the *push-forward* of the time derivative of the Green-Lagrange strain tensor $\dot{\mathbf{E}} \equiv \frac{1}{2} \dot{\mathbf{C}}$ and denote the push-forward operation

$$\mathbf{d} = \varphi_*(\dot{\mathbf{E}}) = \varphi_*\left(\frac{1}{2} \dot{\mathbf{C}}\right) := \mathbf{F}^{-T} \cdot \left(\frac{1}{2} \dot{\mathbf{C}}\right) \cdot \mathbf{F}^{-1}, \quad (5.1.56)$$

or in indicial notation

$$d_{ab} = F_{Aa}^{-1} \frac{1}{2} \dot{C}_{AB} F_{Bb}^{-1}.$$

Analogously, we have the identity

$$\frac{1}{2} \dot{\mathbf{C}} = \dot{\mathbf{E}} = \mathbf{F}^T \cdot \mathbf{d} \cdot \mathbf{F} \quad (5.1.57)$$

where $\frac{1}{2} \dot{\mathbf{C}} = \dot{\mathbf{E}}$ is the *pull-back* of the rate of deformation tensor $\mathbf{d}$

$$\dot{\mathbf{E}} = \frac{1}{2} \dot{\mathbf{C}} = \varphi^*(\mathbf{d}) = \mathbf{F}^T \cdot \mathbf{d} \cdot \mathbf{F}, \quad (5.1.58)$$

or in indicial notation

$$\dot{E}_{AB} = \frac{1}{2} \dot{C}_{AB} = F_{aA} d_{ab} F_{bB}.$$

The Almansi strain tensor and the Green-Lagrange strain tensor are also related through the pull-back and push-forward maps, i.e. $\mathbf{E} = \mathbf{F}^T \cdot \mathbf{e} \cdot \mathbf{F}$. The Almansi strain tensor $\mathbf{e}$ is the *push-forward* of the Green-Lagrange strain tensor $\mathbf{E}$

$$\mathbf{e} = \varphi_*(\mathbf{E}) := \mathbf{F}^{-T} \cdot \mathbf{E} \cdot \mathbf{F}^{-1}, \quad e_{ab} = F_{Aa}^{-1} E_{AB} F_{Bb}^{-1}. \quad (5.1.59)$$

Analogously, E is the *pull-back* of the Almansi strain tensor e

$$E = \varphi^*(e) = F^T \cdot e \cdot F, \quad E_{AB} = F_{aA} e_{ab} F_{bB}. \quad (5.1.60)$$

Similar pull-back and push-forward relations exist also among the different stress measures that will be discussed in the next chapter.

5.1.8 Lie Derivative of Spatial Fields

The Lie derivative of a spatial field, say $f(x, t)$, is defined as

$$\mathcal{L}_v f(x, t) := \varphi_* \left\{ \frac{d}{dt} [\varphi^*(f(x, t))] \right\}, \quad (5.1.61)$$

which is none other than the push-forward of the time derivative of the pull-back of the spatial field $f(x, t)$.

For instance, it can be readily shown that the rate of deformation tensor d is the Lie derivative of the Almansi strain tensor e . Since we have shown that the rate of deformation tensor d is push-forward of $\dot{E}$, $d = \varphi_*(\dot{E})$ and the Green-Lagrange strain tensor is pull-back of the Almansi strain tensor $E = \varphi^*(e)$, we have

$$\mathcal{L}_v e := \varphi_* \left\{ \frac{d}{dt} [\varphi^*(e)] \right\} = \varphi_* \left\{ \frac{d}{dt} [E] \right\} = \varphi_*(\dot{E}) = d. \quad (5.1.62)$$

The Lie (Oldroyd) derivatives constitute the basic class of objective time derivatives of Eulerian objects.

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CONTINUUM MECHANICS

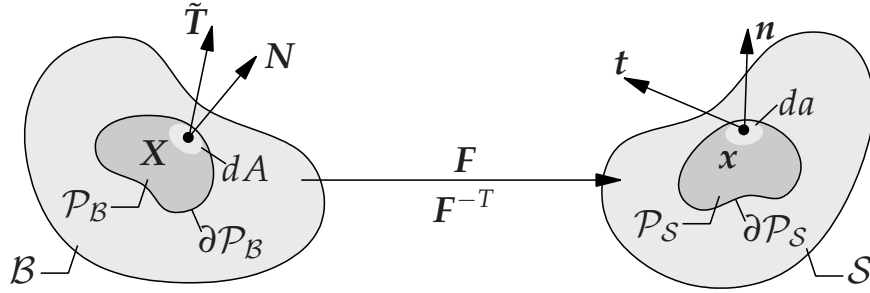
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5.2 Constitutive Theory

5.2.1 Cauchy's Stress Theorem and Fundamental Stress Measures

Consider a part $\mathcal{P}_B \subset \mathcal{B}$ cut off from the reference body $\mathcal{B}$ and its spatial counterpart $\mathcal{P}_S \subset \mathcal{S}$ closed by the respective boundaries $\partial\mathcal{P}_B$ and $\partial\mathcal{P}_S$.



In the deformed configuration, we introduce the *stress vector* $\mathbf{t}$ that acts on the surface element da of $\partial\mathcal{P}_S$ and represents the force action of the rest of the body at the vicinity $\mathcal{P}_S \setminus \mathcal{S}$ on $\partial\mathcal{P}_S$. The *Cauchy stress theorem* states that the spatial traction vector $\mathbf{t} \in T_x\mathcal{S}$ linearly depends on the spatial normal $\mathbf{n} \in T_x^*\mathcal{S}$ of the surface $\partial\mathcal{P}_S$, i.e.

$$\mathbf{t}(x, t; \mathbf{n}) := \boldsymbol{\sigma}(x, t) \cdot \mathbf{n} , \quad (5.2.1)$$

through the *Cauchy (true) stress tensor* $\boldsymbol{\sigma}$. Cauchy's stress theorem can be proven based on the force equilibrium on a tetrahedron. In the geometrical framework outlined so far, the Cauchy stress tensor can be understood as a contravariant mapping transforming normals $\mathbf{n} \in T_x^*\mathcal{S}$ onto tangent vectors $\mathbf{t} \in T_x\mathcal{S}$

$$\boldsymbol{\sigma} := \begin{cases} T_x^*\mathcal{S} \rightarrow T_x\mathcal{S} , \\ \mathbf{n} \mapsto \mathbf{t} = \boldsymbol{\sigma} \cdot \mathbf{n} . \end{cases} \quad (5.2.2)$$

Another spatial stress measure, the *Kirchhoff stress tensor*, also known as the *weighted Cauchy stress tensor*, is defined as

$$\boldsymbol{\tau} := J\boldsymbol{\sigma} \quad (5.2.3)$$

and widely used in the spatial description of stress power terms in the reference volume. Owing to the scalar scaling by the Jacobian J , the Kirchhoff stresses retain the geometrical transformation characteristics of the Cauchy stress.

Now let us consider another spatial traction vector $\boldsymbol{T} \in T_x \mathcal{S}$ defined through the force equality $\boldsymbol{T} dA := \boldsymbol{t} da$ by scaling the spatial force term $(\boldsymbol{t} da)$ through the reference area element dA . Based on this definition, we introduce the *first Piola-Kirchhoff stress tensor* by the reference Cauchy theorem $\boldsymbol{T} := \boldsymbol{P} \cdot \boldsymbol{N}$ leading to $\boldsymbol{P} \cdot \boldsymbol{N} dA = \boldsymbol{\sigma} \cdot \boldsymbol{n} da$. Using the area map $\boldsymbol{n} da = J\boldsymbol{F}^{-T} \cdot \boldsymbol{N} dA$, we obtain the relation $\boldsymbol{P} = \boldsymbol{\tau} \cdot \boldsymbol{F}^{-T}$ between the first Piola-Kirchhoff stress tensor and the spatial stress measures. Notice that $\boldsymbol{P}$ is a two-point tensor possessing the geometrical mapping properties

$$\boldsymbol{P} := \begin{cases} T_X^* \mathcal{B} \rightarrow T_x \mathcal{S}, \\ \boldsymbol{N} \mapsto \boldsymbol{T} = \boldsymbol{P} \cdot \boldsymbol{N}. \end{cases} \quad (5.2.4)$$

The transformation $(\blacksquare) = J(\bullet)\boldsymbol{F}^{-T}$ devised in obtaining the first Piola-Kirchhoff stress tensor from the Cauchy stress tensor is called the *Piola transformation*. It is widely employed in transforming the objects acting on a *spatial surface* onto their material counterparts. The immediate outcome of the Piola transformation is the *Piola Identity*

$$J \operatorname{div}(\bullet) = \operatorname{Div}(\blacksquare) = \operatorname{Div}(J(\bullet)\boldsymbol{F}^{-T}) \quad (5.2.5)$$

that also implies the equality $\operatorname{Div}(J\boldsymbol{F}^{-T}) = \mathbf{0}$.

The third fundamental stress measure, the *second Piola-Kirchhoff stress tensor* S , is then defined by $\tilde{T} := S \cdot N$ yielding

$$S := \begin{cases} T_X^* \mathcal{B} \rightarrow T_X \mathcal{B} , \\ N \mapsto \tilde{T} = S \cdot N . \end{cases} \quad (5.2.6)$$

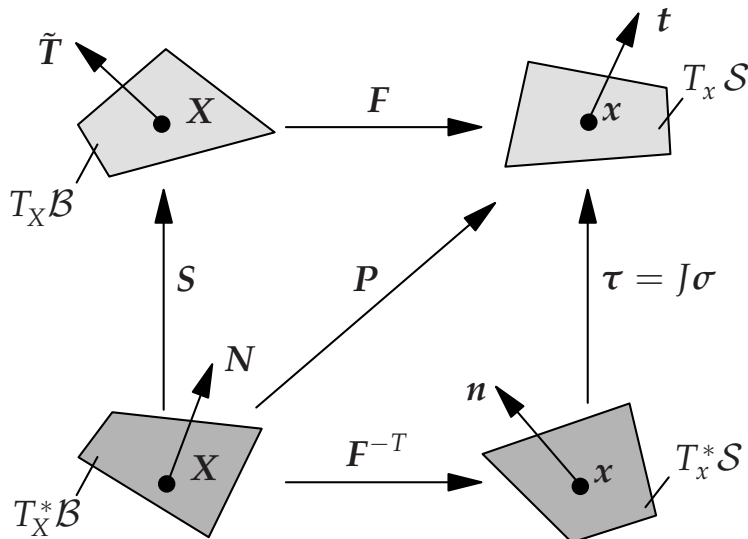
Incorporating the definitions (5.2.2)–(5.2.4) in (5.2.6), we can express the second Piola-Kirchhoff stress tensor in terms of the other stress tensors

$$S := \varphi^*(P) = F^{-1} \cdot P \quad , \quad S_{AB} = (F^{-1})_{Aa} P_{aB} ,$$

$$S := \varphi^*(\tau) = F^{-1} \cdot \tau \cdot F^{-T} \quad , \quad S_{AB} = (F^{-1})_{Aa} \tau_{ab} (F^{-1})_{Bb}$$

as the pull-back of the contravariant two-point and spatial stress tensors. Apparently the reverse push-forward relations do also hold for the spatial stress tensors

$$\tau = J\sigma = \varphi_*(P) = P \cdot F^T \quad \text{and} \quad \tau = \varphi_*(S) = F \cdot S \cdot F^T ,$$



5.2.2 Closure Problem

The fundamental balance equations are summarized as

Balance of	Spatial	Material
Mass	$\dot{\rho}_t + \rho_t \operatorname{div}(\mathbf{v}) = 0$	$\dot{\rho}_0 = 0$
Linear M.	$\rho_t \mathbf{a} = \operatorname{div}(\boldsymbol{\sigma}) + \rho_t \mathbf{b}$	$\rho_0 \mathbf{a} = \operatorname{Div}(\mathbf{P}) + \rho_0 \mathbf{b}$
Angular M.	$\boldsymbol{\sigma} = \boldsymbol{\sigma}^T, \boldsymbol{\tau} = \boldsymbol{\tau}^T$ $\mathbf{P} \cdot \mathbf{F}^T = \mathbf{F} \cdot \mathbf{P}^T$	$\mathbf{S} = \mathbf{S}^T$ $\mathbf{F}^{-1} \cdot \mathbf{P} = \mathbf{P}^T \cdot \mathbf{F}^{-T}$
Energy	$\rho_t \frac{di}{dt} = \boldsymbol{\sigma} : \mathbf{d} - \operatorname{div} \mathbf{q} + R_t$	$\rho_0 \frac{di}{dt} = \mathbf{P} : \dot{\mathbf{F}} - \operatorname{Div} \mathbf{Q} + R_0$

where $\rho_0 = J\rho_t$, $R_0 = JR_t$ and $\mathbf{Q} = J\mathbf{q} \cdot \mathbf{F}^{-T}$.

The number of equations provided by these balance laws and the number of unknowns can be identified as

Balance of	#	Unknown	#
Mass	1	density ρ_t	1
Linear M.	3	Placement $\varphi_t(\mathbf{X})$	3
Angular M.	3	Stresses $\boldsymbol{\sigma}$	9
Energy	1	Temperature θ	1
		Heat Flux $\mathbf{q}$	3
Σ	8	Σ	17

where the volume-specific body forces $\mathbf{b}$ and the heat source R_t are assumed to be given. Comparison of the number of unknowns with the number of equations indicates that nine additional equations are needed to solve the problem. These additional equations are called the *constitutive equations* which describe the stresses $\hat{\boldsymbol{\sigma}} = \hat{\boldsymbol{\sigma}}^T$ (six equations), and the heat flux $\hat{\mathbf{q}}$ (three equations).

In common practice the placement $\varphi_t(\mathbf{X})$ and the absolute temperature θ are considered as the primary unknowns that are solved from the balance of linear momentum and the balance of energy equations.

It is important to notice that the *stress power* $\mathcal{W} := \mathbf{P} : \dot{\mathbf{F}}$ per unit reference volume appearing in the balance of energy equation can also be expressed in terms of other stress measures by using the push-forward and pull-back relations among the fundamental stress tensors and the rate of deformation tensors.

$$\mathcal{W} := \mathbf{P} : \dot{\mathbf{F}} = \boldsymbol{\tau} : \mathbf{d} = \mathbf{S} : \dot{\mathbf{E}}.$$

The alternative representations of the reference stress power $\mathcal{W}$ manifest the distinct work conjugate couples

$$(\mathbf{P}; \dot{\mathbf{F}}) \quad , \quad (\boldsymbol{\tau}; \mathbf{d}) \quad , \quad (\mathbf{S}; \dot{\mathbf{E}}).$$

5.2.3 Methodology of Coleman and Gurtin

Let us focus on a problem of thermoelasticity for a homogeneous material, for which the local dissipation $\mathcal{D}_{loc}$ vanishes identically. Being consistent with the *principle of locality*, we assume that the free energy ψ depend upon field variables defined in the neighborhood of the material point $\mathbf{X}$

$$\psi = \hat{\psi}(\mathbf{F}, \theta, \boldsymbol{\mathcal{G}}) \tag{5.2.7}$$

where $\boldsymbol{\mathcal{G}} := \nabla_{\mathbf{X}}\theta$ denotes the material gradient of the temperature field. Based on this assumption, we can substitute the time derivative of the free energy

$$\dot{\psi} = \partial_{\mathbf{F}}\psi : \dot{\mathbf{F}} + \partial_{\theta}\psi : \dot{\theta} + \partial_{\boldsymbol{\mathcal{G}}}\psi : \dot{\boldsymbol{\mathcal{G}}}$$

in the Clausius-Planck inequality

$$J\mathcal{D}_{loc} := (\mathbf{P} - \partial_F \psi) : \dot{\mathbf{F}} - (\eta + \partial_\theta \psi) \dot{\theta} - \partial_{\mathcal{G}} \psi \cdot \dot{\mathcal{G}} = 0. \quad (5.2.8)$$

Following the celebrated reasoning of Coleman & Noll (1963) and Coleman & Gurtin (1967) within the framework of thermodynamics with internal variables, we contend that the thermodynamic restriction should be fulfilled for an arbitrary rate of the deformation gradient, temperature and temperature gradient. Therefore, (5.2.8) implies a particular form of constitutive equations such that

$$\mathbf{P} := \partial_F \psi, \quad \eta := -\partial_\theta \psi \quad \text{and} \quad \partial_{\mathcal{G}} \psi = \mathbf{0}. \quad (5.2.9)$$

The first two equations of (5.2.9) state that the free energy acts as a potential for the stresses and the entropy while (5.2.9)₃ implies that the free energy does not depend on the temperature gradient $\mathcal{G}$, i.e. $\psi = \hat{\psi}(F, \theta)$.

5.2.4 Principle of Material Frame Invariance

We consider a rigid body motion $\tilde{\zeta}(\mathbf{x}, t) : \mathcal{S} \times \mathcal{R}_+ \longrightarrow \mathcal{R}^3$ superimposed onto a non-linear motion $\varphi_t(\mathbf{X})$, i.e.

$$\tilde{\zeta}(\mathbf{x}, t) := \mathbf{c}(t) + \mathbf{Q}(t) \cdot \mathbf{x}$$

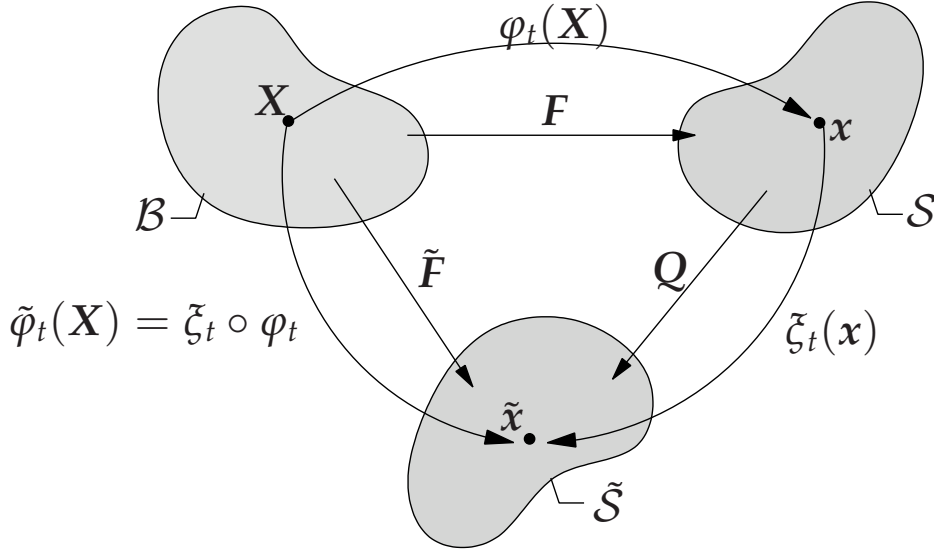
where $\mathbf{Q}$ is a proper orthogonal tensor. This definition yields

$$\tilde{\varphi}(\mathbf{X}, t) := \tilde{\zeta} \circ \varphi = \mathbf{c}(t) + \mathbf{Q}(t) \cdot \varphi(\mathbf{X}, t).$$

Under this superimposed motion, we observe the following transformation relations for:

• Fundamental Maps:

$$\tilde{\mathbf{F}} := \mathbf{Q} \cdot \mathbf{F}, \quad \tilde{\mathbf{F}}^{-T} = \mathbf{Q} \cdot \mathbf{F}^{-T}, \quad \tilde{\mathbf{J}} = \mathbf{J}$$



• **Lagrangian Strain Measures:**

$$\tilde{C} = \tilde{F}^T \cdot \tilde{F} = F^T \cdot Q^T \cdot Q \cdot F = C$$

• **Eulerian Strain Measures:**

$$\tilde{b} = \tilde{F} \cdot \tilde{F}^T = Q \cdot F^T \cdot F \cdot Q^T = Q \cdot b \cdot Q^T$$

$$\tilde{c} = \tilde{b}^{-1} = Q \cdot c \cdot Q^T$$

We call these type of transformation operations *objective transformation of second order tensors*. It is important to note that there also exist spatial tensors that do not fulfill the objective transformation rules, such as the spatial velocity v , the spatial velocity gradient l , to name a few.

One of the fundamental principles of continuum mechanics is the *principle of material frame invariance* that requires the invariance of the energy stored under rigid body rotations superimposed on the current spatial configuration.

Therefore, we locally demand

$$\psi(F, \theta) = \psi(\tilde{F}, \theta) \quad (5.2.10)$$

Since the right Cauchy-Green tensor $\mathbf{C} := \mathbf{F}^T \cdot \mathbf{F} = \tilde{\mathbf{F}}^T \cdot \tilde{\mathbf{F}}$ satisfies this condition, a storage function $\hat{\psi}$ constructed in terms of $\mathbf{C} = \mathbf{F}^T \cdot \mathbf{F}$ is *a priori* objective and the form $\hat{\psi}(\mathbf{C}, \theta) = \hat{\psi}(\mathbf{F}^T \mathbf{F}, \theta)$ represents its reduced form.

$$\hat{\psi}(\mathbf{C}, \theta) = \hat{\psi}(\mathbf{F}^T \cdot \mathbf{F}, \theta). \quad (5.2.11)$$

Based on this restriction, we can rewrite the term

$$(\mathbf{P} - \partial_{\mathbf{F}}\psi) : \dot{\mathbf{F}} \quad \text{as} \quad (\mathbf{S} - 2\partial_{\mathbf{C}}\hat{\psi}) : \frac{1}{2}\dot{\mathbf{C}}$$

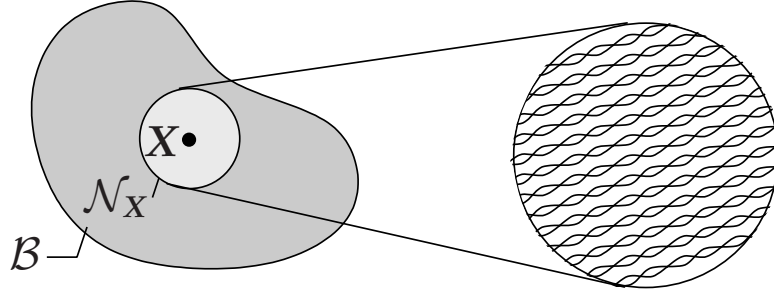
due to the equivalent stress power expressions. This leads us to the functional definition of the second Piola-Kirchhoff stress tensor

$$\mathbf{S} = 2\partial_{\mathbf{C}}\hat{\psi}(\mathbf{C}, \theta). \quad (5.2.12)$$

5.2.5 Concept of Material Symmetry

When a material has a particular micro-structure that has to be taken into account in constitutive equations. As we discussed before, in the context continuum modeling of materials, the scale difference between the micro and macro levels is so large that the specific discrete micro-structure can be incorporated in a smeared way. The micro-structure might have an *anisotropic (direction-dependent)* character, which is observed, for example, in crystals, nano-composites, biological tissues to name a few. *Isotropic* material response, however, corresponds to the case where the micro-structure has no direction-dependent pattern.

A particular micro-structure can be classified through *symmetry groups* of rotations, which do not disturb the pattern

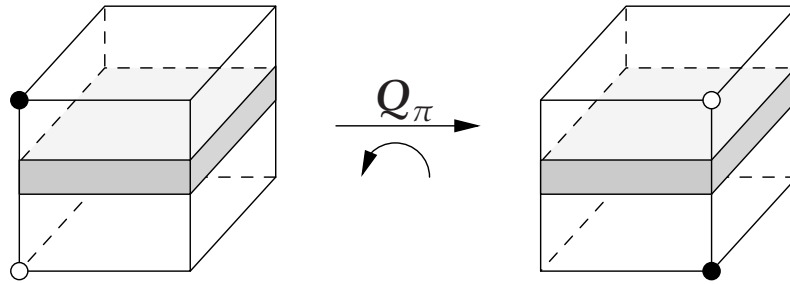


of the micro-structure under consideration. Elementary examples for the symmetry groups are *transverse isotropy* and *orthotropy*. A material symmetry group G is then defined in terms of rotations preserving the material micro-structure

$$G := \{Q_1, Q_2, \dots\} \subset \mathcal{SO}(3) \quad (5.2.13)$$

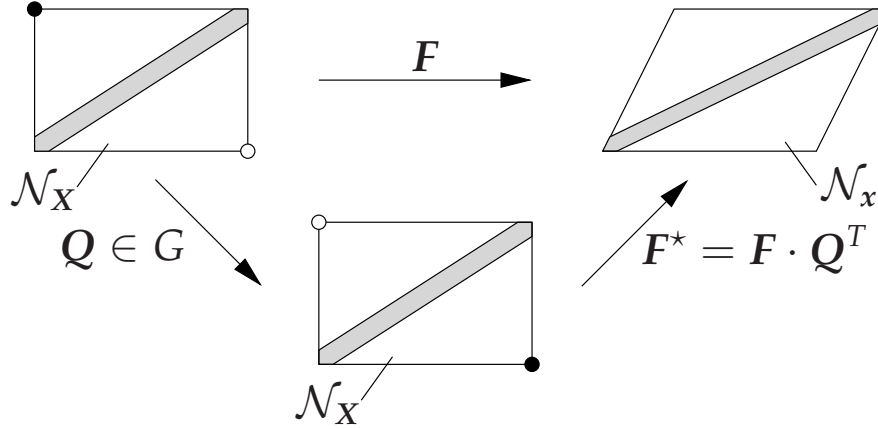
where $\mathcal{SO}(3)$ denotes the *special orthogonal group* defined by

$$\mathcal{SO}(3) := \{Q | Q \cdot Q^T = \mathbf{1} \wedge \det(Q) = 1\} \quad (5.2.14)$$



Since the rotations belonging to the symmetry group G do not alter the material architecture, we require the free energy must also be invariant with respect to these rotations superimposed on the reference neighborhood of the material $\mathcal{N}_X$.

$$\hat{\psi}(F \cdot Q^T) = \hat{\psi}(F) \quad \forall Q \in G \subset \mathcal{SO}(3). \quad (5.2.15)$$



5.2.6 Isotropic Hyperelasticity

Isotropic materials do not possess a distinct micro-structure. Therefore, the energy storage function, free energy, must be invariant with respect to all rotations. The symmetry group of an isotropic material is the full special orthogonal group

$$G_{iso} \equiv \mathcal{SO}(3) . \quad (5.2.16)$$

This property leads us to the simple representation of isotropic free energy functions in terms of principal stretches or invariants. To show this, we consider the spectral representation of the deformation gradient $F = \sum_{\alpha=1}^3 \lambda_{\alpha} \mathbf{n}_{\alpha} \otimes \mathbf{N}_{\alpha}$ in the free energy function

$$\begin{aligned} \psi(F) &= \hat{\psi}(\sum_{\alpha=1}^3 \lambda_{\alpha} \mathbf{n}_{\alpha} \otimes \mathbf{N}_{\alpha}) \\ \psi(F \cdot Q^T) &= \hat{\psi}(\sum_{\alpha=1}^3 \lambda_{\alpha} \mathbf{n}_{\alpha} \otimes Q \cdot \mathbf{N}_{\alpha}) \end{aligned} \quad (5.2.17)$$

and we require the equality

$$\hat{\psi}(\sum_{\alpha=1}^3 \lambda_{\alpha} \mathbf{n}_{\alpha} \otimes \mathbf{N}_{\alpha}) = \hat{\psi}(\sum_{\alpha=1}^3 \lambda_{\alpha} \mathbf{n}_{\alpha} \otimes Q \cdot \mathbf{N}_{\alpha}) \quad \forall Q \in \mathcal{SO}(3) . \quad (5.2.18)$$

with the Lagrangean eigenvectors N_α mapped to an arbitrary vector $\mathbf{Q} \cdot N_\alpha$. This implies that the free energy $\hat{\psi}$ cannot depend upon N_α . We also recall that the principle of objectivity requires

$$\begin{aligned} \hat{\psi}(\mathbf{Q} \cdot \mathbf{F}) &= \hat{\psi}(\mathbf{F}) , \\ \hat{\psi}(\sum_{\alpha=1}^3 \lambda_\alpha \mathbf{Q} \cdot \mathbf{n}_\alpha \otimes \mathbf{N}_\alpha) &= \hat{\psi}(\sum_{\alpha=1}^3 \lambda_\alpha \mathbf{n}_\alpha \otimes \cdot \mathbf{N}_\alpha) \end{aligned} \quad (5.2.19)$$

for all $\mathbf{Q} \in \mathcal{SO}(3)$. In addition to the material symmetry condition, the objectivity principle excludes the functional dependency of the free energy is dependent on the Eulerian eigenvectors $\mathbf{n}_\alpha$. Combination of the conclusions we drawn from (5.2.18) and (5.2.19) leads us to simple representation of the free energy of isotropic materials

$$\psi = \bar{\psi}(\lambda_1, \lambda_2, \lambda_3) \quad (5.2.20)$$

Since the principal invariants of the right and left Cauchy Green tensors $\mathbf{C}$ and $\mathbf{b}$ are symmetric functions of the principal stretches, i.e.

$$\begin{aligned} I_1(\mathbf{C}) = I_1(\mathbf{b}) &= \lambda_1^2 + \lambda_2^2 + \lambda_3^2 , \\ I_2(\mathbf{C}) = I_2(\mathbf{b}) &= \lambda_1^2 \lambda_2^2 + \lambda_2^2 \lambda_3^2 + \lambda_1^2 \lambda_3^2 , \\ I_3(\mathbf{C}) = I_3(\mathbf{b}) &= \lambda_1^2 \lambda_2^2 \lambda_3^2 , \end{aligned} \quad (5.2.21)$$

an alternative representation of isotropic elasticity is obtained in terms of the principal invariants

$$\psi = \tilde{\psi}(I_1, I_2, I_3) . \quad (5.2.22)$$

At this point, it is worth noting that the left Cauchy Green tensor remains unaltered under arbitrary rotations of the reference neighborhood $\mathcal{N}_X$

$$\mathbf{b}^* = \mathbf{F} \cdot \mathbf{Q}^T \cdot \mathbf{Q} \cdot \mathbf{F}^T = \mathbf{F} \cdot \mathbf{F}^T = \mathbf{b} .$$

This suggests that a free energy of hyperelasticity depending solely on $\mathbf{b}$, i.e. $\hat{\psi}(\mathbf{b})$, is restricted to the isotropic elastic material response.

5.2.6.1 Compressible Neo-Hookean Material

A version of compressible Neo-Hookean material model of hyperelasticity is described by the following free energy

$$\psi = \hat{\psi}(I_1, J) = \frac{\Lambda}{4}(J^2 - 1) - \left(\mu + \frac{\Lambda}{2}\right) \ln J + \frac{\mu}{2}(I_1 - 3)$$

where Λ and μ are the material parameters, $I_1 := \mathbf{C} : \mathbf{1}$ and $J^2 = \det(\mathbf{C})$. The second Piola-Kirchhoff stress tensor can be computed by using the result $\mathbf{S} = 2\partial_{\mathbf{C}}\psi$

$$\mathbf{S} = 2\frac{\partial\hat{\psi}}{\partial\mathbf{C}} = \frac{\partial\hat{\psi}}{\partial I_1} 2\frac{\partial I_1}{\partial\mathbf{C}} + \frac{\partial\hat{\psi}}{\partial J} 2\frac{\partial J}{\partial\mathbf{C}}$$

Since $\partial_{\mathbf{C}}\det(\mathbf{C}) = \det(\mathbf{C})\mathbf{C}^{-1}$ and $\partial_{\mathbf{C}}I_1 = \mathbf{1}$, the tensorial derivatives can be shown to be

$$2\partial_{\mathbf{C}}I_1 = 2\mathbf{1} \quad \text{and} \quad 2\partial_{\mathbf{C}}J = J\mathbf{C}^{-1}.$$

Derivatives of the free energy function with respect to the deformation measures I_1 and J are

$$\partial_{I_1}\hat{\psi} = \frac{\mu}{2} \quad \text{and} \quad \partial_J\hat{\psi} = \frac{\Lambda}{2}J - \left(\mu + \frac{\Lambda}{2}\right)J^{-1}.$$

Insertion yields the explicit form of $\mathbf{S}$

$$\mathbf{S} = 2\frac{\partial\hat{\psi}}{\partial\mathbf{C}} = \mu(\mathbf{1} - \mathbf{C}^{-1}) + \frac{\Lambda}{2}(J^2 - 1)\mathbf{C}^{-1}.$$

Having this result at hand, the first Piola-Kirchhoff stress tensor

$$\mathbf{P} = \mathbf{F} \cdot \mathbf{S} = \mu(\mathbf{F} - \mathbf{F}^{-T}) + \frac{\Lambda}{2}(J^2 - 1)\mathbf{F}^{-T}$$

and the Kirchhoff stresses

$$\boldsymbol{\tau} = \mathbf{P} \cdot \mathbf{F}^T = \mathbf{F} \cdot \mathbf{S} \cdot \mathbf{F}^T = \mu(\mathbf{b} - \mathbf{1}) + \frac{\Lambda}{2} (J^2 - 1) \mathbf{1}$$

are computed through the push-forward operations.

5.2.6.2 Computation of Homogeneous Stress-Stretch Response

Identification of material parameters of a phenomenological material model is commonly carried out by fitting experimental data. These data often consist of stress-stretch curves acquired from homogeneous experiments, such as uniaxial, biaxial, simple shear deformations. *Purely* homogeneous deformations correspond to the cases where the eigenvectors (both spatial and material) do not evolve during the deformation, but remain parallel to the chosen reference rectangular coordinate system. That is, in purely homogeneous deformations, the deformation gradient $\mathbf{F}$ has a diagonal representation

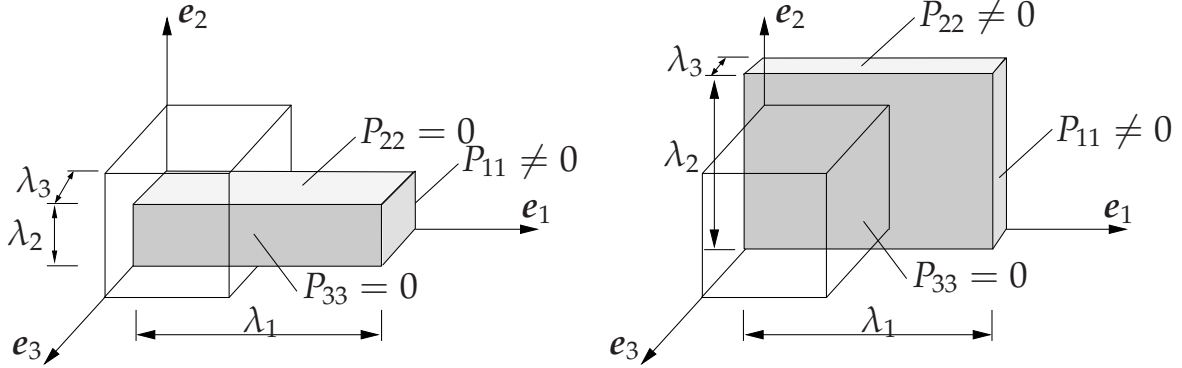
$$\mathbf{F} = \text{diag}[F_{11}, F_{22}, F_{33}] = \text{diag}[\lambda_1, \lambda_2, \lambda_3] \quad (5.2.23)$$

in terms of principal stretches with respect to the chosen coordinate system. This form of the deformation gradient results in also diagonal form of the first Piola-Kirchhoff stress for an *isotropic* material

$$\mathbf{P} = \hat{\mathbf{P}}(\mathbf{F}) = \text{diag}[P_{11}(\mathbf{F}), P_{22}(\mathbf{F}), P_{33}(\mathbf{F})] . \quad (5.2.24)$$

In different homogeneous deformations, at least one component of the deformation gradient is prescribed and the other components are to be computed from the traction (stress)

boundary conditions associated with the particular experiment considered. The stress component of interest is generally the one in the direction in which we prescribe the deformation. The forthcoming examples illustrate boundary conditions for the uniaxial and biaxial tests.



Component(s) of the stress tensor in the direction(s) where the stretch value is prescribed is/are unknown. On the other hand, stresses in the directions that are free to deform are zero. In a general inhomogeneous boundary value problem, these define the Dirichlet (essential) and Neumann (natural) boundaries whose intersection is an empty set.

Since the first Piola-Kirchhoff stress tensor and the deformation gradient have diagonal representations, we can store them in 3×1 vector format

$$[F] = \begin{bmatrix} F_{11} \\ F_{22} \\ F_{33} \end{bmatrix} = \begin{bmatrix} F_u \\ F_k \end{bmatrix} \quad \text{and} \quad [P] = \begin{bmatrix} P_{11} \\ P_{22} \\ P_{33} \end{bmatrix} = \begin{bmatrix} P_k \\ P_u \end{bmatrix} \quad (5.2.25)$$

where the subscripts k and u denote the *known* and *unknown* components, respectively. The known components of the

stress tensor $\mathbf{P}_k$ are generally zero, $\mathbf{P}_k = \mathbf{0}$. Therefore, these equations serve as a residual vector. That is, for a given $\mathbf{F}_k$, the following equality

$$\mathbf{P}_k = \hat{\mathbf{P}}_k(\mathbf{F}_u) = \mathbf{0} \quad (5.2.26)$$

has to be solved for $\mathbf{F}_u$. In general, the residual is a non-linear function of $\mathbf{F}_u$ and therefore should be solved iteratively. To this end, for a given $\mathbf{F}_k$ we linearize $\mathbf{P}_k$ at some intermediate value of $\mathbf{F}_u$ during iteration, say $\bar{\mathbf{F}}_u$,

$$\text{Lin } \mathbf{P}_k|_{\bar{\mathbf{F}}_u} = \mathbf{P}_k|_{\bar{\mathbf{F}}_u} + \mathbf{A}_{ku} \cdot (\mathbf{F}_u - \bar{\mathbf{F}}_u) = \mathbf{0} \quad (5.2.27)$$

where $\mathbf{A}_{ku} := \partial_{\mathbf{F}_u} \mathbf{P}_k|_{\bar{\mathbf{F}}_u}$ stands for the tangent matrix containing appropriate components of the fourth-order elasticity tensor (moduli) $\mathbb{A} := \partial_{\mathbf{F}} \mathbf{P} = \partial_{\mathbf{F}\mathbf{F}}^2 \psi$. Solution of the linearized residual expression (5.2.27) for $\mathbf{F}_u$ leads us to the update equation of $\mathbf{F}_u$

$$\mathbf{F}_u \longleftarrow \bar{\mathbf{F}}_u - \mathbf{A}_{ku}^{-1} \cdot \mathbf{P}_k|_{\bar{\mathbf{F}}_u} , \quad (5.2.28)$$

which is performed until the desired accuracy $\|\mathbf{P}_k\| < \text{tol}$ is obtained. The following box summarizes the algorithm

-
1. For a given $\mathbf{F}_k$, set $\mathbf{F}_u = \bar{\mathbf{F}}_u (= \mathbf{1})$
 2. Compute the residual $\mathbf{P}_k|_{\bar{\mathbf{F}}_u}$ and the tangent $\mathbf{A}_{ku} := \partial_{\mathbf{F}_u} \mathbf{P}_k|_{\bar{\mathbf{F}}_u}$
 3. Update $\mathbf{F}_u \longleftarrow \bar{\mathbf{F}}_u - \mathbf{A}_{ku}^{-1} \cdot \mathbf{P}_k|_{\bar{\mathbf{F}}_u}$
 4. Check IF $\|\mathbf{P}_k\| < \text{tol}$
 5. NO set $\bar{\mathbf{F}}_u = \mathbf{F}_u$ GOTO 2.
 6. ELSE compute $\mathbf{P}_u(\mathbf{F})$ and
 7. Assign new value to $\mathbf{F}_k$ and GOTO 1.
-

Example: Uniaxial Tension at Finite Strains

We now consider a uniaxial extension of a hyperelastic material in x_1 direction. For this purpose, we prescribe $\mathbf{F}_k \equiv F_{11}(t)$ as some function of time and want to compute the engineering stress $\mathbf{P}_u \equiv P_{11}$ in that direction for a given value of F_{11} . The lateral faces of the specimen are stress-free, i.e. $\mathbf{P}_k \equiv [P_{22} \ P_{33}]^T = \mathbf{0}$. Therefore, the stretches in x_2 and x_3 directions are unknown, $\mathbf{F}_u \equiv [F_{22} \ F_{33}]^T$.

Recall the compressible neo-Hookean material model we discussed in Section 5.2.6.1

$$P_{aA} = \mu(F_{aA} - F_{Aa}^{-1}) + \frac{\Lambda}{2} (J^2 - 1) F_{Aa}^{-1}.$$

Having the first Piola-Kirchhoff stresses at hand, the associated moduli can be computed as

$$\begin{aligned} \mathbb{A}_{aAbB} := \partial_{F_{bB}} P_{aA} &= \mu \delta_{ab} \delta_{AB} + \left(\mu - \frac{\Lambda}{2} (J^2 - 1) \right) F_{Ab}^{-1} F_{Ba}^{-1} \\ &\quad + \Lambda J^2 F_{Aa}^{-1} F_{Bb}^{-1}. \end{aligned}$$

The residual vector and the tangent matrix that we need during the Newton iterations can then be shown to be

$$\mathbf{P}_k(\mathbf{F}) = \begin{bmatrix} P_{22} \\ P_{33} \end{bmatrix} = \begin{bmatrix} \mu(F_{22} - F_{22}^{-1}) + \frac{\Lambda}{2} (J^2 - 1) F_{22}^{-1} \\ \mu(F_{33} - F_{33}^{-1}) + \frac{\Lambda}{2} (J^2 - 1) F_{33}^{-1} \end{bmatrix}$$

$$\mathbf{A}_{ku}(\mathbf{F}) = \partial_{\mathbf{F}_u} \mathbf{P}_k = \frac{\partial [P_{22} \ P_{33}]^T}{\partial [F_{22} \ F_{33}]^T} = \begin{bmatrix} \mathbb{A}_{2222} & \mathbb{A}_{2233} \\ \mathbb{A}_{3322} & \mathbb{A}_{3333} \end{bmatrix}$$

for the uniaxial tension.

```
#include <stdio.h>
#include <stdlib.h>
#include <math.h>
/*-----

ME 338A Continuum Mechanics
Stanford University
E. Kuhl, S. Goktepe
Driver routine for uniaxial tension
-----*/

int main()
{

    // open file for output
    FILE *fp;
    fp = fopen("data.out","w");
    if (fp==NULL){
        printf("Cannot create the output file!!\n");
        exit(1);
    }

    int i,j;
    // solution parameters
    double dlam=1.e-2, lam_max=4., nres=0.,tol=1.e-12;
    int niter, nitermax=50;
    //material parameters
    double Lam=0.3, mu=0.8;
    //deformation measures, stresses and moduli
    double J,F[3],P[3],AA[3][3];
    //residual, tangent for iterations
    double res[2],tang[2][2],det,temp;

    //initialize F
    for(i=0;i<3;i++) F[i]=1.;

    // loop for loading
    do {
        niter = 0;
        F[0] +=dlam;
        printf("STRETCH=%4e\n", F[0]);
        // loop of iterations
        do {
            niter +=1 ;
        } while (1);
    } while (1);
    // compute Jacobi, det(F)
```

```
J = F[0]*F[1]*F[2];

// MATERIAL MODEL specific computation of stresses and moduli
//-----
// Compressible Neo-Hookean Material
for (i=0;i<3;i++) {
    P[i] = mu*(F[i]-1./F[i]) + Lam*(J*J-1.)/F[i]/2;
    for (j=0;j<3;j++) {
        AA[i][j] = Lam*J*J/F[i]/F[j];
        if(i==j) AA[i][j] += (mu - (-mu+Lam*(J*J-1.)/2.)/F[i]/F[j]);
    }
}
//-----
}
//compute residual vector and its norm
res[0] = P[1]; res[1] = P[2];
nres = sqrtl(res[0]*res[0] + res[1]*res[1]);

//construct tangent matrix
tang[0][0] = AA[1][1]; tang[0][1] = AA[1][2];
tang[1][0] = AA[2][1]; tang[1][1] = AA[2][2];

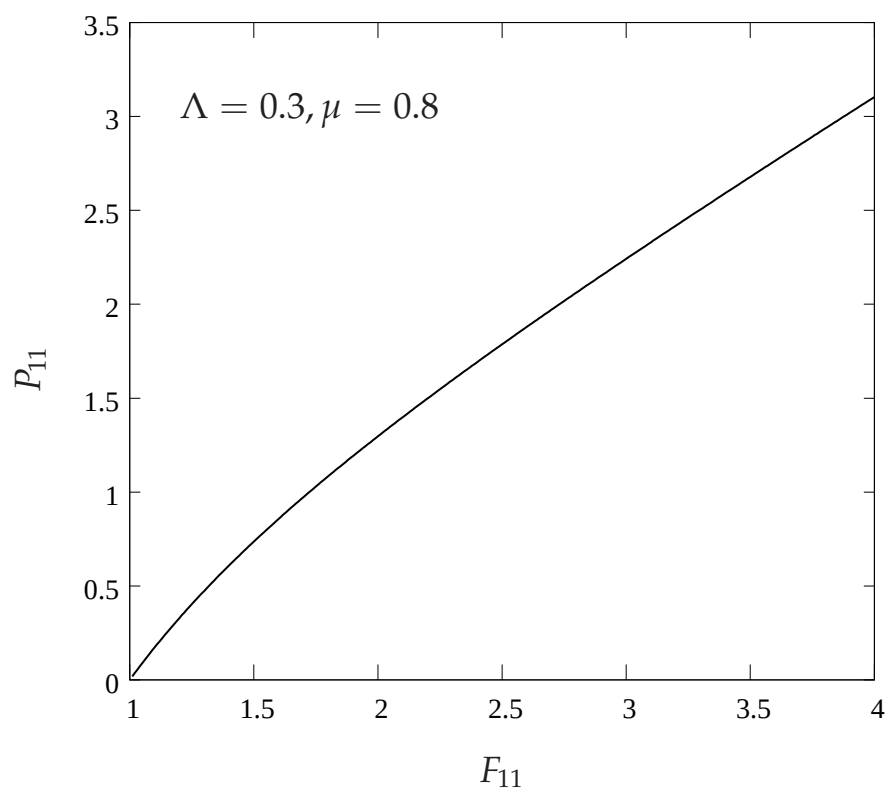
// invert tangent matrix
det = tang[0][0] *tang[1][1] - tang[0][1] *tang[1][0];
temp = tang[0][0];
tang[0][0] = tang[1][1]/det;
tang[1][1] = temp/det;
tang[0][1] = -tang[0][1]/det;
tang[1][0] = -tang[1][0]/det;

// update lateral stretches
F[1] = F[1] - (tang[0][0]*res[0]+tang[0][1]*res[1]);
F[2] = F[2] - (tang[1][0]*res[0]+tang[1][1]*res[1]);

// print iteration counter and norm of residual
printf("Iteration %4d, residual=%4e\n", niter,nres);
    } while (tol<nres && niter<nitermax); // Check convergence
    if (niter==nitermax) {printf("No convergence!\n"); break;};

    // print stretch and PK1 stresses
    fprintf(fp,"%4e, %4e\n",F[0],P[0]);

    } while ((lam_max-F[0])>1.e-8); // check loading
    fclose(fp); //close file
    return 0;
}
```



ME338A
CONTINUUM MECHANICS

lecture notes 17

tuesday, march 9, 2010

5.2.7 Incompressible Elasticity

5.2.7.1 Nearly incompressible elasticity

Many engineering materials exhibit distinct behavior under volumetric and isochoric deformations. Common examples include the nearly incompressible elasticity of rubber-like materials and soft biological tissues, shear-dominated ductile viscoplasticity of amorphous glassy polymers, and volume-preserving plasticity of metals. To this end, we decompose the deformation gradient F multiplicatively into the volumetric (spherical) F_{vol} and isochoric (unimodular) $\bar{F}$ parts

$$F = \bar{F} \cdot F_{vol} \quad (5.2.29)$$

where $F_{vol} := J^{1/3} \mathbf{1}$ and $\bar{F} := J^{-1/3} F$.

This implies that the volume map is solely described by the volumetric part F_{vol} so that the identity $\det(F_{vol}) = J$ holds. Therefore, the deformation under the unimodular part $\bar{F}$ with $\det \bar{F} = 1$ does not include any volume change but purely isochoric deformations. Having this multiplicative kinematic split at hand, the free energy of a hyperelastic material can be additively decomposed into the volumetric $U(J)$ and isochoric $\bar{\psi}$ parts

$$\psi = U(J) + \bar{\psi}(\bar{F}). \quad (5.2.30)$$

Owing to the material frame objectivity requirement, the isochoric part of the free energy can also be expressed as $\bar{\psi} = \hat{\psi}(\bar{F}^T \cdot \bar{F}) = \hat{\psi}(\bar{C})$, i.e.

$$\psi = U(J) + \hat{\psi}(\bar{C}) \quad (5.2.31)$$

in terms of $\bar{\mathbf{C}} := \bar{\mathbf{F}}^T \cdot \bar{\mathbf{F}} = J^{-2/3} \mathbf{C}$.

The additive decomposition of the free energy results in also the additively split 'stress response

$$\mathbf{S} = 2\partial_{\mathbf{C}}\psi = \mathbf{S}_{vol} + \mathbf{S}_{iso} \quad (5.2.32)$$

in terms of the volumetric part

$$\mathbf{S}_{vol} := 2\partial_{\mathbf{C}}U(J) = U'(J)2\partial_{\mathbf{C}}J = JU'(J)\mathbf{C}^{-1} \quad (5.2.33)$$

and the isochoric part

$$\mathbf{S}_{iso} := 2\partial_{\bar{\mathbf{C}}}\hat{\psi}(\bar{\mathbf{C}}) = 2\partial_{\bar{\mathbf{C}}}\hat{\psi}(\bar{\mathbf{C}}) : \partial_{\bar{\mathbf{C}}}\bar{\mathbf{C}} = \bar{\mathbf{S}} : \mathbb{Q} \quad (5.2.34)$$

of the second Piola-Kirchhoff stress tensor. In (5.2.34), the *chain-rule* stresses $\bar{\mathbf{S}}$ and the fourth order *Lagrangian* deviatoric projection tensor $\mathbb{Q}$ are defined as

$$\bar{\mathbf{S}} := 2\partial_{\bar{\mathbf{C}}}\hat{\psi}(\bar{\mathbf{C}})$$

and

$$\begin{aligned} \mathbb{Q} &:= \partial_{\mathbf{C}}\bar{\mathbf{C}} = \partial_{\mathbf{C}}(J^{-2/3}\mathbf{C}) = J^{-2/3}\partial_{\mathbf{C}}\mathbf{C} + \mathbf{C} \otimes \partial_{\mathbf{C}}(J^{-2/3}) \\ &= J^{-2/3}\mathbb{I} + \mathbf{C} \otimes \left(-\frac{2}{3}J^{-2/3-1} \frac{1}{2}J\mathbf{C}^{-1}\right) \\ &= J^{-2/3} \left[\mathbb{I} - \frac{1}{3}\mathbf{C} \otimes \mathbf{C}^{-1}\right] \end{aligned}$$

respectively. The overall expression of the second Piola-Kirchhoff stresses can then be obtained as

$$\mathbf{S} = JU'(J)\mathbf{C}^{-1} + \bar{\mathbf{S}} : J^{-2/3} \left[\mathbb{I} - \frac{1}{3}\mathbf{C} \otimes \mathbf{C}^{-1}\right]. \quad (5.2.35)$$

The complete push forward of the second Piola-Kirchhoff stress tensor yields the Kirchhoff stress tensor

$$\boldsymbol{\tau} = \mathbf{F} \cdot \mathbf{S} \cdot \mathbf{F}^T = JU'(J)\mathbf{1} + \bar{\boldsymbol{\tau}} : \mathbb{P} \quad (5.2.36)$$

where $\bar{\tau} := \bar{\mathbf{F}} \cdot \bar{\mathbf{S}} \cdot \bar{\mathbf{F}}^T$ and $\mathbb{P} := \mathbb{I} - \frac{1}{3} \mathbf{1} \otimes \mathbf{1}$ denotes the *spatial* deviatoric projection tensor such that $\text{tr}(\bar{\tau} : \mathbb{P}) = 0$.

Then, the Cauchy stress tensor reads

$$\boldsymbol{\sigma} = J^{-1} \boldsymbol{\tau} = U'(J) \mathbf{1} + J^{-1} \bar{\tau} : \mathbb{P} . \quad (5.2.37)$$

Employing the definition of the pressure, we obtain

$$p(J) := -\frac{1}{3} \text{tr}(\boldsymbol{\sigma}) = -U'(J) . \quad (5.2.38)$$

This result leads us to the following decomposed representation of the Cauchy stresses

$$\boldsymbol{\sigma} = -p(J) \mathbf{1} + \boldsymbol{\sigma}_{dev} \quad \text{with} \quad \boldsymbol{\sigma}_{dev} := J^{-1} \bar{\tau} : \mathbb{P} . \quad (5.2.39)$$

In modeling of *nearly* incompressible response of materials in three-dimensional boundary-value problems, the pressure $p(J)$ serves as a *penalty* parameter employed to enforce the incompressibility condition. However, straightforward implementation of this penalty formulation in standard displacement finite elements often brings about undesired *volumetric locking* effects. In order to overcome this drawback, one of the remedies is the use of a three-field Hu-Washizu functional-based mixed finite element formulation that considers the pressure $\bar{p}$, the Jacobian Θ , and the deformation map $\boldsymbol{\varphi}$ as independent fields, albeit with different continuity requirements. These numerical treatments, however, are beyond the scope of this class.

5.2.7.2 Strictly incompressible elasticity

The *strictly* incompressible materials can be conceived as substances that are capable of carrying loads by undergoing solely volume conserving (isochoric) deformations; that is, the value of the volume map J is always restricted to unity

$$J := \det(\mathbf{F}) \doteq 1 . \quad (5.2.40)$$

If the type of loading has a purely spherical character, i.e. pure pressure or pure suction, an incompressible material behaves rigidly and hence does not store energy. This means, the pressure cannot be determined from the free energy function, e.g. by evaluating the function $U'(J)$ as we did in the preceding section. Therefore, pressure should be considered as unknown in the stress expressions

$$\mathbf{S} = -pJ\mathbf{C}^{-1} + 2\partial_{\mathbf{C}}\psi \quad \text{or} \quad \mathbf{P} = -pJ\mathbf{F}^{-T} + \partial_{\mathbf{F}}\psi \quad (5.2.41)$$

This formula holds only for the cases where the constraint (5.2.40) is strictly fulfilled. This implies that for a given deformation $\mathbf{F}$, one of the principal stretches, say λ_3 , can be expressed in terms of the other principal stretches λ_1, λ_2 ; that is,

$$\det(\mathbf{F}) = \lambda_1\lambda_2\lambda_3 \doteq 1 \quad \rightsquigarrow \quad \lambda_\beta = \prod_{\alpha \neq \beta}^3 \lambda_\alpha^{-1} . \quad (5.2.42)$$

For $\beta = 3$, we have $\lambda_3 = \lambda_1^{-1}\lambda_2^{-1}$. For an isotropic, incompressible material undergoing purely homogeneous deformations such as uniaxial and biaxial tests, components of the deformation gradient $\mathbf{F}$ are completely known. The following are the examples for the uniaxial, pure shear, and

equi-biaxial cases with corresponding traction boundary conditions:

$$\text{Uniaxial} \quad F = \text{diag}[\lambda, \lambda^{-\frac{1}{2}}, \lambda^{-\frac{1}{2}}], \quad P = \text{diag}[P_{11}, 0, 0]$$

$$\text{Pure Shear} \quad F = \text{diag}[\lambda, \lambda^{-1}, 1], \quad P = \text{diag}[P_{11}, 0, P_{33}]$$

$$\text{Equi-biaxial} \quad F = \text{diag}[\lambda, \lambda, \lambda^{-2}], \quad P = \text{diag}[P_{11}, P_{22}, 0]$$

The stress boundary conditions provide us with an additional equation that is solved to compute the unknown pressure.

5.2.7.3 Stretch-Based Elasticity Models

In Section 5.2.6, we concluded that the free energy of an isotropic material can be expressed as a function of principal stretches

$$\psi = \psi(\lambda_1, \lambda_2, \lambda_3) .$$

In order to compute the second Piola-Kirchhoff stresses from the stretch-based free energy function, we need to determine the derivative of the principal stretches with respect to C . To this end, we consider the third invariant

$$I_3 = J^2 = \det(C) = \lambda_1^2 \lambda_2^2 \lambda_3^2 .$$

The derivative of I_3 with respect to C can be shown to be

$$\partial_C I_3 = \sum_{\alpha=1}^3 \partial_{\lambda_\alpha^2} I_3 \partial_C \lambda_\alpha^2 = I_3 \sum_{\alpha=1}^3 \lambda_\alpha^{-2} \partial_C \lambda_\alpha^2 .$$

We also know that $\text{cof}(C) = \partial_C I_3$ whose spectral representation is given by

$$\partial_C I_3 = I_3 C^{-1} = I_3 \sum_{\alpha=1}^3 \lambda_\alpha^{-2} N_\alpha \otimes N_\alpha .$$

Comparing these two results, we deduce that

$$\partial_C \lambda_\alpha^2 = \mathbf{N}_\alpha \otimes \mathbf{N}_\alpha \quad \text{and} \quad 2\partial_C \lambda_\alpha = \lambda_\alpha^{-1} \mathbf{N}_\alpha \otimes \mathbf{N}_\alpha .$$

We can then compute the second Piola-Kirchhoff stress tensor in the spectral form

$$\mathbf{S} = 2\partial_C \psi(\lambda_1, \lambda_2, \lambda_3) = \sum_{\alpha=1}^3 \partial_{\lambda_\alpha^2} \psi \, 2\partial_C \lambda_\alpha^2 = \sum_{\alpha=1}^3 2\partial_{\lambda_\alpha^2} \psi \, \mathbf{N}_\alpha \otimes \mathbf{N}_\alpha .$$

Substituting $S_\alpha := 2\partial_{\lambda_\alpha^2} \psi = \lambda_\alpha^{-1} \partial_{\lambda_\alpha} \psi$, we end up with the compact spectral representation of $\mathbf{S}$

$$\mathbf{S} = \sum_{\alpha=1}^3 S_\alpha \, \mathbf{N}_\alpha \otimes \mathbf{N}_\alpha \quad \text{with} \quad S_\alpha := \lambda_\alpha^{-1} \partial_{\lambda_\alpha} \psi$$

standing for the eigenvalues of the second Piola-Kirchhoff stress tensor. From this result, we immediately observe that the second Piola-Kirchhoff stress tensor $\mathbf{S}$ possesses the same eigenvectors as $\mathbf{C} = \sum_{\alpha=1}^3 \lambda_\alpha^2 \mathbf{N}_\alpha \otimes \mathbf{N}_\alpha$. This is a typical result obtained from an isotropic free energy function. When two second order tensors have the same eigenvectors, they are said to be *co-axial* and therefore they *commute*, i.e.

$$\mathbf{S} \cdot \mathbf{C} = \mathbf{C} \cdot \mathbf{S} .$$

The first Piola-Kirchhoff stress tensor $\mathbf{P}$ and the Kirchhoff stress tensor $\boldsymbol{\tau}$ are then readily computed through the push forward operation

$$\mathbf{P} = \mathbf{F} \cdot \mathbf{S} = \sum_{\alpha=1}^3 P_\alpha \, \mathbf{n}_\alpha \otimes \mathbf{N}_\alpha \quad \text{with} \quad P_\alpha := \lambda_\alpha S_\alpha = \partial_{\lambda_\alpha} \psi$$

$$\boldsymbol{\tau} = \mathbf{P} \cdot \mathbf{F}^T = \sum_{\alpha=1}^3 \tau_\alpha \, \mathbf{n}_\alpha \otimes \mathbf{n}_\alpha \quad \text{with} \quad \tau_\alpha := \lambda_\alpha P_\alpha = \lambda_\alpha \partial_{\lambda_\alpha} \psi$$

Example: The Ogden Model of Incompressible Elasticity

We now consider the well-known Ogden model of incompressible elasticity

$$\psi = \psi(\lambda_1, \lambda_2, \lambda_3) = \sum_{n=1}^3 \frac{\mu_n}{\beta_n} (\lambda_1^{\beta_n} + \lambda_2^{\beta_n} + \lambda_3^{\beta_n} - 3) .$$

For exact incompressibility ($J=1$), we recall the expression for $\mathbf{P}$ from (5.2.41)

$$\mathbf{P} = -p \mathbf{F}^{-T} + \partial_{\mathbf{F}} \psi ,$$

or in the spectral form

$$\mathbf{P} = \sum_{\alpha=1}^3 P_{\alpha} \mathbf{n}_{\alpha} \otimes \mathbf{N}_{\alpha} = \sum_{\alpha=1}^3 (-p \lambda_{\alpha}^{-1} + \partial_{\lambda_{\alpha}} \psi) \mathbf{n}_{\alpha} \otimes \mathbf{N}_{\alpha} .$$

The eigenvalues of $\mathbf{P}$ are then

$$P_{\alpha} = -p \lambda_{\alpha}^{-1} + \partial_{\lambda_{\alpha}} \psi \quad \text{for } \alpha = 1, 2, 3 .$$

For the Ogden model, the derivative of the free energy with respect to each principal stretch has the following form

$$\partial_{\lambda_{\alpha}} \psi = \sum_{n=1}^3 \mu_n \lambda_{\alpha}^{\beta_n-1} \quad \text{for } \alpha = 1, 2, 3 .$$

This leads us to the model-specific expressions for the principal stresses

$$P_{\alpha} = -p \lambda_{\alpha}^{-1} + \sum_{n=1}^3 \mu_n \lambda_{\alpha}^{\beta_n-1} \quad \text{for } \alpha = 1, 2, 3 .$$

We can now derive the analytic stress expressions for three distinct cases of homogeneous deformations:

- **Uniaxial** $F = \text{diag}[\lambda, \lambda^{-\frac{1}{2}}, \lambda^{-\frac{1}{2}}], \quad P = \text{diag}[P_{11}, 0, 0]$

For the stress-free lateral faces, we compute the pressure

$$P_{\{2,3\}} = -p \lambda_{\{2,3\}}^{-1} + \sum_{n=1}^3 \mu_n \lambda_{\{2,3\}}^{\beta_n - 1} = 0 \rightsquigarrow p = \sum_{n=1}^3 \mu_n \lambda_{\{2,3\}}^{\beta_n}.$$

Insertion of this result into P_1 yields

$$P_1 = \sum_{n=1}^3 \mu_n (\lambda_1^{\beta_n - 1} - \lambda_{\{2,3\}}^{\beta_n} \lambda_1^{-1}).$$

For $\lambda_1 = \lambda, \lambda_2 = \lambda_3 = \lambda^{-\frac{1}{2}}$, we obtain

$$P_1 = \sum_{n=1}^3 \mu_n (\lambda^{\beta_n - 1} - \lambda^{-\beta_n/2 - 1}).$$

- **Pure Shear** $F = \text{diag}[\lambda, \lambda^{-1}, 1], \quad P = \text{diag}[P_{11}, 0, P_{33}]$

Following the analogous steps

$$P_2 = -p \lambda_2^{-1} + \sum_{n=1}^3 \mu_n \lambda_2^{\beta_n - 1} = 0 \rightsquigarrow p = \sum_{n=1}^3 \mu_n \lambda_2^{\beta_n}.$$

$$P_1 = \sum_{n=1}^3 \mu_n (\lambda_1^{\beta_n - 1} - \lambda_2^{\beta_n} \lambda_1^{-1}) = \sum_{n=1}^3 \mu_n (\lambda^{\beta_n - 1} - \lambda^{-(\beta_n + 1)})$$

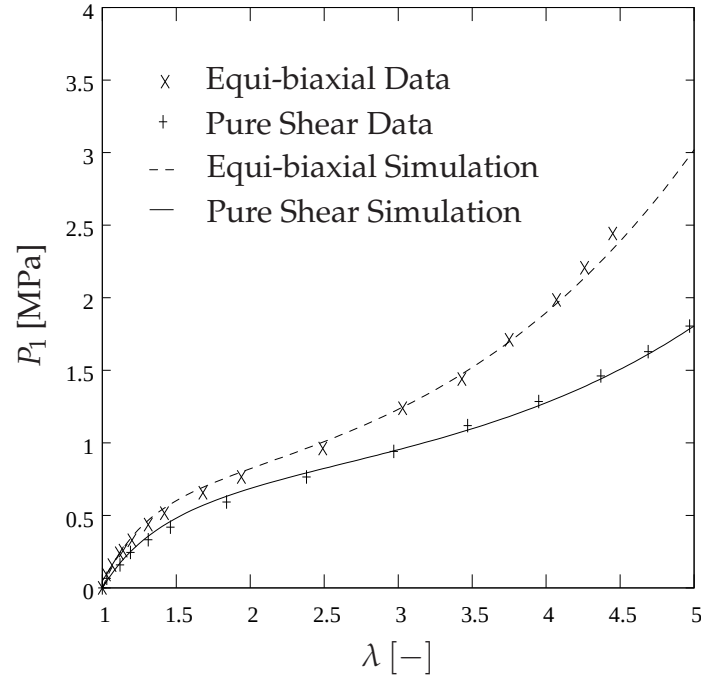
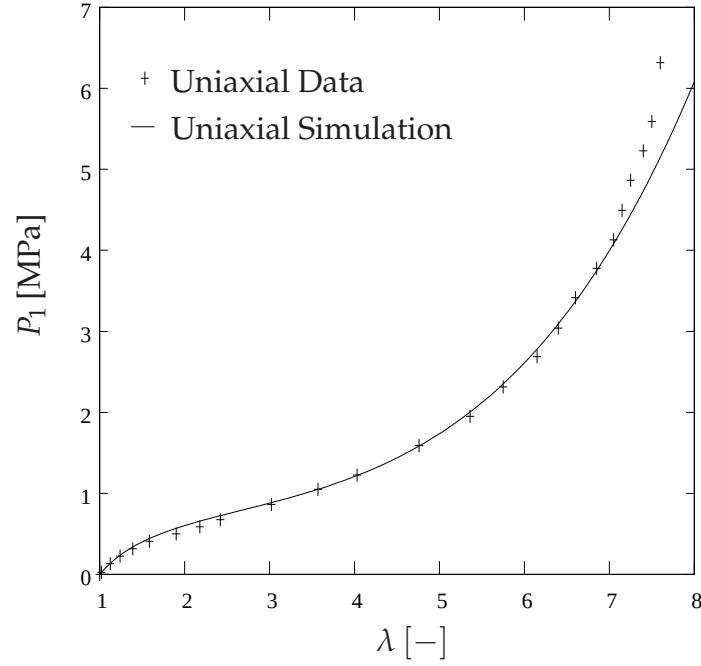
- **Equi-biaxial** $F = \text{diag}[\lambda, \lambda, \lambda^{-2}], P = \text{diag}[P_{11}, P_{22}, 0]$

Similarly,

$$P_3 = -p \lambda_3^{-1} + \sum_{n=1}^3 \mu_n \lambda_3^{\beta_n - 1} = 0 \rightsquigarrow p = \sum_{n=1}^3 \mu_n \lambda_3^{\beta_n}.$$

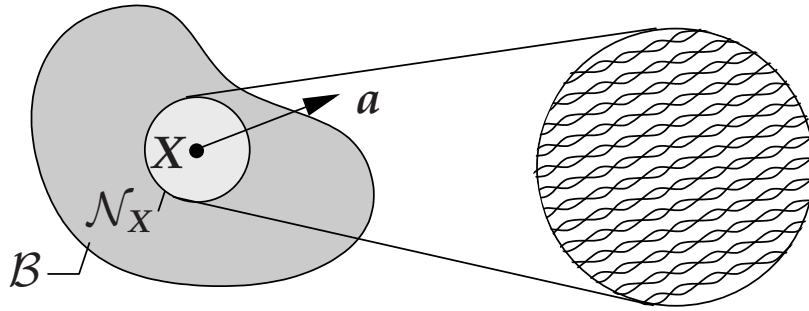
$$P_{\{1,2\}} = \sum_{n=1}^3 \mu_n (\lambda_{\{1,2\}}^{\beta_n - 1} - \lambda_3^{\beta_n} \lambda_{\{1,2\}}^{-1}) = \sum_{n=1}^3 \mu_n (\lambda^{\beta_n - 1} - \lambda^{-(2\beta_n + 1)})$$

For the material parameters $\mu_1 = 0.63$ MPa, $\mu_2 = 0.0012$ MPa, $\mu_3 = -0.01$ MPa and $\beta_1 = 1.3, \beta_2 = 5, \beta_3 = -2$, the Ogden model captures the classical data of Treloar (1944) acquired from vulcanized unfilled natural rubber.



5.2.8 Transversely Isotropic Elasticity

This section is devoted to the coordinate-free representation of *transversely isotropic* elasticity, which is probably the simplest kind of anisotropy one can conceive. The coordinate-independent framework is obtained in terms of *structural tensors* that describe the microstructure of the material under consideration.



The microstructure of a transversely isotropic material can be considered as a continuum reinforced by micro-fibers. Common examples cover man-made fiber-reinforced composites such as carbon nanotube loaded materials, automobile tires, and natural collagenous biological tissues, to mention a few.

The fibrous microstructure of a transversely isotropic material possesses a well-defined *preferred* direction, which we denote with the unit vector $\mathbf{a}$, $\|\mathbf{a}\|=1$. This is the key information that helps us construct the symmetry group of a transversely isotropic matter. As we mentioned in Section 5.2.5, a symmetry group is defined by a set of rotations that preserve the structural characteristics of the material on

micro-level. In the case of transverse isotropy, the symmetry group contains all rotations that do not alter the preferred orientation $\mathbf{a}$. These include the rotations of an arbitrary amount about the axis $\mathbf{a}$, i.e. $\mathbf{Q}_{\parallel \mathbf{a}}$ and the rotations about the axis perpendicular to $\mathbf{a}$ by an amount of π , i.e. $\mathbf{Q}_{\perp \mathbf{a}}^\pi$, that flip $\mathbf{a}$ horizontally. We then define the transversely isotropic symmetry group defined as

$$G_{\text{transiso}} := \left\{ \mathbf{Q}_{\parallel \mathbf{a}}, \mathbf{Q}_{\perp \mathbf{a}}^\pi \right\} . \quad (5.2.43)$$

The underlying key idea of coordinate-free representation of anisotropic materials is the *isotropic tensor functions* with extended set of arguments such that they remain invariant under arbitrary rotations $\mathbf{Q} \in \mathcal{SO}(3)$. This opens up a possibility to recast an anisotropic constitutive function into an isotropic one by means of the so-called *structural tensors*. For the case of transverse isotropy, we have

$$\psi = \hat{\psi}(\mathbf{C}, \mathbf{M}) \quad \text{with} \quad \mathbf{M} := \mathbf{a} \otimes \mathbf{a} . \quad (5.2.44)$$

denoting the *constant* rank-one structural tensor. The free energy function with the extended list of arguments is then required to be an isotropic function, i.e. invariant under arbitrary rotations $\mathbf{Q} \in \mathcal{SO}(3)$

$$\hat{\psi}(\mathbf{Q} \cdot \mathbf{C} \cdot \mathbf{Q}^T, \mathbf{Q} \cdot \mathbf{M} \cdot \mathbf{Q}^T) = \hat{\psi}(\mathbf{C}, \mathbf{M}) \quad \forall \mathbf{Q} \in \mathcal{SO}(3) . \quad (5.2.45)$$

Since the rotations belonging to the symmetric group G_{transiso} preserve the direction $\mathbf{a}$, the structural tensor $\mathbf{M}$ remains invariant under these rotations

$$\mathbf{Q} \cdot \mathbf{M} \cdot \mathbf{Q}^T = \mathbf{M} \quad \forall \mathbf{Q} \in G_{\text{transiso}} . \quad (5.2.46)$$

Therefore, the new isotropic free energy also fulfills

$$\hat{\psi}(\mathbf{Q} \cdot \mathbf{C} \cdot \mathbf{Q}^T, \mathbf{M}) = \hat{\psi}(\mathbf{C}, \mathbf{M}) \quad \forall \mathbf{Q} \in G_{\text{transiso}} \subset \mathcal{SO}(3). \quad (5.2.47)$$

The representation theorem of general isotropic tensor functions of several arguments is based on the concept of *integrity bases*, which defines a minimum number of invariants for a particular set of arguments of the energy storage function. For two generic *symmetric* second order tensors, say $\mathbf{A}_1$ and $\mathbf{A}_2$, the irreducible integrity bases are given by

$$\begin{aligned} \mathcal{I} = \{ & J_1(\mathbf{A}_1), J_2(\mathbf{A}_1), J_3(\mathbf{A}_1), J_1(\mathbf{A}_2), J_2(\mathbf{A}_2), J_3(\mathbf{A}_2) \\ & J_4(\mathbf{A}_1, \mathbf{A}_2), J_5(\mathbf{A}_1, \mathbf{A}_2), J_6(\mathbf{A}_1, \mathbf{A}_2), J_7(\mathbf{A}_1, \mathbf{A}_2) \} \end{aligned} \quad (5.2.48)$$

where

$$J_1(\mathbf{A}_i) := \text{tr}(\mathbf{A}_i), \quad J_2(\mathbf{A}_i) := \text{tr}(\mathbf{A}_i^2), \quad J_3(\mathbf{A}_i) := \text{tr}(\mathbf{A}_i^3)$$

are the *basic invariants* of respective tensors for $i = 1, 2$ and

$$\begin{aligned} J_4 &:= \text{tr}(\mathbf{A}_1 \cdot \mathbf{A}_2), & J_5 &:= \text{tr}(\mathbf{A}_1^2 \cdot \mathbf{A}_2), \\ J_6 &:= \text{tr}(\mathbf{A}_1 \cdot \mathbf{A}_2^2), & J_7 &:= \text{tr}(\mathbf{A}_1^2 \cdot \mathbf{A}_2^2) \end{aligned}$$

denote the *mixed invariants*.

In transversely isotropic elasticity, the free energy function depends on the symmetric tensors $\mathbf{A}_1 = \mathbf{C}$ and $\mathbf{A}_2 = \mathbf{M}$. Since the structural tensor $\mathbf{M}$ is also constant and $\mathbf{M}^n = \mathbf{M}$, the following identities

$$\begin{aligned} J_3(\mathbf{M}) &= \text{tr}(\mathbf{M}^3) = \text{tr}(\mathbf{M}^2) = \text{tr}(\mathbf{M}) \rightsquigarrow J_3(\mathbf{M}) = J_2(\mathbf{M}) = J_1(\mathbf{M}) \\ J_6 &= \text{tr}(\mathbf{C} \cdot \mathbf{M}^2) = \text{tr}(\mathbf{C} \cdot \mathbf{M}) \rightsquigarrow J_6 = J_4 \\ J_7 &= \text{tr}(\mathbf{C}^2 \cdot \mathbf{M}^2) = \text{tr}(\mathbf{C}^2 \cdot \mathbf{M}) \rightsquigarrow J_7 = J_5 \end{aligned}$$

reduce the integrity bases of transverse isotropy to

$$\mathcal{I} = \{J_1(\mathbf{C}), J_2(\mathbf{C}), J_3(\mathbf{C}), J_4(\mathbf{C}, \mathbf{M}), J_5(\mathbf{C}, \mathbf{M}); J_1(\mathbf{M})\} \quad (5.2.49)$$

Since we also know that $\|\mathbf{a}\| = 1$, the basic invariant(s) of the structural tensor are none other than $J_1(\mathbf{M}) = 1$. Therefore, it has no effect on the energy storage function. The effective list of deformation-dependent invariants then boils down to

$$\mathcal{I} = \{J_1(\mathbf{C}), J_2(\mathbf{C}), J_3(\mathbf{C}), J_4(\mathbf{C}, \mathbf{M}), J_5(\mathbf{C}, \mathbf{M})\} .a \quad (5.2.50)$$

Since the basic invariants of $\mathbf{C}$ can be expressed in terms of the principal invariants

$$J_1 = I_1, \quad J_2 = I_1^2 - 2I_2, \quad J_3 = I_1^3 - 3I_1I_2 + 3I_3 ,$$

the set in (5.2.50) may be recast as

$$\mathcal{I} = \{I_1(\mathbf{C}), I_2(\mathbf{C}), I_3(\mathbf{C}), J_4(\mathbf{C}, \mathbf{M}), J_5(\mathbf{C}, \mathbf{M})\} . \quad (5.2.51)$$

5.2.8.1 Extended Neo-Hookean Model of Transverse Isotropy

We consider a model problem of transverse isotropy. The free energy is assumed to be given as

$$\psi = \hat{\psi}_I(I_1, J) + \hat{\psi}_A(I_1, I_2, J_4, J_5) , \quad (5.2.52)$$

which is split into the purely isotropic ψ_I and the anisotropic ψ_A parts. This leads us to the additive form of the stresses

$$\mathbf{S} = 2\partial_{\mathbf{C}}\psi = \mathbf{S}_I + \mathbf{S}_A \quad \text{and} \quad \mathbf{P} = \mathbf{P}_I + \mathbf{P}_A , \quad (5.2.53)$$

where $\mathbf{S}_I := 2\partial_{\mathbf{C}}\hat{\psi}_I$, $\mathbf{S}_A := 2\partial_{\mathbf{C}}\hat{\psi}_A$ and $\mathbf{P}_I := \mathbf{F} \cdot \mathbf{S}_I$, $\mathbf{P}_A := \mathbf{F} \cdot \mathbf{S}_A$.

The isotropic free energy function is taken to be identical to the compressible neo-Hookean model we discussed before

$$\hat{\psi}_I(I_1, J) = \frac{\Lambda}{4}(J^2 - 1) - \left(\mu + \frac{\Lambda}{2}\right) \ln J + \frac{\mu}{2}(I_1 - 3) . \quad (5.2.54)$$

We then retrieve the stress expressions for the isotropic part

$$\begin{aligned} \mathbf{S}_I &= 2 \frac{\partial \hat{\psi}_I}{\partial \mathbf{C}} = \mu(\mathbf{1} - \mathbf{C}^{-1}) + \frac{\Lambda}{2} (J^2 - 1) \mathbf{C}^{-1} , \\ \mathbf{P}_I &= \mathbf{F} \cdot \mathbf{S}_I = \mu(\mathbf{F} - \mathbf{F}^{-T}) + \frac{\Lambda}{2} (J^2 - 1) \mathbf{F}^{-T} . \end{aligned} \quad (5.2.55)$$

For the anisotropic part of the free energy we consider the following form

$$\begin{aligned} \hat{\psi}_A(I_1, I_2, J_4, J_5) &= \alpha_1 (J_4 - 1)^2 + \alpha_2 [J_5 - I_1 J_4 + I_2 - 1] \\ &\quad + \alpha_2 (J_4 - I_1 + 2) , \end{aligned} \quad (5.2.56)$$

where α_1 and α_2 are additional material parameters. Geometrical interpretation of the kinematic terms appearing in this free energy will make the meaning of these additional material parameters more transparent. The square of the stretch in the direction $\mathbf{a}$ is given by J_4 , i.e.

$$\lambda_a^2 := ||\mathbf{F} \cdot \mathbf{a}||^2 = (\mathbf{F} \cdot \mathbf{a}) \cdot (\mathbf{F} \cdot \mathbf{a}) = \mathbf{C} : (\mathbf{a} \otimes \mathbf{a}) = \mathbf{C} : \mathbf{M} \equiv J_4 .$$

Thus, the first non-linear term of the free energy governs the amount energy stored due to the stretching of fibers, and the material parameter α_1 is closely related to the stiffness of these fibers. The second term in square brackets, however, measures the area stretch ν_a in the direction $\mathbf{a}$. We compute

the square of the area stretch through the area map $\text{cof}(F) := JF^{-T}$

$$\nu_a^2 := ||\text{cof}(F) \cdot \mathbf{a}||^2 = J^2(F^{-T} \cdot \mathbf{a}) \cdot (F^{-T} \cdot \mathbf{a}) = J^2 \mathbf{C}^{-1} : \mathbf{M} .$$

In order to see the relation between this result and the second term of $\hat{\psi}_A$, we recall the Cayley-Hamilton theorem

$$\mathbf{C}^3 - I_1 \mathbf{C}^2 + I_2 \mathbf{C} - I_3 \mathbf{1} = \mathbf{0} ,$$

Multiplying this equation with $\mathbf{C}^{-1}$ and solving the result for $I_3 \mathbf{C}^{-1}$, we obtain

$$I_3 \mathbf{C}^{-1} = J^2 \mathbf{C}^{-1} = \mathbf{C}^2 - I_1 \mathbf{C} + I_2 \mathbf{1} .$$

Contraction of this result with the structural tensor $\mathbf{M}$ then gives the square of the area stretch in the preferred direction

$$\begin{aligned} \nu_a^2 &= J^2 \mathbf{C}^{-1} : \mathbf{M} = \mathbf{C}^2 : \mathbf{M} - I_1 \mathbf{C} : \mathbf{M} + I_2 \mathbf{1} : \mathbf{M} \\ &= J_5 - I_1 J_4 + I_2 . \end{aligned}$$

Therefore, the material parameter α_2 can be conceived as the stiffness of the isotropic matrix that surrounds the fibers. The third term $\alpha_2(J_4 - I_1 + 2)$ in $\hat{\psi}_A$ is introduced to have a stress-free state in the undeformed configuration. Computation of the stress contributions from the anisotropic part requires the knowledge of the following derivatives

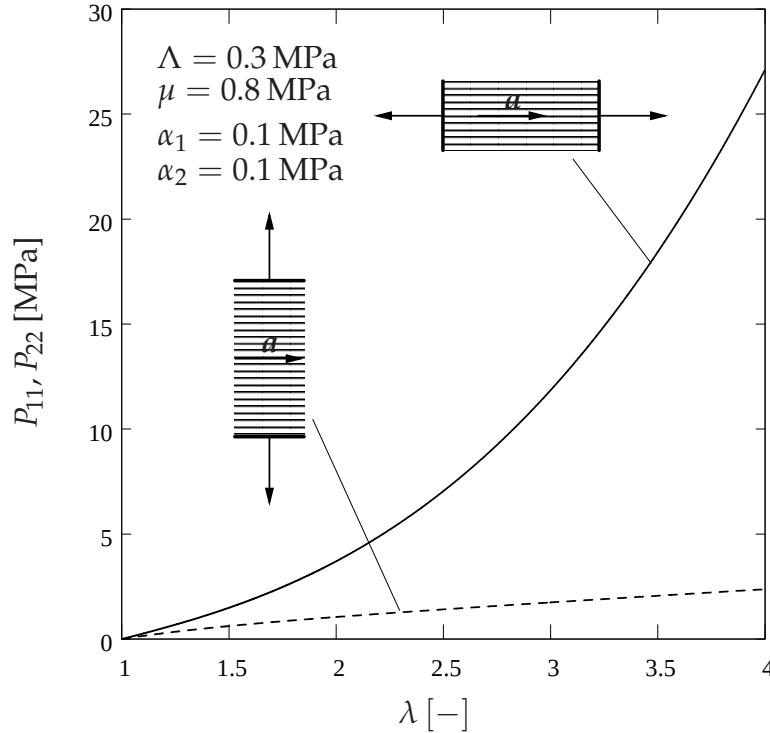
$$\begin{aligned} 2\partial_{\mathbf{C}} I_1 &= 2\partial_{\mathbf{C}}(\mathbf{C} : \mathbf{1}) = 2 \mathbf{1} , \\ 2\partial_{\mathbf{C}} I_2 &= 2\partial_{\mathbf{C}} \frac{1}{2} (I_1^2 - \mathbf{C}^2 : \mathbf{1}) = 2 (I_1 \mathbf{1} - \mathbf{C}) , \\ 2\partial_{\mathbf{C}} J_4 &= 2\partial_{\mathbf{C}}(\mathbf{C} : \mathbf{M}) = 2 \mathbf{M} , \\ 2\partial_{\mathbf{C}} J_5 &= 2\partial_{\mathbf{C}}(\mathbf{C}^2 : \mathbf{M}) = 2 (\mathbf{M} \cdot \mathbf{C} + \mathbf{C} \cdot \mathbf{M}) . \end{aligned}$$

The anisotropic part of the second Piola-Kirchhoff stresses is shown to be

$$\begin{aligned} \mathbf{S}_A = & 4 \alpha_1 (J_4 - 1) \mathbf{M} \\ & + 2 \alpha_2 [(\mathbf{M} \cdot \mathbf{C} + \mathbf{C} \cdot \mathbf{M}) - \mathbf{C} + (I_1 - J_4 - 1) \mathbf{1} - (I_1 - 1) \mathbf{M}] , \end{aligned}$$

from which the first Piola-Kirchhoff stresses can readily be computed through the push forward operation $\mathbf{P}_A = \mathbf{F} \cdot \mathbf{S}_A$.

Uniaxial stress-stretch response of the transversely isotropic model in two perpendicular directions is depicted below. Observe the fairly stiffer behavior in the direction coinciding with the preferred direction $\mathbf{a}$.



ME 338A – Continuum Mechanics

Tue/Thu 11:00-12:15, 530-127

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Although the basic concepts of continuum mechanics have been established more than five decades ago, the 21st century faces many new and exciting potential applications of continuum mechanics that go way beyond the standard classical theory. When applying continuum mechanics to these challenging new phenomena, it is important to understand the main three ingredients of continuum mechanics: the kinematic equations, the balance equations and the constitutive equations. After a brief repetition of the relevant equations in tensor algebra and analysis, this class will introduce the basic concepts of small strain kinematics. We will then discuss the balance equations for mass, momentum, moment of momentum, energy and entropy. While all these equations are general and valid for any kind of material, the last set of equations, the constitutive equations, specifies particular subclasses of materials. Towards the end of the quarter, we will illustrate how the three sets of equations will change in nonlinear continuum mechanics. Throughout this class, we will discuss the mechanics of the heart with its strains, fiber stretches, stresses, forces, and constitutive equations, to illustrate how continuum mechanics can be applied to relevant clinical problems.

Grading

Homework	30 %	three homework assignments, 10% each
Midterm	40 %	closed book, closed notes, one single page cheat sheet
Final Project	30 %	written evaluation of a manuscript and its discussion in class

Tue 01/05 **Tensor calculus I – Vector algebra**

Scalar product
Vector product
Scalar triple product

Thu 01/07 **Tensor calculus II – Tensor algebra**

Scalar product
Dyadic product
Scalar triple product

Tue 01/12 **Tensor calculus III – Tensor algebra**

Invariants, trace, determinant, inverse
Spectral decomposition
Symmetric – skew-symmetric decomposition
Volumetric – deviatoric decomposition
Orthogonal tensors

- Thu 01/14 **Tensor calculus IV – Tensor analysis**
Frechet and Gateaux derivatives
Gradient, Divergence, Laplace operator
Integral transformations – Green theorem, Gauss theorem
- Tue 01/19 **Kinematics I – Motion**
Motion, rates and gradients of kinematic quantities
Symmetric – skew-symmetric decomposition
Strain and rotation
- Thu 01/21 **Kinematics II – Strain**
Volumetric – deviatoric decomposition
Strain vector
Normal-shear decomposition
Principal strains – stretch
Compatibility
Plane strain
- Tue 01/26 **Balance equations I – Contact fluxes**
Concept of mass and heat flux
Concept of stress
Cauchy's postulate, lemma, and theorem
Stress tensor
- Thu 01/28 **Balance equations II – Concept of stress**
Volumetric – deviatoric decomposition
Normal – shear decomposition
Principal stresses
Plane stress
- Tue 02/02 **Balance equations III – Mass and momentum**
Global and local balance of mass
Global and local balance of momentum
- Thu 02/04 **Balance equations IV – Angular momentum and energy**
Global and local balance of angular momentum
Global and local balance of energy
First law of thermodynamics
- Tue 02/09 **Balance equations V - Entropy**
Global and local balance of entropy
Second law of thermodynamics
Global and local generic balance equation
Thermodynamic potentials

- Thu 02/11 **Constitutive equations I – Linear equations**
 Mass flux – Fick's law
 Momentum flux – Hooke's law
 Heat flux – Fourier's law
- Tue 02/16 **Constitutive equations II - Hyperelasticity**
 Specific stored energy
 Specific complementary energy
 Isotropic hyperelasticity
- Thu 02/18 **Constitutive equations III – Isotropic hyperelasticity**
 Specific stored energy
 Volumetric – deviatoric decomposition
 Specific complementary energy
 Volumetric – deviatoric decomposition
 Elastic constants
- Tue 02/23 **Prep midterm / Problems**
 Kinematics
 Balance equations
 Constitutive equations
- Thu 02/25 **Midterm**
- Tue 03/02 **Intro to nonlinear continuum mechanics I – Finite strain kinematics**
 Motion, configurations
 Velocity, acceleration
 Geometric maps
 Push forward and pull back
- Thu 03/04 **Intro to nonlinear continuum mechanics II – Stress measures**
 First and second Piola Kirchhoff, Cauchy, and Kirchhoff stresses
 Closure problem
 Frame invariance and material symmetry
- Tue 03/09 **Intro to nonlinear continuum mechanics III – Constitutive equations**
 Isotropic hyperelasticity
 Incompressible isotropic elasticity, example: Ogden model
 Transversely isotropic elasticity
- Thu 03/11 **Journal club – Final project**
 Final presentations / discussion
 Written projects due